GRADE 8 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 9) • Forward 2 (Grade 10)

58 question sets • 1450 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

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SET 1 • 8.NS.A.1 • THE NUMBER SYSTEM

Classify numbers as rational or irrational

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Write a fraction as a terminating or repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Write rational numbers as quotients of integers	**7.NS.A.2.b**
FORWARD 1	Decide whether sums and products are rational or irrational	**HSN-RN.B.3**
FORWARD 2	Extend the number system to complex numbers using i	**HSN-CN.A.1**

SET 1 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

Which number is irrational?

A. $\sqrt{49}$

B. $\frac{-7}{3}$

C. 0.1212... (the digits 12 repeat)

D. $\sqrt{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

8

QUESTION ID

S1-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. √15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q1/5

Is 0.375 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

9

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rational. It equals 3/8, a quotient of integers (its decimal ends).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q2/5

Is √50 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

10

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

Irrational. 50 is not a perfect square, so √50 cannot be written as a quotient of integers; its decimal neither ends nor repeats.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q3/5

True or false?

Every number whose decimal expansion repeats is rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

11

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q4/5

True or false?

π is rational because it is often written as 3.14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

12

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3.14 is only an approximation; π is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 1

Write a fraction as a terminating or repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

14

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.625

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q1/5

Write $\frac{2}{3}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

15

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{3}{5}$

B. $\frac{1}{6}$

C. $\frac{1}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

16

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{11}$ = 0.3636... (the digits 36 repeat)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

17

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{1}{3} = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

18

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/3 = 0.333..., a repeating decimal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 8.NS.A.1

BACKWARD 2

Write rational numbers as quotients of integers • Grade 7 • The Number System • 7.NS.A.2.b

Write $-2\frac{1}{4}$ as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

20

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−9/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q1/5

Write 0.7 as a quotient of two integers.

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

21

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q2/5

Which number is equal to $\frac{-5}{6}$?

A. $\frac{5}{6}$

B. $\frac{-5}{-6}$

C. $\frac{6}{-5}$

D. $\frac{5}{-6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

22

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5/(−6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q3/5

True or false?

-12 can be written as $\frac{-12}{1}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

23

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{-3}{0}$ is a rational number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

24

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 1

Decide whether sums and products are rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 4 + √3 rational or irrational? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

26

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational. A rational number plus an irrational number is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q1/5

Is √8 × √2 rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

27

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; √16 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q2/5

Which product is irrational?

A. $\sqrt{3} \times \sqrt{12}$

B. $3\sqrt{5}$

C. $0 \times \sqrt{7}$

D. $\frac{2}{3} \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

28

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $3\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

29

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q4/5

True or false?

The product of two irrational numbers is always irrational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

30

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{2} \times \sqrt{2} = 2$, which is rational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 8.NS.A.1

FORWARD 2

Extend the number system to complex numbers using i • High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1

The number i has $i^2 = -1$.

Write $\sqrt{-16}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

32

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q1/5

The number i has $i^2 = -1$.

What is i^4?

$i^4 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

33

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q2**

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q2/5

The number i has $i^2 = -1$.

Which number is a solution of $x^2 = -9$?

A. 3

B. -3

C. 9i

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

34

QUESTION ID

S1-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number a is also a complex number, a + 0i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

35

QUESTION ID

S1-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

$\sqrt{-25}$ is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

36

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{-25} = 5i$, which is not real.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 2 • Q5/5

SET 2 • 8.NS.A.1 • THE NUMBER SYSTEM

Convert a repeating decimal into a fraction

MAIN	Grade 8 main question	**8.NS.A.1**
BACKWARD 1	Use long division to write a fraction as a repeating decimal	**7.NS.A.2.d**
BACKWARD 2	Multiply decimals by powers of 10	**5.NBT.A.2**
FORWARD 1	Solve linear equations whose solutions are fractions	**HSA-REI.B.3**
FORWARD 2	Write a geometric sequence from a repeating decimal	**HSF-BF.A.2**

SET 2 • 8.NS.A.1

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.444... (the 4 repeats) as a fraction. Show your work.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

39

QUESTION ID

S2-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

4/9

GRADING NOTE REQUIRED WORK: Any valid method. Example: let x = 0.444...; then 10x = 4.444...; subtracting gives 9x = 4, so x = 4/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q1/5

Write 0.2727... (the digits 27 repeat) as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

40

QUESTION ID

S2-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/11

GRADING NOTE 27/99 is equal but is not in simplest form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q2/5

Which fraction is equal to 0.1666... (the 6 repeats)?

A. $\frac{1}{16}$

B. $\frac{1}{6}$

C. $\frac{16}{100}$

D. $\frac{6}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

41

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q3/5

True or false?

2.333... (the 3 repeats) is equal to $\frac{7}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

42

QUESTION ID

S2-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q4/5

True or false?

0.0909... (the digits 09 repeat) is equal to $\frac{9}{100}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

43

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is equal to 9/99 = 1/11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 1

Use long division to write a fraction as a repeating decimal • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{7}{9}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

45

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.777... (the 7 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q1/5

Write $\frac{5}{12}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

46

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.41666... (the 6 repeats)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{3}{11}$?

A. 0.27 (the decimal ends)

B. 0.3

C. 0.2727... (the digits 27 repeat)

D. 0.311

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

47

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.2727... (the digits 27 repeat)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{9} = 0.111...$ (the 1 repeats)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

48

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q4/5

True or false?

$\frac{5}{6} = 0.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

49

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5/6 = 0.8333...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 8.NS.A.1

BACKWARD 2

Multiply decimals by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

0.454 × 100

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

51

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

45.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q1/5

Multiply.

3.27 × 10

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

52

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

32.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q2/5

Multiply.

0.0081 × 1,000

A. 81

B. 8.1

C. 0.81

D. 0.081

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

53

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 \times 100 = 250$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

54

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$0.36 \times 10 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

55

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 1

Solve linear equations whose solutions are fractions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Write the answer as a fraction in simplest form.

$100x - x = 45$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

57

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5/11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$10x = x + 6.3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

58

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 7/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q2/5

Solve for x.

$1{,}000x - 10x = 270$

A. $x = \frac{3}{11}$

B. $x = \frac{1}{37}$

C. $x = \frac{27}{100}$

D. $x = 27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

59

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 3/11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q3/5

True or false?

$x = \frac{4}{9}$ is the solution of $10x - x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

60

QUESTION ID

S2-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

$x = 0.12$ is the solution of $100x - x = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

61

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is 12/99 = 4/33.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 8.NS.A.1

FORWARD 2

Write a geometric sequence from a repeating decimal • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

0.333... = 0.3 + 0.03 + 0.003 + ... The terms 0.3, 0.03, 0.003, ... form a geometric sequence.

What is the common ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

63

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.1

GRADING NOTE 1/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q1/5

Write an explicit formula for the nth term of the sequence 0.3, 0.03, 0.003, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

64

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.3(0.1)^(n − 1)

GRADING NOTE Equivalent forms such as 3(0.1)^n are correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q2**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q2/5

Which recursive formula gives the sequence 0.27, 0.0027, 0.000027, ...?

A. $a(1) = 0.27;\ a(n) = 100 \cdot a(n-1)$

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

C. $a(1) = 0.27;\ a(n) = a(n-1) - 0.01$

D. $a(1) = 0.0027;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

65

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $a(1) = 0.27;\ a(n) = 0.01 \cdot a(n-1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q3/5

True or false?

The sequence 0.5, 0.05, 0.005, ... is geometric with a common ratio of $\frac{1}{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

66

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q4/5

True or false?

The sequence 0.4, 0.04, 0.004, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

67

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The terms have a common ratio, not a common difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 2 • FORWARD 2 • Q5/5

SET 3 • 8.NS.A.2 • THE NUMBER SYSTEM

Estimate the value of an irrational number

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Find squares of whole numbers and decimals	**6.EE.A.1**
BACKWARD 2	Round decimals to the nearest tenth	**5.NBT.A.4**
FORWARD 1	Solve quadratic equations by taking square roots, exactly and ap…	**HSA-REI.B.4.b**
FORWARD 2	Find arc lengths and sector areas exactly in terms of π and appro…	**HSG-C.B.5**

SET 3 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √40?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

70

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

6 and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q1/5

Estimate √20 to the nearest tenth. Explain how you know.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

71

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

4.5. $4.4^2 = 19.36$ and $4.5^2 = 20.25$, and 20 is closer to 20.25.

GRADING NOTE Grade both: 4.5, and a reason that compares squares of tenths.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q2/5

Which is the best estimate of √75 to the nearest tenth?

A. 8.7

B. 8.6

C. 37.5

D. 9.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

72

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q3/5

True or false?

√2 is between 1.4 and 1.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

73

QUESTION ID

S3-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q4/5

True or false?

√90 is between 8 and 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

74

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $9^2 = 81$ and $10^2 = 100$, so √90 is between 9 and 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 1

Find squares of whole numbers and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

7^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

76

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q1/5

Find the value.

1.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

77

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q2/5

Find the value.

12^2

A. 24

B. 121

C. 122

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

78

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$8^2 = 64$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

79

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

$3^2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

80

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 8.NS.A.2

BACKWARD 2

Round decimals to the nearest tenth • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 4.472 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

82

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q1/5

Round 8.660 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

83

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q2/5

Round 3.162 to the nearest tenth.

A. 3.1

B. 3.0

C. 3.2

D. 3.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

84

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q3/5

True or false?

6.325 rounded to the nearest tenth is 6.3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

85

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q4/5

True or false?

2.449 rounded to the nearest tenth is 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

86

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 1

Solve quadratic equations by taking square roots, exactly and approximately • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 30$. Give the exact solutions, then round each to the nearest tenth.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

88

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{30}$ and $x = -\sqrt{30}$; about 5.5 and −5.5

GRADING NOTE Both solutions are required, exact and rounded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q1/5

Solve $x^2 - 7 = 0$. Give the exact solutions.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

89

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{7}$ and $x = -\sqrt{7}$

GRADING NOTE Both solutions are required; $x = \pm\sqrt{7}$ is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q2/5

Solve $(x - 1)^2 = 5$.

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

B. $x = 1 + \sqrt{5}$ only

C. $x = 6$ or $x = -4$

D. $x = \sqrt{6}$ or $x = -\sqrt{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

90

QUESTION ID

S3-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $x = 1 + \sqrt{5}$ or $x = 1 - \sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S3-F1-Q3**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q3/5

True or false?

The solutions of $x^2 = 12$ are $x = 2\sqrt{3}$ and $x = -2\sqrt{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

91

QUESTION ID

S3-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q4/5

True or false?

$x^2 = -9$ has the real solutions 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

92

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No real number squared is negative, so there is no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 3 • FORWARD 1 • Q5/5

SET 3 • 8.NS.A.2

FORWARD 2

Find arc lengths and sector areas exactly in terms of π and approximately • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5

A circle has a radius of 6 cm.

Find the length of a 90° arc exactly in terms of π, then to the nearest tenth.

Arc length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5

94

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3π cm, about 9.4 cm

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q1/5

A circle has a radius of 4 inches.

Find the area of a 45° sector exactly in terms of π, then to the nearest tenth.

Sector area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5

95

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2π square inches, about 6.3 square inches

GRADING NOTE Both the exact and the rounded value are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q2/5

A circle has a radius of 3 m.

To the nearest tenth, how long is a 60° arc?

A. 6.3 m

B. 9.4 m

C. 3.1 m

D. 18.8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5

96

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.1 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q3/5

True or false?

The arc of a semicircle with a radius of 1 is π units long, about 3.14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5

97

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q4/5

True or false?

A 180° sector of a circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5

98

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its area is half of 4π, which is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.B.5 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 8.NS.A.2 • THE NUMBER SYSTEM

Locate and compare irrational numbers on a number line

MAIN	Grade 8 main question	**8.NS.A.2**
BACKWARD 1	Read inequalities as positions on a number line	**6.NS.C.7.a**
BACKWARD 2	Position decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Report irrational values to a level of accuracy that fits the situati…	**HSN-Q.A.3**
FORWARD 2	Use coordinates to find exact lengths and perimeters with radicals	**HSG-GPE.B.7**

SET 4 • 8.NS.A.2

MAIN

Grade 8 main question • Grade 8 • The Number System • 8.NS.A.2

Plot √5 on the number line. Use an estimate to the nearest tenth.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

101

QUESTION ID

S4-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A point at about 2.2: between 2 and 2.5, closer to 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q1/5

Which list is in order from least to greatest?

A. 2, √5, √3, 1.8

B. √3, 1.8, 2, √5

C. √3, 2, 1.8, √5

D. 1.8, √3, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

102

QUESTION ID

S4-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. √3, 1.8, 2, √5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q2/5

Write <, >, or =.

π ___ √10

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

103

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

<

GRADING NOTE π is about 3.14 and √10 is about 3.16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q3/5

Which number is the best name for point P: √2, √3, or √6?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

104

QUESTION ID

S4-M-Q4

TYPE

Short answer

CORRECT ANSWER

√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q4/5

True or false?

√30 is to the left of 5.5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

105

QUESTION ID

S4-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE √30 is about 5.48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 4 • MAIN • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 1

Read inequalities as positions on a number line • Grade 6 • The Number System • 6.NS.C.7.a

2.4 is to the right of 2.25 on a number line.

Write an inequality that shows this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

107

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.4 > 2.25

GRADING NOTE 2.25 < 2.4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q1/5

Which number is farthest to the left on a number line?

A. 0.5

B. −1.9

C. −2.05

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

108

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. −2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q2/5

What does 1.7 < 1.75 tell you about the positions of the two numbers on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

109

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

1.7 is to the left of 1.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q3/5

True or false?

$-1.4 > -1.41$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

110

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q4/5

True or false?

3.09 > 3.1

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

111

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 8.NS.A.2

BACKWARD 2

Position decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot 1.4 and 2.75 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

113

QUESTION ID

S4-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

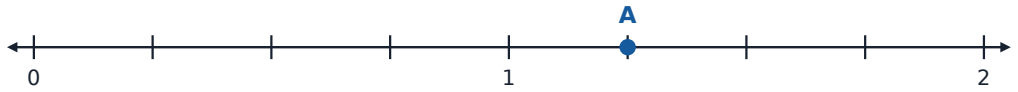
Points at 1.4 (a little past the 1.25 mark) and 2.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q1/5

What decimal is at point A?



Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

114

QUESTION ID

S4-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q2/5

Which decimal is between 1.7 and 1.8?

A. 1.81

B. 1.73

C. 1.69

D. 1.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

115

QUESTION ID

S4-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q3/5

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

116

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q4/5

True or false?

0.9 is to the right of 1.1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

117

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 1

Report irrational values to a level of accuracy that fits the situation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A square garden has an area of 50 square meters, so its side is √50 m. A tape measure reads to the nearest centimeter.

Report the side length to that accuracy, in meters.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

119

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7.07 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q1/5

A coin has a diameter of 4 cm. Its circumference is 4π cm.

Report the circumference to the nearest tenth of a centimeter.

Circumference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

120

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q2/5

Which value of √2 is given to the nearest thousandth?

A. 1.41

B. 1.414

C. 1.4142

D. 1.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

121

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.414

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q3/5

True or false?

Reporting √3 as 1.7 is accurate to the nearest tenth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

122

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q4/5

True or false?

Writing more digits of √5 always makes a measured length more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

123

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A reported length cannot be more accurate than the measuring tool.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 8.NS.A.2

FORWARD 2

Use coordinates to find exact lengths and perimeters with radicals • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the exact perimeter of the triangle with vertices (0, 0), (2, 0), and (0, 2).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

125

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 2√2 units

GRADING NOTE About 6.8 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q1/5

Find the exact length of the segment from (1, 1) to (4, 3).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

126

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

√13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q2/5

A square has vertices (0, 0), (1, 2), (−1, 3), and (−2, 1).

What is its perimeter?

A. 2√5 units

B. 20 units

C. 4√5 units

D. 4√3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

127

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4√5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S4-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q3/5

True or false?

The distance from $(-1, -1)$ to $(2, 3)$ is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

128

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices (0, 0), (3, 0), (3, 1), and (0, 1) has a diagonal 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

129

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diagonal is √10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 4 • FORWARD 2 • Q5/5

SET 5 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the product and quotient properties of integer exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Evaluate whole-number exponents	**6.EE.A.1**
BACKWARD 2	Write powers of 10 with whole-number exponents	**5.NBT.A.2**
FORWARD 1	Multiply and divide monomials using exponent properties	**HSA-SSE.A.2**
FORWARD 2	Use exponent properties to rewrite and interpret exponential funct	**HSF-IF.C.8.b**

SET 5 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$3^4 \times 3^5$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

132

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

3^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q1/5

Write as a single power.

$7^{10} \div 7^{4}$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

133

QUESTION ID

S5-M-Q2

TYPE

Short answer

CORRECT ANSWER

7^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q2/5

Which expression is equal to 2^6?

A. $2^{12} \div 2^2$

B. 2×6

C. $2^2 \times 2^3$

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

134

QUESTION ID

S5-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2^2 \times 2^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q3/5

True or false?

$5^3 \times 5^2 = 5^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

135

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q4/5

True or false?

$4^8 \div 4^2 = 4^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

136

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^8 \div 4^2 = 4^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 5 • MAIN • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 1

Evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

2^5

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

138

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q1/5

Find the value.

10^4

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

139

QUESTION ID

S5-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q2/5

Find the value.

3^4

A. 81

B. 64

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

140

QUESTION ID

S5-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$6^2 = 36$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

141

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q4/5

True or false?

$2^3 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

142

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 8.EE.A.1

BACKWARD 2

Write powers of 10 with whole-number exponents • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

144

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10^4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q1/5

Write 1,000,000 as a power of 10.

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

145

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q2/5

What is the value of 10^5?

A. 10,000

B. 50

C. 100,000

D. 1,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

146

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 100,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q3/5

True or false?

$10^3 = 1{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

147

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q4/5

True or false?

$10^2 \times 10 = 10^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

148

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^2 \times 10 = 1{,}000 = 10^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 1

Multiply and divide monomials using exponent properties • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Simplify.

$(3x^4)(5x^2)$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

150

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$15x^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q1/5

Simplify. Assume $x \neq 0$.

$$\frac{12x^7}{4x^3}$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

151

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$3x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q2/5

Simplify.

$x^5 \cdot x \cdot x^2$

A. x^7

B. $3x^8$

C. x^{10}

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

152

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q3/5

True or false?

$(2a^3)(4a^2) = 8a^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

153

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q4/5

True or false?

Assume $y \neq 0$. $\frac{y^9}{y^3} = y^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

154

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $y^9 \div y^3 = y^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 8.EE.A.1

FORWARD 2

Use exponent properties to rewrite and interpret exponential functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b

f(t) = 2^(t + 3)

Use a property of exponents to write f(t) in the form a · 2^t.

f(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

156

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8 · 2^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q1/5

A population is P(t) = 200 · 2^t after t years.

By what factor does the population grow each year, and what is P(5)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

157

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It doubles (factor 2) each year;
P(5) = 6,400

GRADING NOTE Both parts are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q2/5

$g(t) = 5^{(t - 1)}$

Which expression is equal to g(t)?

A. $5^t - 5$

B. $5 \div 5^t$

C. $5^t \div 5$

D. $5^t - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

158

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $5^t \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q3/5

True or false?

If f(x) = 5 · 2^x, then f(x + 1) = 2 · f(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

159

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q4/5

True or false?

If f(x) = 2^x, then f(3) · f(4) = f(12).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

160

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(3) · f(4) = $2^3 \cdot 2^4 = 2^7$ = f(7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 5 • FORWARD 2 • Q5/5

SET 6 • 8.EE.A.1 • EXPRESSIONS & EQUATIONS

Use the power-of-a-power property and zero and negative exponents

MAIN	Grade 8 main question	**8.EE.A.1**
BACKWARD 1	Find powers of fractions and decimals	**6.EE.A.1**
BACKWARD 2	Multiply signed numbers	**7.NS.A.2.a**
FORWARD 1	Explain rational exponents with the properties of exponents	**HSN-RN.A.1**
FORWARD 2	Rewrite exponential expressions with the properties of exponents	**HSA-SSE.B.3.c**

SET 6 • 8.EE.A.1

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power.

$(5^2)^3$

Power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

163

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

5^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q1/5

Find the value.

4^{-2}

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

164

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q2/5

Which expression is equal to 1?

A. 0^9

B. 9^0

C. $1^{-9} \times 9$

D. 9^{-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9^0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q3/5

True or false?

$3^{-2} \times 3^{5} = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

166

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q4/5

True or false?

$(2^3)^2 = 2^5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

167

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(2^3)^2 = 2^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 6 • MAIN • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 1

Find powers of fractions and decimals • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

$(\frac{1}{2})^3$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

169

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q1/5

Find the value.

0.3^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

170

QUESTION ID

S6-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.09

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q2/5

Find the value.

$(\frac{2}{3})^2$

A. $\frac{4}{3}$

B. $\frac{4}{9}$

C. $\frac{2}{9}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

171

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$0.1^3 = 0.001$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

172

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q4/5

True or false?

$(\frac{3}{4})^2 = \frac{6}{8}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

173

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3/4)^2 = 9/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 8.EE.A.1

BACKWARD 2

Multiply signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$(-2)(-2)(-2)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

175

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q1/5

Multiply.

$(-3)(-3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

176

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q2/5

Multiply.

$(-1)(-1)(-1)(-1)$

A. -1

B. -4

C. 1

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

177

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q3/5

True or false?

$(-5)(-5) = 25$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

178

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q4/5

True or false?

$(-2)(-2)(-2)(-2) = -16$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

179

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 1

Explain rational exponents with the properties of exponents • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1

$(5^{(1/2)})^2 = 5^{((1/2) \cdot 2)} = 5.$

Which radical is equal to $5^{(1/2)}$?

Radical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

181

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q1/5

Write 7^(1/3) as a radical.

Radical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

182

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q2/5

Which number is equal to 9^(3/2)?

A. 6

B. 27

C. 13.5

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

183

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q3/5

True or false?

Because $(8^{(1/3)})^3 = 8$, $8^{(1/3)} = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

184

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

x^(1/2) means x ÷ 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

185

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^(1/2) is the square root of x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 8.EE.A.1

FORWARD 2

Rewrite exponential expressions with the properties of exponents • High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c

Rewrite 1.21^t as a power of 1.1.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

187

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.1^(2t)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q1/5

A value grows by a factor of 1.04 each month, so after t years it has grown by (1.04^12)^t.

Write the yearly growth factor, rounded to the nearest hundredth.

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

188

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.60

GRADING NOTE 1.04^12 ≈ 1.601, about 60% growth per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q2/5

Which expression is equivalent to 8^x?

A. 2^x ÷ 3

B. 3^(2x)

C. 2^(3x)

D. 2^(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

189

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2^(3x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q3/5

True or false?

$(0.5)^{(-t)} = 2^t$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

190

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q4/5

True or false?

$4^x \times 4^x = 16^{(2x)}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

191

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^x \times 4^x = 4^{(2x)} = 16^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 10 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.c • Set 6 • FORWARD 2 • Q5/5

SET 7 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Solve equations of the form $x^2 = p$ and $x^3 = p$

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Solve one-step multiplication equations	**6.EE.B.7**
BACKWARD 2	Find squares and cubes of whole numbers	**6.EE.A.1**
FORWARD 1	Solve quadratic equations by taking square roots	**HSA-REI.B.4.b**
FORWARD 2	Find a dimension from a volume with square and cube roots	**HSG-GMD.A.3**

SET 7 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve for x. Give all solutions.

$x^2 = 49$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

194

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

x = 7 and x = −7

GRADING NOTE Both solutions are required; x = ±7 is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q1/5

Solve for x.

$x^3 = 64$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

195

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q2/5

Solve for x.

$x^2 = \frac{9}{16}$

A. $x = \frac{3}{4}$ only

B. $x = \frac{3}{4}$ or $x = -\frac{3}{4}$

C. $x = \frac{9}{32}$

D. $x = \frac{81}{256}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

196

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3/4 or x = −3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q3/5

Solve for x.

$x^3 = -27$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

197

QUESTION ID

S7-M-Q4

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q4/5

True or false?

$x^2 = 10$ has the two solutions $x = \sqrt{10}$ and $x = -\sqrt{10}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

198

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 7 • MAIN • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 1

Solve one-step multiplication equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

4x = 64

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

200

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q1/5

Solve for x.

$\frac{x}{3} = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

201

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q2/5

Solve for x.

9x = 45

A. x = 36

B. x = 5

C. x = 405

D. x = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q3/5

True or false?

x = 12 is the solution of 3x = 36.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

203

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q4/5

True or false?

x = 5 is the solution of 7x = 42.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

204

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 8.EE.A.2

BACKWARD 2

Find squares and cubes of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

9^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

206

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q1/5

Find the value.

5^3

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

207

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which whole number cubed is 8?

A. 3

B. 4

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

208

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q3/5

True or false?

$10^4 = 10{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

209

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q4/5

True or false?

$4^2 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

210

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 1

Solve quadratic equations by taking square roots • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x. Give all solutions.

$2x^2 = 50$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

212

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 5 and x = −5

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q1/5

Solve for x. Give all solutions.

$(x + 2)^2 = 36$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

213

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 4 and x = −8

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$x^2 - 9 = 0$

A. x = 9 or x = −9

B. x = 3 only

C. x = 3 or x = −3

D. x = 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

214

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 3 or x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q3/5

True or false?

The solutions of $3x^2 = 27$ are 3 and −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

215

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q4/5

True or false?

The solutions of $(x - 1)^2 = 16$ are 5 and 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

216

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solutions are 5 and −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 8.EE.A.2

FORWARD 2

Find a dimension from a volume with square and cube roots • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A cube has a volume of 343 cubic centimeters.

How long is each edge?

Edge: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

218

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q1/5

A sphere has a volume of 36π cubic centimeters.

($V = \frac{4}{3}\pi r^3$)

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

219

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q2/5

A cylinder has a volume of 200π cubic units and a height of 8 units.

What is its radius?

A. 10 units

B. 25 units

C. 5 units

D. √200 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

220

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q3/5

True or false?

A cone with a volume of 12π cubic units and a height of 4 units has a radius of 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

221

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q4/5

True or false?

A cube with a volume of 1,000 cubic centimeters has edges 100 cm long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

222

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each edge is 10 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 7 • FORWARD 2 • Q5/5

SET 8 • 8.EE.A.2 • EXPRESSIONS & EQUATIONS

Evaluate square roots and cube roots and recognize irrational roots

MAIN	Grade 8 main question	**8.EE.A.2**
BACKWARD 1	Find factor pairs and square numbers	**4.OA.B.4**
BACKWARD 2	Find the volume of a cube	**5.MD.C.5.b**
FORWARD 1	Simplify square roots by factoring out perfect squares	**HSN-RN.A.2**
FORWARD 2	Solve quadratic equations by completing the square	**HSA-REI.B.4.a**

SET 8 • 8.EE.A.2

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.2

Find the value.

√144

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

225

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q1/5

Find the value.

$\sqrt[3]{125}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

226

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q2/5

Which square root is irrational?

A. $\sqrt{\frac{1}{4}}$

B. $\sqrt{0.09}$

C. $\sqrt{20}$

D. $\sqrt{36}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. √20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q3/5

True or false?

$\sqrt[3]{(-8)} = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

228

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q4/5

True or false?

√2 can be written as a fraction of two whole numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

229

QUESTION ID
S8-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE √2 is irrational.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • MAIN • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 1

Find factor pairs and square numbers • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factor pairs of 16.

Factor pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

231

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 × 16, 2 × 8, and 4 × 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q1/5

What whole number times itself is 49?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

232

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q2/5

Which number has a factor pair made of two equal factors?

A. 30

B. 24

C. 18

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

233

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q3/5

True or false?

81 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

234

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q4/5

True or false?

12 has a factor pair made of two equal factors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

235

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 8.EE.A.2

BACKWARD 2

Find the volume of a cube • Grade 5 • Measurement & Data • 5.MD.C.5.b

A cube has edges 3 units long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

237

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q1/5

A cube has edges 5 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

238

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

125 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q2/5

Which cube has a volume of 64 cubic units?

A. A cube with 32-unit edges

B. A cube with 4-unit edges

C. A cube with 16-unit edges

D. A cube with 8-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

239

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A cube with 4-unit edges

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q3/5

True or false?

A cube with 2-unit edges has a volume of 8 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

240

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q4/5

True or false?

A cube with 10-unit edges has a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

241

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 1

Simplify square roots by factoring out perfect squares • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify.

√48

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

243

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q1/5

Simplify.

√200

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

244

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q2/5

Which expression is equal to √18?

A. 2√3

B. 9√2

C. 3√2

D. 6√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

245

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3√2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q3/5

True or false?

$\sqrt{50} = 5\sqrt{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

246

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q4/5

True or false?

$\sqrt{27} = 9\sqrt{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

247

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{27} = 3\sqrt{3}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q5**

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 8.EE.A.2

FORWARD 2

Solve quadratic equations by completing the square • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

What number goes in the blank?

$x^2 + 6x + 9 = (x + ___)^2$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

249

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q1**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve by completing the square. Give exact solutions.

$x^2 + 4x = 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

250

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

x = −2 + √5 and
x = −2 − √5

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q2/5

Which number completes the square?

$x^2 - 10x + ___$

A. 5

B. −25

C. 100

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

251

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q3/5

True or false?

$x^2 + 8x + 16 = (x + 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

252

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q4/5

True or false?

Completing the square on $x^2 + 2x = 3$ gives $(x + 1)^2 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

253

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives $(x + 1)^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 8 • FORWARD 2 • Q5/5

SET 9 • 8.EE.A.3 • EXPRESSIONS & EQUATIONS

Estimate very large and very small quantities as a single digit times a power of 10

MAIN	Grade 8 main question	**8.EE.A.3**
BACKWARD 1	Multiply and divide by powers of 10	**5.NBT.A.2**
BACKWARD 2	Round whole numbers to any place	**4.NBT.A.3**
FORWARD 1	Choose a level of accuracy for very large and very small quantities	**HSN-Q.A.3**
FORWARD 2	Use base-10 logarithms to describe orders of magnitude	**HSF-LE.A.4**

SET 9 • 8.EE.A.3

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.3

Estimate 412,000,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

256

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

4×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q1/5

Estimate 0.00079 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

257

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

8×10^{-4}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q2/5

A country has about 3×10^8 people. A city has about 6×10^5 people.

About how many times as many people live in the country?

A. 2

B. 50

C. 500

D. 5,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

258

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q3/5

A whale has a mass of about 2×10^5 kg. A mouse has a mass of about 2×10^{-2} kg.

About how many times as large is the whale's mass?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

259

QUESTION ID

S9-M-Q4

TYPE

Short answer

CORRECT ANSWER

10,000,000

GRADING NOTE 10^7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q4/5

True or false?

6×10^{-3} is greater than 9×10^{-4}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

260

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 1

Multiply and divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

4×10^3

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

262

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q1/5

Divide.

$7 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

263

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.07

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q2/5

What is the value of 3×10^5?

A. 15

B. 3,000,000

C. 300,000

D. 30,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

264

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 300,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$6 \div 10^3 = 0.006$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

265

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$8 \times 10^2 = 8{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

266

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 8.EE.A.3

BACKWARD 2

Round whole numbers to any place • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3

Round 412,000,000 to the nearest hundred million.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

268

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

400,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q1/5

Round 6,750 to the nearest thousand.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

269

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q2/5

Round 58,300 to the nearest ten thousand.

A. 50,000

B. 59,000

C. 60,000

D. 58,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

270

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

245,000 rounded to the nearest hundred thousand is 200,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

271

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

8,500 rounded to the nearest thousand is 8,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

272

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 1

Choose a level of accuracy for very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

Earth's mean radius is about 6,371 km.

Write it to one significant digit as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

274

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^3 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q1/5

A bacterium is 0.0000023 m long.

Write its length in micrometers. (1 μm = 10^{-6} m)

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

275

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2.3 μm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q2/5

A sheet of paper is about 0.0001 m thick.

Which unit best expresses this thickness?

A. Millimeters (0.1 mm)

B. Megameters

C. Meters

D. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

276

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Millimeters (0.1 mm)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reporting 7,900,000,000 people as 8×10^9 is accurate to one significant digit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

277

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A distance known to be about 3×10^8 m is reported more accurately as 300,000,000.0 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

278

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Writing more digits does not make an estimate more accurate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 8.EE.A.3

FORWARD 2

Use base-10 logarithms to describe orders of magnitude • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Find the value.

$\log_{10}(1{,}000{,}000)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

280

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q1**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q1/5

Solve for x.

10^x = 0.001

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

281

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q2**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q2/5

Between which two whole numbers is $\log_{10}(4{,}500)$?

A. 2 and 3

B. 45 and 46

C. 4 and 5

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

282

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F2-Q3**

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q3/5

True or false?

$\log_{10}(10^{-5}) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

283

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q4/5

True or false?

$\log_{10}(50)$ is between 5 and 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

284

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10^1 = 10$ and $10^2 = 100$, so $\log_{10}(50)$ is between 1 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 9 • FORWARD 2 • Q5/5

SET 10 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Multiply and divide numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Multiply decimals	**5.NBT.B.7**
BACKWARD 2	Multiply by powers of 10	**5.NBT.A.2**
FORWARD 1	Use units to guide multistep calculations in scientific notation	**HSN-Q.A.1**
FORWARD 2	Model exponential growth and decay with scientific notation	**HSF-LE.A.1.c**

SET 10 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Multiply. Write the answer in scientific notation.

$(4 \times 10^{6})(2.5 \times 10^{3})$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

287

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

1×10^{10}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q1/5

Divide. Write the answer in scientific notation.

$(6.3 \times 10^{8}) \div (9 \times 10^{2})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

288

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

7×10^{5}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q2/5

Light travels about 3×10^5 km per second.

How far does it travel in 5×10^2 seconds?

A. 1.5×10^7 km

B. 6×10^2 km

C. 1.5×10^8 km

D. 8×10^7 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

289

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.5×10^8 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q3/5

True or false?

$(2 \times 10^{-3})(3 \times 10^{-4}) = 6 \times 10^{-7}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

290

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q4/5

True or false?

$(8 \times 10^9) \div (4 \times 10^3) = 2 \times 10^3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

291

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 2×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 10 • MAIN • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 1

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.5 × 4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

293

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q1/5

Multiply.

1.2 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

294

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q2/5

Multiply.

6×0.5

A. 6.5

B. 3

C. 0.3

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

295

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$2.4 \times 0.5 = 1.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

296

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$3.5 \times 2 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

297

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 8.EE.A.4

BACKWARD 2

Multiply by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Multiply.

2.5×10^4

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

299

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q1/5

Multiply.

3.1×10^2

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

300

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

310

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q2/5

Multiply.

0.6×10^3

A. 60

B. 0.0006

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

301

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$7 \times 10^3 = 7{,}000$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

302

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

303

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 1

Use units to guide multistep calculations in scientific notation • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A drive transfers 2×10^8 bytes per second.

How many seconds does it take to transfer 6×10^{10} bytes?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

305

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q1/5

Light travels 3×10^8 meters per second.

How many meters does it travel in 1 hour (3.6×10^3 seconds)? Use scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

306

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.08×10^{12} m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S10-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q2/5

The Sun is about 1.5×10^{11} m from Earth. Light travels 3×10^{8} m per second.

About how long does sunlight take to reach Earth?

A. 500 seconds

B. 4.5×10^{19} seconds

C. 5×10^{19} seconds

D. 50 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

307

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 500 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q3/5

True or false?

A flow of 5×10^3 liters per minute is 3×10^5 liters per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

308

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q4/5

True or false?

6×10^{-3} km is equal to 6×10^{-6} m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

309

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6×10^{-3} km = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 8.EE.A.4

FORWARD 2

Model exponential growth and decay with scientific notation • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A colony of 2×10^3 bacteria triples every hour.

How many bacteria are there after 4 hours? Use scientific notation.

Bacteria: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

311

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.62×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q1/5

A quantity of 5×10^9 is cut in half each year.

What is it after 3 years? Use scientific notation.

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

312

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6.25×10^8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q2/5

A population of 4×10^6 grows by 10% each year.

What is the population after 2 years?

A. 4.84×10^6

B. 8×10^6

C. 4.4×10^6

D. 4.8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

313

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4.84×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q3/5

True or false?

Doubling 3×10^5 ten times gives about 3.07×10^8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q4/5

True or false?

A quantity that grows by a constant percent rate increases by the same amount each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

315

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 10 • FORWARD 2 • Q5/5

SET 11 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Add and subtract numbers written in scientific notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Add decimals	**5.NBT.B.7**
BACKWARD 2	Write a number times a power of 10 in standard form	**5.NBT.A.2**
FORWARD 1	Add and subtract polynomials	**HSA-APR.A.1**
FORWARD 2	Add and subtract functions	**HSF-BF.A.1.b**

SET 11 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

Add. Write the answer in scientific notation.

$(3.2 \times 10^5) + (4.5 \times 10^4)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

318

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

3.65×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q1/5

Subtract. Write the answer in scientific notation.

$(9 \times 10^{7}) - (6 \times 10^{6})$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

319

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

8.4×10^{7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q2/5

Add. Write the answer in scientific notation.

$(1.5 \times 10^{-3}) + (2 \times 10^{-4})$

A. 1.7×10^{-4}

B. 3.5×10^{-3}

C. 1.7×10^{-3}

D. 3.5×10^{-7}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

320

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1.7×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q3/5

True or false?

$(5 \times 10^8) + (5 \times 10^8) = 1 \times 10^9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

321

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(4 \times 10^6) - (3 \times 10^5) = 1 \times 10^1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

322

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3.7×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • MAIN • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 1

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.2 + 0.45

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q1/5

Add.

8.4 + 1.6

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

325

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q2/5

Add.

1.5 + 0.2

A. 0.17

B. 1.7

C. 1.52

D. 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

326

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q3/5

True or false?

2.75 + 0.25 = 3

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

327

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q4/5

True or false?

$4.6 + 0.04 = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

328

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 8.EE.A.4

BACKWARD 2

Write a number times a power of 10 in standard form • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Write 6×10^6 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

330

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q1/5

Write 0.6×10^7 in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

331

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6,000,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q2/5

Which number is equal to 45×10^4?

A. 450,000

B. 4,500

C. 45,000

D. 4,500,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

332

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 450,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q3/5

True or false?

$3.2 \times 10^2 = 320$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

333

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

$0.5 \times 10^3 = 50$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

334

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 1

Add and subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Add.

$(3x^2 + 5x) + (4x^2 - 2x)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

336

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$7x^2 + 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q1/5

Subtract.

$(6a^3 - 2a + 1) - (2a^3 + 5a - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

337

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$4a^3 - 7a + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q2/5

Add.

$(x^2 - 4) + (3x^2 + x)$

A. $4x^2 + x - 4$

B. $3x^4 + x - 4$

C. $4x^2 - 3$

D. $4x^2 + x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

338

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x^2 + x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q3/5

True or false?

$(2y^2 + 3) - (y^2 - 3) = y^2 + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

339

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q4/5

True or false?

$(5n - 2) - (3n - 4) = 2n - 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

340

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 2n + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 8.EE.A.4

FORWARD 2

Add and subtract functions • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b

Revenue is R(x) = 25x and cost is C(x) = 4,000 + 9x for x items.

Write the profit function P(x) = R(x) − C(x).

P(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5

342

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 4,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q1/5

The profit function is P(x) = 16x − 4,000.

Find P(1,000).

P(1,000) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5

343

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q2/5

$f(x) = 3x^2$ and $g(x) = x^2 - 5$.

Which expression is $(f + g)(x)$?

A. $2x^2 + 5$

B. $4x^2 - 5$

C. $3x^4 - 5$

D. $4x^2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5

344

QUESTION ID

S11-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4x^2 - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q3/5

True or false?

If f(x) = 2x and g(x) = x + 1, then (f − g)(3) = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5

345

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x + 4$ and $g(x) = 2x$, then $(f + g)(x) = 2x^2 + 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5

346

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(f + g)(x) = 3x + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.1.b • Set 11 • FORWARD 2 • Q5/5

SET 12 • 8.EE.A.4 • EXPRESSIONS & EQUATIONS

Choose units for scientific notation and read calculator notation

MAIN	Grade 8 main question	**8.EE.A.4**
BACKWARD 1	Convert metric units	**5.MD.A.1**
BACKWARD 2	Divide by powers of 10	**5.NBT.A.2**
FORWARD 1	Convert units of very large and very small quantities	**HSN-Q.A.1**
FORWARD 2	Define the quantities and units a model needs	**HSN-Q.A.2**

SET 12 • 8.EE.A.4

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A calculator shows 4.7E8.

Write the number in scientific notation and in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

349

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.7×10^8

$= 470{,}000{,}000$

GRADING NOTE Grade both forms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q1/5

A calculator shows 2.05E-6.

Write the number in standard form.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

350

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

0.00000205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q2/5

The distance from Earth to the Moon is about 3.84×10^8 m.

Which unit is the most sensible for this distance?

A. Kilometers

B. Centimeters

C. Millimeters

D. Micrometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

351

QUESTION ID

S12-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q3/5

A dust particle has a mass of 7.53×10^{-10} kg.

Write the mass in micrograms. (1 μg = 10^{-9} kg)

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

352

QUESTION ID

S12-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.753 μg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q4/5

True or false?

A calculator display of 3E-4 means −30,000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

353

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3E-4 means $3 \times 10^{-4} = 0.0003$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • MAIN • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 1

Convert metric units • Grade 5 • Measurement & Data • 5.MD.A.1

How many grams are in 3.5 kilograms?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

355

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

How many meters are in 4,200 millimeters?

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

356

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many centimeters are in 6 meters?

A. 60

B. 0.06

C. 6,000

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

357

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q3

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

True or false?

5 km = 5,000 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

358

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

250 cm = 25 m

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

359

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 5 • Measurement & Data • 5.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 8.EE.A.4

BACKWARD 2

Divide by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

Divide.

$3{,}840 \div 10^3$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

361

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

Divide.

$75 \div 10^2$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

362

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

$6 \div 10^4$

A. 0.006

B. 0.00006

C. 60,000

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

363

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.0006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

True or false?

520 ÷ 10 = 52

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

364

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

True or false?

$9 \div 10^3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

365

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 1

Convert units of very large and very small quantities • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

Convert 3.6×10^6 mm to kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

367

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.6 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q1/5

A server sends 4.5×10^9 bytes per minute.

Write this rate in gigabytes per second. (1 GB = 10^9 bytes)

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

368

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.075 GB per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F1-Q2**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q2/5

A virus is 1.2×10^{-7} m wide.

What is its width in nanometers? (1 nm = 10^{-9} m)

A. 1,200 nm

B. 120 nm

C. 12 nm

D. 0.12 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

369

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 120 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q3/5

True or false?

7×10^5 g = 700 kg

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

370

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q4/5

True or false?

2.5×10^{-3} km = 2.5 cm

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

371

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2.5×10^{-3} km = 2.5 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 8.EE.A.4

FORWARD 2

Define the quantities and units a model needs • High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2

You want to estimate how much water a city of 2×10^6 people uses each day.

Which quantities do you need, and with what units?

Quantities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

373

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The number of people and the water used per person per day (liters per person per day).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q1/5

2×10^6 people each use about 3×10^2 liters of water per day.

About how many liters does the city use per day? Use scientific notation.

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

374

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6×10^8 liters per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q2/5

Which unit is the best choice for the mass of a large ship?

A. Metric tons

B. Grams

C. Ounces

D. Milligrams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

375

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Metric tons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q3/5

True or false?

Choosing units that keep numbers between about 1 and 1,000 can make a model easier to read.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

376

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q4/5

True or false?

Kilometers are the best unit for the width of a cell.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

377

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 10 in this bank) • Number & Quantity: Quantities • HSN-Q.A.2 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Graph a proportional relationship and interpret the unit rate as the slope

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Find the constant of proportionality from a graph	**7.RP.A.2.b**
FORWARD 1	Calculate and interpret an average rate of change	**HSF-IF.B.6**
FORWARD 2	Compare properties of functions represented in different ways	**HSF-IF.C.9**

SET 13 • 8.EE.B.5

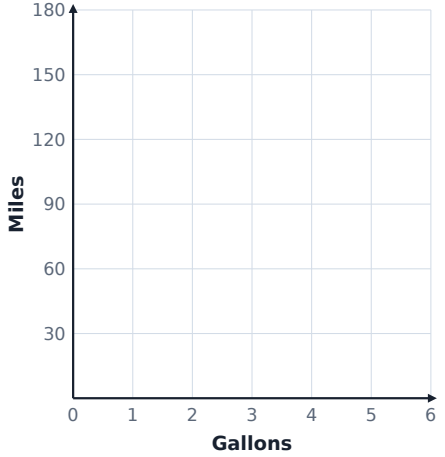
MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

A car uses gas at a constant rate: 3 gallons for every 90 miles.

Graph the relationship between gallons (x) and miles (y). What is the slope, and what does it mean?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

380

QUESTION ID

S13-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 0) and points such as (1, 30) and (3, 90)

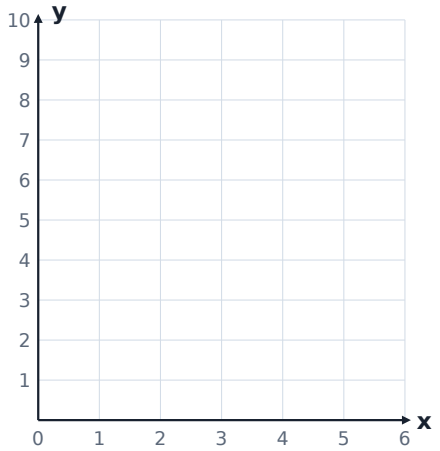
Written: 30; the car travels 30 miles per gallon.

GRADING NOTE Grade both: the line through the origin with slope 30, and the slope with its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q1/5

Graph y = 1.5x on the grid for x from 0 to 6.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

381

QUESTION ID

S13-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A straight line through (0, 0), (2, 3), (4, 6), and (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

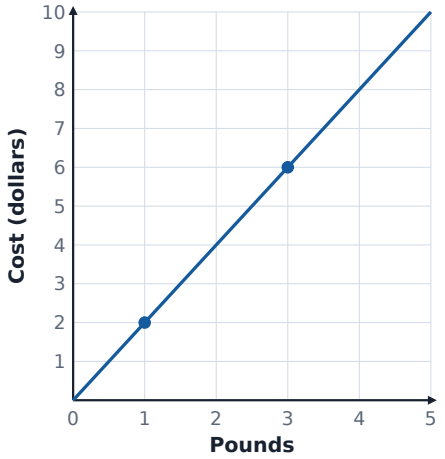
ID S13-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q2/5

The graph shows the cost of apples.

What is the slope, and what does it represent?

Slope and meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

382

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

2; the apples cost $2 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q3/5

The graph of a proportional relationship passes through (4, 10).

What is the unit rate?

A. 14

B. 2.5

C. 0.4

D. 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

383

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q4/5

True or false?

On graphs of distance (y) against time (x) that pass through the origin, a steeper line means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

384

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 13 • MAIN • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of hours (x) and pay in dollars (y) passes through (1, 15).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

386

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The pay is $15 for 1 hour (the unit rate).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q1/5

A proportional graph shows cups of flour (x) and cookies (y).

What does the point (0, 0) mean?

A. Each cup makes 0 cookies.

B. 0 cups of flour make 0 cookies.

C. The recipe starts with some cookies.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

387

QUESTION ID

S13-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 0 cups of flour make 0 cookies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and pages read (y) passes through (5, 20).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

388

QUESTION ID

S13-B1-Q3

TYPE

Short answer

CORRECT ANSWER

20 pages are read in 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) tells you the unit rate is r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

389

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q4/5

True or false?

On a graph of hours (x) and miles (y), the point (3, 12) means 3 miles in 12 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

390

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 12 miles in 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 8.EE.B.5

BACKWARD 2

Find the constant of proportionality from a graph • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A proportional graph passes through (0, 0) and (5, 20).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

392

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q1/5

Which point shows k = 3 on the graph of a proportional relationship?

A. (3, 6)

B. (6, 2)

C. (3, 1)

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

393

QUESTION ID

S13-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q2/5

A proportional graph passes through (8, 2).

What is the constant of proportionality?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

394

QUESTION ID

S13-B2-Q3

TYPE

Short answer

CORRECT ANSWER

0.25

GRADING NOTE 1/4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q3/5

True or false?

A line through (0, 0) and (10, 25) has a constant of proportionality of 2.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

395

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q4/5

True or false?

A line through (0, 0) and (6, 3) has a constant of proportionality of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

396

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $k = 3 \div 6 = 0.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 1

Calculate and interpret an average rate of change • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 2$

What is the average rate of change of f from $x = 1$ to $x = 5$?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

398

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q1/5

The table shows a trip.

What is the average rate of change from 2 hours to 5 hours?

Hours	0	2	5
Miles	0	110	260

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

399

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q2/5

$h(t) = t^2$

What is the average rate of change of h from t = 1 to t = 4?

A. 5

B. 3

C. 16

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

400

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q3/5

True or false?

For a linear function, the average rate of change is the same over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

401

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

402

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 8.EE.B.5

FORWARD 2

Compare properties of functions represented in different ways • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 2.5x. The graph of g is a line through (0, 0) and (4, 12).

Which function has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

404

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g (its rate is 3, and f's is 2.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q1/5

f(x) = 40 + 5x. The table shows g.

Which function has the greater initial value, and which has the greater rate of change?

x	0	1	2
g(x)	30	37	44

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

405

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f has the greater initial value (40 > 30); g has the greater rate of change (7 > 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q2/5

f(x) = 2^x and g(x) = 10x.

For which x-value is f(x) > g(x)?

A. x = 3

B. x = 4

C. x = 6

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

406

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q3/5

True or false?

A linear function with a slope of 4 eventually grows more slowly than f(x) = 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q4/5

True or false?

Two functions with the same y-intercept must be the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

408

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.9 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 8.EE.B.5 • EXPRESSIONS & EQUATIONS

Compare two proportional relationships represented in different ways

MAIN	Grade 8 main question	**8.EE.B.5**
BACKWARD 1	Write an equation for a proportional relationship	**7.RP.A.2.c**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Create and compare linear equations in two variables	**HSA-CED.A.2**
FORWARD 2	Compare linear and exponential growth	**HSF-LE.A.3**

SET 14 • 8.EE.B.5

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.5

Runner A's distance is d = 6.2t (miles, hours). Runner B's graph is a line through (0, 0) and (2, 13).

Which runner is faster?

A. Runner A

B. Runner B

C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

411

QUESTION ID

S14-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Runner B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q1/5

Plan A costs c = 0.12m dollars for m minutes. The table shows Plan B.

Which plan costs less per minute, and by how much?

Minutes	100	250	400
Cost ($)	15	37.5	60

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

412

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

Plan A, by $0.03 per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q2/5

Pump A fills 45 gallons in 3 minutes at a constant rate.
Pump B fills according to g = 14m.

Which pump is faster, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

413

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

Pump A, by 1 gallon per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q3/5

True or false?

The graph of y = 0.8x is less steep than the graph of a proportional relationship through (5, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

414

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q4/5

True or false?

A proportional relationship through (3, 7) has a greater unit rate than y = 2.5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

415

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 7 ÷ 3 ≈ 2.33, which is less than 2.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 14 • MAIN • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost $0.45 each.

Write an equation for the cost c of n pencils.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

417

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 0.45n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	2	4	6
y	7	14	21

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

418

QUESTION ID

S14-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q2/5

A car travels 55 miles per hour.

Which equation gives the distance d after t hours?

A. $d = \frac{t}{55}$

B. $d = 55t$

C. $t = 55d$

D. $d = t + 55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. d = 55t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q3/5

True or false?

A recipe uses 6 cups of water for every cup of rice. The equation $y = 6x$ gives the cups of water y for x cups of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

420

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q4/5

True or false?

A printer prints 30 pages in 2 minutes. The equation p = 30m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

421

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 15m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 8.EE.B.5

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ty walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

How many miles per hour is this?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

423

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q1/5

$\frac{2}{5}$ pound of nuts costs $2.

What is the price per pound?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

424

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q2/5

A hose fills $\frac{1}{2}$ gallon in $\frac{1}{6}$ minute.

How many gallons per minute is this?

A. $\frac{1}{12}$

B. 6

C. $\frac{1}{3}$

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

425

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

Walking $\frac{1}{3}$ mile in $\frac{1}{6}$ hour is a speed of 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

426

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q4/5

True or false?

A drink uses $\frac{3}{4}$ cup of juice for every $\frac{1}{2}$ cup of water. That is $\frac{3}{8}$ cup of juice per cup of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

427

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3/2 cups of juice per cup of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 1

Create and compare linear equations in two variables • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Tank A has 20 gallons and gains 5 gallons per minute. Tank B starts empty and gains 7 gallons per minute.

Write an equation for each tank, with y gallons after x minutes.

Equations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

429

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Tank A: y = 5x + 20;

Tank B: y = 7x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q1/5

Tank A holds y = 5x + 20 gallons and Tank B holds y = 7x gallons after x minutes.

When do they hold the same amount?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

430

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

After 10 minutes (70 gallons each)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q2/5

A car starts 30 miles from home and drives away at 50 miles per hour.

Which equation gives its distance d from home after t hours?

A. d = 50t − 30

B. d = 50t + 30

C. d = 30t + 50

D. d = 80t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

431

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. d = 50t + 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q3/5

True or false?

The graphs of y = 5x + 20 and y = 7x intersect at (10, 70).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q4/5

True or false?

$y = 3x + 4$ and $y = 3x - 2$ have different rates of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

433

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both have a rate of change of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 8.EE.B.5

FORWARD 2

Compare linear and exponential growth • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

f(x) = 100 + 50x and g(x) = 100(1.5)^x.

Which is greater at x = 5? Give both values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

435

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

g: g(5) ≈ 759.4, while f(5) = 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q1/5

Which function eventually has the greater values:
y = 1,000x or y = 2^x?

A. It depends only on the y-intercepts

B. y = 1,000x

C. y = 2^x

D. Neither; they are always equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

436

QUESTION ID

S14-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q2/5

Which function in the table is exponential? Explain.

x	0	1	2	3
f(x)	2	4	8	16
g(x)	2	4	6	8

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

437

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

f, because it doubles each time x increases by 1 (g increases by 2 each time).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

A quantity that increases exponentially eventually exceeds a quantity that increases linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

438

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

At x = 2, 3^x is greater than 10x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

439

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$, which is less than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 8.EE.B.6 • EXPRESSIONS & EQUATIONS

Use similar triangles to explain slope and derive y = mx and y = mx + b

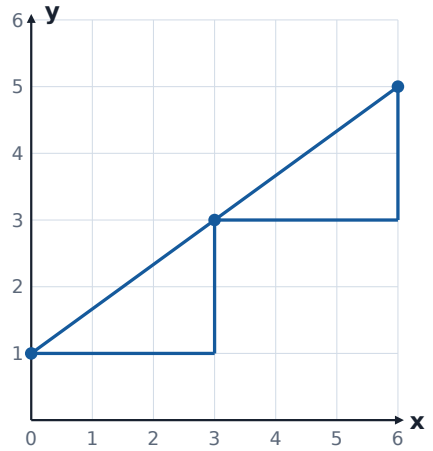
MAIN	Grade 8 main question	**8.EE.B.6**
BACKWARD 1	Use a scale factor to find a length in a scaled copy	**7.G.A.1**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Use similarity transformations to decide whether triangles are sim…	**HSG-SRT.A.2**
FORWARD 2	Use slopes of parallel and perpendicular lines	**HSG-GPE.B.5**

SET 15 • 8.EE.B.6

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.B.6

Two slope triangles are drawn under the line. Explain why they give the same slope.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

442

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

The triangles are similar (they have the same angles), so their rise-to-run ratios are equal: 2/3 = 2/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q1/5

A line passes through the origin and (4, 6). By similar triangles, any point (x, y) on the line has y ÷ x = 6 ÷ 4.

Write the equation of the line.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

443

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

y = 1.5x

GRADING NOTE y = (3/2)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q2/5

A line has a slope of 2 and crosses the y-axis at (0, 3).

Which equation describes it?

A. y = 2x

B. y = 2x + 3

C. y = 3x + 2

D. y = x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q3/5

A line crosses the y-axis at (0, −2) and passes through (5, 8).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

445

QUESTION ID

S15-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q4/5

True or false?

Any two slope triangles drawn on the same non-vertical line are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

446

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 15 • MAIN • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 1

Use a scale factor to find a length in a scaled copy • Grade 7 • Geometry • 7.G.A.1

A right triangle has legs of 2 and 3. In a scaled copy, the leg of 2 becomes 8.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

448

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q1/5

A right triangle has legs of 4 and 6. A scaled copy has a leg of 6 matching the leg of 4.

How long is the other leg in the copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

449

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q2/5

Which pair of right triangles are scaled copies of each other (matching the shorter legs)?

A. Legs 2 and 5; legs 4 and 7

B. Legs 2 and 5; legs 6 and 12

C. Legs 2 and 5; legs 3 and 6

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

450

QUESTION ID

S15-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Legs 2 and 5; legs 4 and 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q3/5

True or false?

A scaled copy with a scale factor of 3 has every length 3 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

451

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q4/5

True or false?

Right triangles with legs 3 and 4 and legs 6 and 9 are scaled copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

452

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 8.EE.B.6

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	6	7	8

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

454

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	1	2	3
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

455

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q2/5

Which equation matches the table?

x	0	1	2
y	1	3	5

A. y = 2x

B. y = x + 1

C. y = 3x

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

456

QUESTION ID

S15-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q3/5

True or false?

For y = 2x + 4, y = 10 when x = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

457

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q4/5

True or false?

For y = x + 7, y = 14 when x = 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

458

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 3, 4, and 5. Triangle DEF has sides 6, 8, and 10.

Are the triangles similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

460

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; the scale factor is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q1/5

Explain how a dilation shows that two slope triangles on the same line are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

461

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A dilation centered at a point on the line maps one triangle onto the other. Dilations keep angle measures, so the triangles are similar.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q2/5

Which information is enough to show that two triangles are similar?

A. One pair of equal sides

B. Two pairs of equal angles

C. Equal perimeters

D. One pair of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

462

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Two pairs of equal angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q3/5

True or false?

If one triangle is a dilation of another, their corresponding sides are proportional.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

463

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q4/5

True or false?

Any two right triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

464

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 8.EE.B.6

FORWARD 2

Use slopes of parallel and perpendicular lines • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Line p has a slope of $\frac{3}{4}$.

What is the slope of a line perpendicular to p?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

466

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−4/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q1/5

Write an equation of the line through (0, 2) that is parallel to y = −3x + 7.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

467

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = −3x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q2/5

Which line is perpendicular to $y = 2x - 5$?

A. $y = -2x - 5$

B. $y = 2x + 1$

C. $y = -\frac{1}{2}x + 1$

D. $y = \frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

468

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = -(1/2)x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

Parallel lines that are not vertical have equal slopes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

469

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q4/5

True or false?

Lines with slopes 3 and −3 are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

470

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1; 3 × (−3) = −9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a • EXPRESSIONS & EQUATIONS

Decide whether a linear equation has one solution, no solution, or infinitely many

MAIN	Grade 8 main question	**8.EE.C.7.a**
BACKWARD 1	Expand linear expressions	**7.EE.A.1**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Solve systems and recognize when they have no solution or infinit…	**HSA-REI.C.6**
FORWARD 2	Use the form $(x - p)^2 = q$ to find how many real solutions a quadra…	**HSA-REI.B.4.a**

SET 16 • 8.EE.C.7.a

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$4(x + 3) = 4x + 12$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

473

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q1/5

How many solutions does this equation have?

$5x - 2 = 5x + 7$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

474

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q2/5

Which equation has exactly one solution?

A. 3(x + 1) = 3x + 3

B. 2x − 2x = 0

C. 3x + 1 = 2x + 9

D. 3x + 1 = 3x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

475

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3x + 1 = 2x + 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q3/5

What number in the blank makes the equation have no solution?

$6x + 5 = ___x + 1$

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

476

QUESTION ID

S16-M-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q4/5

True or false?

$2(x - 4) = 2x - 4$ has infinitely many solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

477

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It simplifies to $-8 = -4$, so it has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 16 • MAIN • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 1

Expand linear expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

4(x + 3)

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

479

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x + 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q1/5

Expand.

$-2(3x - 5)$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

480

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-6x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q2/5

Which expression is equal to 5(2x − 1)?

A. 10x + 5

B. 10x − 1

C. 10x − 5

D. 7x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

481

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 10x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$3(x + 2) = 3x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

482

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$-(x - 4) = -x - 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

483

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 7 = 22

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

485

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q1/5

Solve for x.

$-2x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

486

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q2/5

Solve for x.

$5x - 6 = 19$

A. $x = \frac{13}{5}$

B. $x = -5$

C. $x = 5$

D. $x = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

487

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x = 2$ is the solution of $4x + 1 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

488

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q4/5

True or false?

$x = 3$ is the solution of $2x - 5 = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

489

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 1

Solve systems and recognize when they have no solution or infinitely many • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

How many solutions does the system have?

$y = 2x + 3$

$y = 2x - 1$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

491

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q1/5

Solve the system.

$x + y = 7$

$x - y = 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

492

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 4, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q2/5

How many solutions does the system have?

$2x + 4y = 8$

$x + 2y = 4$

A. Two

B. Infinitely many

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

493

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Infinitely many

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q3/5

True or false?

A system of two different parallel lines has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

494

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q4/5

True or false?

The system $y = x + 1$ and $y = -x + 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

495

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines meet at (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 8.EE.C.7.a

FORWARD 2

Use the form $(x - p)^2 = q$ to find how many real solutions a quadratic equation has • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a

How many real solutions does $(x - 3)^2 = 0$ have? Give any solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

497

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

One: x = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q1/5

How many real solutions does $(x + 1)^2 = -4$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

498

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

None. A square of a real number cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q2/5

Completing the square on $x^2 + 6x + 13 = 0$ gives $(x + 3)^2 = q$.

What is q, and how many real solutions are there?

A. q = 4; two real solutions

B. q = −4; one real solution

C. q = 13; two real solutions

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

499

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. q = −4; no real solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q3/5

True or false?

$(x - 2)^2 = 9$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

500

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = 5$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q4/5

True or false?

$x^2 - 6x + 9 = 0$ has two different real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

501

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $(x - 3)^2 = 0$, so $x = 3$ is the only solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.a • Set 16 • FORWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Solve multistep linear equations by expanding and collecting like terms

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Solve two-step equations with rational numbers	**7.EE.B.4.a**
FORWARD 1	Solve linear equations with letter coefficients	**HSA-REI.B.3**
FORWARD 2	Justify each step of a solution and find an invalid step	**HSA-REI.A.1**

SET 17 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$3(2x - 4) = 4x + 6$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

504

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q1/5

Solve for x.

$5 - 2(x + 1) = 3x - 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

505

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q2/5

Solve for x.

$4(x - 1) - (x + 2) = 2x + 3$

A. $x = 1$

B. $x = 5$

C. $x = 9$

D. $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

506

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q3/5

Solve. Show each step.

$-3(x + 5) + 2x = 4(x - 10)$

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

507

QUESTION ID

S17-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

5

GRADING NOTE REQUIRED WORK: Any valid sequence of steps. Example: $-3x - 15 + 2x = 4x - 40$; $-x - 15 = 4x - 40$; $25 = 5x$; $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q4/5

True or false?

$x = -1$ is the solution of $2(3x + 1) = 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

508

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • MAIN • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$6x - 4 + 2x + 9$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

510

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q1/5

Simplify.

$-3a + 7 - a$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

511

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-4a + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q2/5

Simplify.

$2(x + 3) + 5x$

A. $10x + 6$

B. $2x + 11$

C. $7x + 3$

D. $7x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

512

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q3/5

True or false?

$4y - y = 3y$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

513

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q4/5

True or false?

$5n + 2 - 2n = 3n$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

514

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5n + 2 - 2n = 3n + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

BACKWARD 2

Solve two-step equations with rational numbers • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$6x - 5 = 31$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

516

QUESTION ID

S17-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q1/5

Solve for x.

$\frac{x}{4} + 3 = 7$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

517

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q2/5

Solve for x.

$-3x + 8 = -10$

A. $x = -6$

B. $x = -\frac{2}{3}$

C. $x = \frac{2}{3}$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

518

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x = -2$ is the solution of $5x + 12 = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

519

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ is the solution of $3x - 2 = 14$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

520

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 1

Solve linear equations with letter coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume a ≠ 3.

ax + 2 = 3x + 8

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

522

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6/(a − 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q1/5

Solve for x. Assume k ≠ 0.

$k(x - 4) = 2k$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

523

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q2/5

Solve for x. Assume b ≠ 0.

$bx - 5b = 2b$

A. $x = 7b$

B. $x = \frac{7}{b}$

C. $x = 7$

D. $x = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

524

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q3/5

True or false?

If $c \neq 1$, the solution of $cx + 1 = x + 1$ is $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

525

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q4/5

True or false?

The solution of $mx = m + 4$ is $x = 5$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

526

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = (m + 4)/m$, which is 5 only when $m = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 8.EE.C.7.b

FORWARD 2

Justify each step of a solution and find an invalid step • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2(x − 3) + 4 = 5x − 11. Give a reason for each step.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

528

QUESTION ID

S17-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

3

GRADING NOTE REQUIRED WORK: Every step with a valid reason. Other valid orders of steps are accepted. A correct answer without reasons is not full credit. Example: distributive property (2x − 6 + 4 = 5x − 11); combine like terms (2x − 2 = 5x − 11); subtract 2x and add 11 to both sides (9 = 3x); divide both sides by 3 (x = 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q1/5

Dee solved $4 - 2(x - 1) = 8$ like this:

Step 1: $4 - 2x - 2 = 8$

Step 2: $2 - 2x = 8$

Step 3: $x = -3$

Which step is wrong? Correct it and give the solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

529

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Step 1: $-2(x - 1) = -2x + 2$, so $6 - 2x = 8$ and $x = -1$.

GRADING NOTE Grade both: Step 1 with the correction, and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q2/5

Which reason justifies rewriting $3x - 5 = 7$ as $3x = 12$?

A. Addition property of equality

B. Multiplication property of equality

C. Commutative property of addition

D. Distributive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

530

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Addition property of equality

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q3/5

True or false?

Each step of a valid solution gives an equation with the same solutions as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

531

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q4/5

True or false?

Multiplying both sides of an equation by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

532

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b • EXPRESSIONS & EQUATIONS

Write and solve linear equations with rational coefficients for real-world problems

MAIN	Grade 8 main question	**8.EE.C.7.b**
BACKWARD 1	Solve word problems that lead to px + q = r	**7.EE.B.4.a**
BACKWARD 2	Multiply rational numbers in context	**7.NS.A.3**
FORWARD 1	Solve systems of linear equations from real-world problems	**HSA-REI.C.6**
FORWARD 2	Create equations and inequalities in one variable to solve proble…	**HSA-CED.A.1**

SET 18 • 8.EE.C.7.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Gym A charges $30 to join plus $12.50 per month. Gym B charges $55 to join plus $10 per month.

Write and solve an equation to find when the total costs are equal.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

535

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 30 + 12.5m = 55 + 10m and the answer m = 10 months

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q1/5

Plant A is $6\frac{1}{2}$ inches tall and grows $\frac{3}{4}$ inch per week.

Plant B is 10 inches tall and grows $\frac{1}{4}$ inch per week.

Write and solve an equation to find when they are the same height.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

536

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 6.5 + 0.75w = 10 + 0.25w and the answer w = 7 weeks

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q2/5

Solve for x.

$\frac{2}{3}x - 4 = \frac{1}{2}x + 1$

A. x = 18

B. $x = \frac{30}{7}$

C. x = 30

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

537

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q3/5

Solve for x.

$0.4(x + 15) = 0.25x + 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

538

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q4/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x + 2 = \frac{1}{2}x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

539

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • MAIN • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 1

Solve word problems that lead to px + q = r • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Ana has $40. She buys some $6.50 tickets and has $1 left.

How many tickets did she buy?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

541

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q1/5

A rental costs $20 plus $8 per hour. The total is $68.

For how many hours was it rented?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

542

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q2/5

Solve for x.

0.5x + 3 = 11

A. x = 7

B. x = 16

C. x = 28

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

543

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q3/5

True or false?

$x = 9$ is the solution of $\frac{1}{3}x + 2 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

544

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2.5x - 1 = 11$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

545

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 4.8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

BACKWARD 2

Multiply rational numbers in context • Grade 7 • The Number System • 7.NS.A.3

A plant grows $\frac{3}{4}$ inch per week.

How much does it grow in 8 weeks?

Growth: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

547

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q1/5

A gym charges $12.50 per month.

How much do 10 months cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

548

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q2/5

A cyclist rides 0.4 mile per minute for 15 minutes.

How far does she ride?

A. 6 miles

B. 60 miles

C. 4 miles

D. 15.4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

549

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3}$ of 9 feet is 6 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

550

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

0.25 of a 12-hour day is 4 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

551

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.25 × 12 = 3 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B2-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 1

Solve systems of linear equations from real-world problems • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve the system.

$y = 12.5x + 30$

$y = 10x + 55$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

553

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 10, y = 155

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q1/5

Adult tickets cost $8 and student tickets cost $5. 120 tickets were sold for $750.

How many adult tickets were sold?

Adult tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

554

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q2/5

Solve the system.

$2x + 3y = 12$

$x - y = 1$

A. $x = 1, y = 0$

B. $x = 3, y = 2$

C. $x = 4, y = 3$

D. $x = 2, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

555

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = 2, y = 5$ is a solution of both $y = 3x - 1$ and $y = x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

556

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q4/5

True or false?

The system x + y = 10 and 2x + 2y = 15 has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

557

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines are parallel, so there is no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 18 • FORWARD 1 • Q5/5

SET 18 • 8.EE.C.7.b

FORWARD 2

Create equations and inequalities in one variable to solve problems • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

Plan A costs $25 plus $0.15 per text. Plan B costs $40 with unlimited texts.

Write and solve an inequality to find when Plan A costs less.

Inequality and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

559

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The inequality 25 + 0.15t < 40 and the answer t < 100 texts (0 to 99 texts)

GRADING NOTE Both parts are required: a valid inequality and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q1/5

Two trains leave a station at the same time in opposite directions, at 55 and 65 miles per hour.

Write and solve an equation to find when they are 300 miles apart.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

560

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 55t + 65t = 300 and the answer t = 2.5 hours

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q2/5

A rectangle's length is 3 more than twice its width w. Its perimeter is 54.

Which equation can find w?

A. 2(2w) + 3 = 54

B. w + 2w + 3 = 54

C. 2w + 2(2w + 3) = 54

D. w(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

561

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

2w + 2(2w + 3) = 54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q3/5

True or false?

A rectangle has a length of 2w + 3 and a perimeter of 54. Its width w is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

562

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q4/5

True or false?

The inequality $25 + 0.15t < 40$ is true when $t = 120$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

563

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $25 + 18 = 43$, which is not less than 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations by graphing

MAIN	Grade 8 main question	**8.EE.C.8.a**
BACKWARD 1	Make a table and graph an equation	**6.EE.C.9**
BACKWARD 2	Graph points in all four quadrants	**6.NS.C.8**
FORWARD 1	Solve f(x) = g(x) from the intersection of two graphs	**HSA-REI.D.11**
FORWARD 2	Graph the solution set of a system of linear inequalities	**HSA-REI.D.12**

SET 19 • 8.EE.C.8.a

MAIN

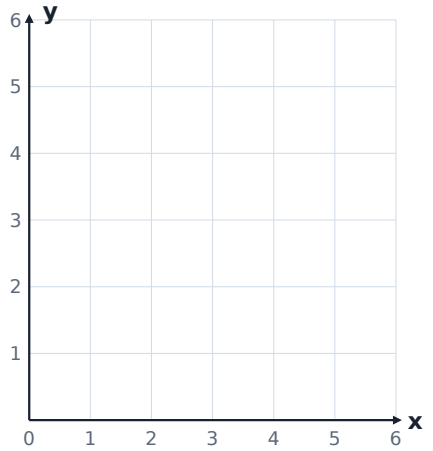
Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph y = x + 1 and
y = −x + 5 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

566

QUESTION ID
S19-M-Q1

TYPE
Drawing + written answer

CORRECT ANSWER

Drawing: The line y = x + 1 through (0, 1) and (4, 5), and the line y = −x + 5 through (0, 5) and (5, 0), crossing at (2, 3)

Written: (2, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (2, 3).

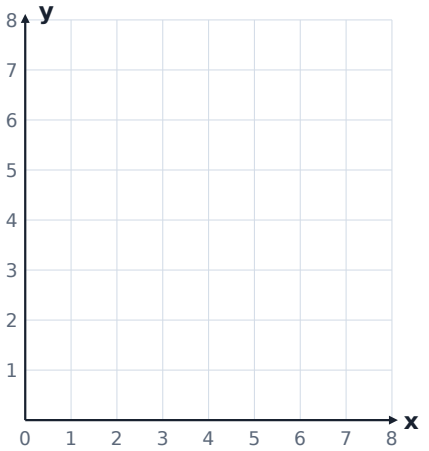
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q1/5

Graph y = 0.5x + 1 and y = −x + 7 on the grid.

What is the solution of the system?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

567

QUESTION ID

S19-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: The line y = 0.5x + 1 through (0, 1) and (6, 4), and the line y = −x + 7 through (0, 7) and (7, 0), crossing at (4, 3)

Written: (4, 3)

GRADING NOTE Grade both: the two lines drawn correctly, and the solution (4, 3).

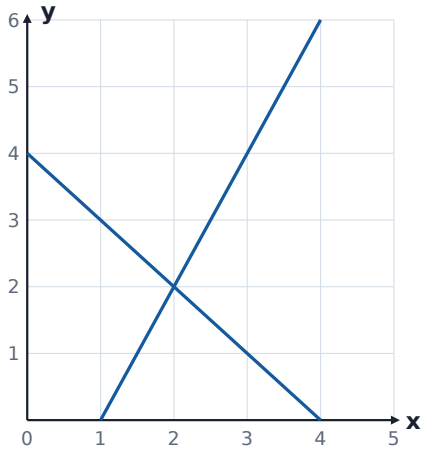
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q2/5

The graph shows the lines $y = -x + 4$ and $y = 2x - 2$.

What is the solution of the system? Check it in both equations.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

568

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 2): 2 = −2 + 4 and 2 = 2(2) − 2.

GRADING NOTE Grade both: (2, 2) and the check in both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q3/5

The graphs of the two equations in a system are parallel lines.

What does this tell you about the system?

A. It has no solution.

B. It has infinitely many solutions.

C. It has one solution.

D. Its solution is the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

569

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q4/5

True or false?

When the graphs of two linear equations are two different lines that cross, the crossing point is the only solution of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

570

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-M-Q5

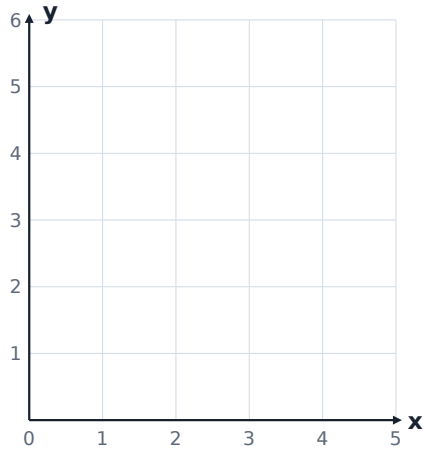
Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 19 • MAIN • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 1

Make a table and graph an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

Make a table for y = x + 2 with x = 0, 1, 2, and 3. Graph the points on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

572

QUESTION ID

S19-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 2), (1, 3), (2, 4), and (3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q1/5

What is the value of y when x = 4 in y = 3x?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

573

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 3?

A. (1, 3)

B. (3, 0)

C. (5, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

574

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q3/5

True or false?

(4, 8) is on the graph of y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

575

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of y = x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

576

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

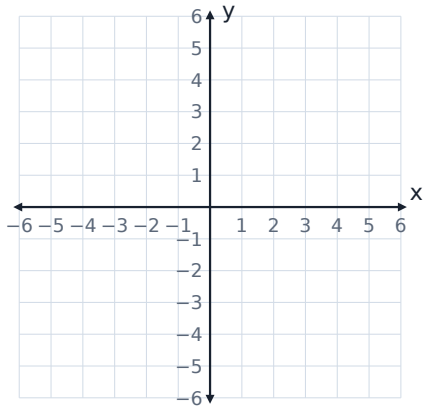
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

BACKWARD 2

Graph points in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Plot A(2, −3) and B(−4, 1) on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

578

QUESTION ID

S19-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

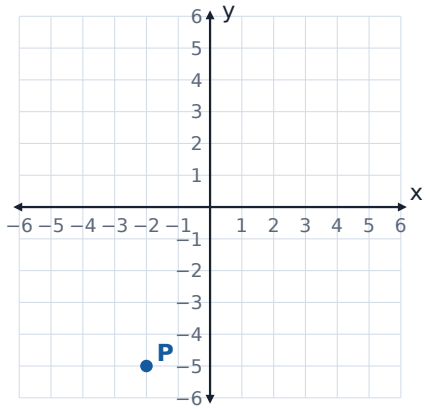
Point A at (2, −3) and point B at (−4, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q1/5

What are the coordinates of point P?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

579

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q2/5

Which point is in Quadrant II?

A. (3, −4)

B. (−3, −4)

C. (3, 4)

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

580

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

(0, −2) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

581

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

(5, −1) is in Quadrant III.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

582

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (5, −1) is in Quadrant IV.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 8.EE.C.8.a

FORWARD 1

Solve f(x) = g(x) from the intersection of two graphs • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of f(x) = x + 1 and g(x) = 7 − x intersect at (3, 4).

What is the solution of x + 1 = 7 − x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

584

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q1/5

Use a table of values for x = 0, 1, 2, and 3 to solve 2x + 1 = 7 − x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

585

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 2 (both sides equal 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q2**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q2/5

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ intersect at (2, 4) and (−1, 1).

What are the solutions of $x^2 = x + 2$?

A. $x = 2$ only

B. $x = -1$ only

C. $x = 4$ and $x = 1$

D. $x = 2$ and $x = -1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

586

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2 and x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q3/5

True or false?

If the graphs of y = f(x) and y = g(x) intersect at (5, 9), then f(5) = g(5) = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

587

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q4/5

True or false?

If the graphs of f and g never meet, then f(x) = g(x) has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

588

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 8.EE.C.8.a

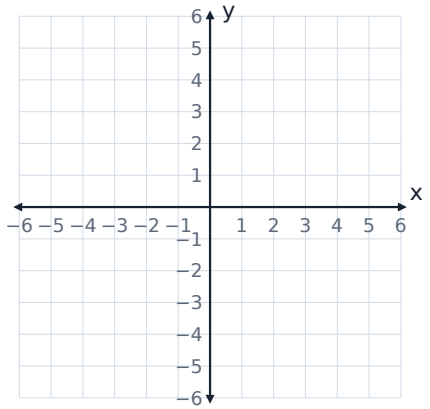
FORWARD 2

Graph the solution set of a system of linear inequalities • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

Graph the system on the coordinate plane.

$y \geq x$

$y < 4$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

590

QUESTION ID

S19-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A solid line y = x and a dashed line y = 4, with the region above or on y = x and below y = 4 shaded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q1/5

Is (1, 3) a solution of the system $y > 2x$ and $y < x + 3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

591

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 3 > 2(1) and 3 < 1 + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q2/5

Which point is a solution of the system $y \leq -x + 4$ and $y > 1$?

A. (1, 2)

B. (4, 3)

C. (3, 2)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

592

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q3/5

True or false?

The boundary line of $y < 2x + 1$ is drawn dashed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

593

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q4/5

True or false?

The solution set of a system of two inequalities is every point that satisfies at least one of them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A solution must satisfy both inequalities (the overlap of the two regions).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve a system of two linear equations algebraically

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve equations with the distributive property	**7.EE.B.4.a**
FORWARD 1	Explain why adding equations (elimination) keeps the same solut…	**HSA-REI.C.5**
FORWARD 2	Solve a system of a linear equation and a quadratic equation	**HSA-REI.C.7**

SET 20 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Solve the system. Show your work.

$y = 3x - 2$

$2x + y = 13$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

597

QUESTION ID

S20-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

x = 3, y = 7

GRADING NOTE REQUIRED WORK: Any valid method (substitution, elimination, or another), with the steps shown. Example: substitute $y = 3x - 2$ into $2x + y = 13$ to get $5x - 2 = 13$, so $x = 3$ and $y = 7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q1/5

Solve the system. Show your work.

$x + 2y = 10$

$3x - 2y = 6$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

598

QUESTION ID

S20-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

x = 4, y = 3

GRADING NOTE REQUIRED WORK: Any valid method, with the steps shown. Example: add the equations to get $4x = 16$, so $x = 4$; then $4 + 2y = 10$, so $y = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q2/5

Solve the system.

$y = x + 4$

$y = -2x + 1$

A. $x = 1, y = 5$

B. $x = 3, y = 7$

C. $x = -1, y = 5$

D. $x = -1, y = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

599

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = −1, y = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system.

$4x + 3y = 18$

$2x - y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

600

QUESTION ID

S20-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 3, y = 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q4/5

True or false?

$x = 2, y = -1$ is a solution of both $3x + y = 5$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

601

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 20 • MAIN • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2x + y when x = 3 and y = 7.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

603

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q1/5

Evaluate 3x − 2 when x = 4.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

604

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q2/5

Evaluate x + 2y when x = 4 and y = 3.

A. 9

B. 10

C. 11

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

605

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q3/5

True or false?

When x = 2, the value of 5x − 1 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

606

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q4/5

True or false?

When y = 3, the value of 2y + 4 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

607

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

BACKWARD 2

Solve equations with the distributive property • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$5x - 2 = 13$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

609

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q1/5

Solve for x.

$2x + 3(x - 2) = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

610

QUESTION ID

S20-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q2/5

Solve for x.

$4x + 2 = -10$

A. $x = 3$

B. $x = -2$

C. $x = -3$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

611

QUESTION ID

S20-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q3/5

True or false?

$x = 4$ is the solution of $3x - 5 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

612

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q4/5

True or false?

$x = -1$ is the solution of $2x + 9 = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

613

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 1

Explain why adding equations (elimination) keeps the same solutions • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

Add the equations x + y = 9 and x − y = 3.

What new equation do you get, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

615

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2x = 12; x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q1/5

Multiply $x + 2y = 7$ by 3, then add the result to $-3x + y = 0$.

What equation results?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

616

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7y = 21 (so y = 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q2/5

In a system, one equation is replaced by its sum with a multiple of the other equation.

What happens to the solutions?

A. The solutions always change.

B. The number of solutions doubles.

C. All solutions are removed.

D. The solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

617

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The solutions stay the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q3/5

True or false?

The system $x + y = 5$ and $x - y = 1$ has the same solution as the system $x + y = 5$ and $2x = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

618

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q4/5

True or false?

Multiplying one equation of a system by 0 keeps the same solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

619

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 20 • FORWARD 1 • Q5/5

SET 20 • 8.EE.C.8.b

FORWARD 2

Solve a system of a linear equation and a quadratic equation • High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve the system. Give all solutions.

$y = x^2$

$y = 4$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

621

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 4) and (−2, 4)

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q1/5

Solve the system. Give all solutions.

$y = x^2 - 1$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

622

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 3) and (−1, 0)

GRADING NOTE Both solutions are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-F2-Q2**

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q2/5

How many solutions does the system have?

$y = x^2 + 3$

$y = 1$

A. Infinitely many

B. Two

C. One

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

623

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q3/5

True or false?

The line $y = 2x$ and the parabola $y = x^2$ meet at (0, 0) and (2, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

624

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q4/5

True or false?

A line and a parabola always meet at exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

625

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 10 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b • EXPRESSIONS & EQUATIONS

Solve simple systems by inspection

MAIN	Grade 8 main question	**8.EE.C.8.b**
BACKWARD 1	Recognize equivalent expressions	**7.EE.A.1**
BACKWARD 2	Decide whether a value makes two equations true	**6.EE.B.5**
FORWARD 1	Solve or classify systems of linear equations	**HSA-REI.C.6**
FORWARD 2	Classify lines as parallel, perpendicular, or neither from their equ..	**HSG-GPE.B.5**

SET 21 • 8.EE.C.8.b

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.b

Without solving, explain why this system has no solution.

$3x + 2y = 5$

$3x + 2y = 6$

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

628

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

The same expression, 3x + 2y, cannot equal both 5 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q1/5

Without solving, how many solutions does this system have? Explain.

$y = 4x - 1$

$y = 4x + 3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

629

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

None. The lines have the same slope and different y-intercepts, so they are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q2/5

Which system has infinitely many solutions?

A. $y = x$ and $y = -x$

B. $x + y = 4$ and $x + y = 8$

C. $x + y = 4$ and $x - y = 4$

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

630

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x + y = 4$ and $2x + 2y = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q3/5

Without solving, how many solutions does this system have? Explain.

$y = -2x + 5$

$y = 3x + 5$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

631

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

One. The slopes are different, so the lines cross once (at (0, 5)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q4/5

True or false?

The system $2x - y = 7$ and $6x - 3y = 21$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

632

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second equation is 3 times the first, so there are infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 21 • MAIN • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 1

Recognize equivalent expressions • Grade 7 • Expressions & Equations • 7.EE.A.1

Is 2x + 2y equivalent to 2(x + y)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

634

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. By the distributive property, 2(x + y) = 2x + 2y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q1/5

Write 3(2x + y) without parentheses.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

635

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6x + 3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q2/5

Which expression is equivalent to 4x + 6y?

A. 6(4x + y)

B. 4(x + 6y)

C. 2(4x + 6y)

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

636

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2(2x + 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q3/5

True or false?

$5x - 5y = 5(x - y)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

637

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q4/5

True or false?

$2(x + 3) = 2x + 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

638

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

BACKWARD 2

Decide whether a value makes two equations true • Grade 6 • Expressions & Equations • 6.EE.B.5

Is x = 2 a solution of both x + 3 = 5 and 2x = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

640

QUESTION ID

S21-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q1/5

Which of the values 1, 2, and 3 makes both 2x = 6 and x + 1 = 4 true?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

641

QUESTION ID

S21-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q2/5

Which value makes both x + 2 = 7 and 3x = 15 true?

A. 2

B. 7

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

642

QUESTION ID

S21-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q3/5

True or false?

No value of x makes both x + 1 = 3 and x + 1 = 4 true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

643

QUESTION ID

S21-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q4/5

True or false?

$x = 4$ makes both $x - 1 = 3$ and $2x = 6$ true.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

644

QUESTION ID

S21-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 1

Solve or classify systems of linear equations • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve the system, or say that it has no solution or infinitely many.

$2x + y = 3$

$4x + 2y = 6$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

646

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions (the second equation is 2 times the first)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q1/5

Solve the system.

$3x - y = 5$

$x + y = 7$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

647

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 3, y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q2/5

Which value of k makes the system have infinitely many solutions?

$x + ky = 4$

$2x + 6y = 8$

A. 4

B. 2

C. 6

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q3/5

True or false?

If the two equations of a system describe the same line, every point on the line is a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

649

QUESTION ID

S21-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q4/5

True or false?

The system y = 3x + 1 and y = 3x + 1 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

650

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F1-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 8.EE.C.8.b

FORWARD 2

Classify lines as parallel, perpendicular, or neither from their equations • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Are the lines y = 2x + 1 and 4x − 2y = 6 parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

652

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Parallel; both have slope 2 (and different y-intercepts).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q1/5

Are the lines $3x + y = 2$ and $x - 3y = 9$ parallel, perpendicular, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

653

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Perpendicular; the slopes are −3 and 1/3, and their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q2/5

One line passes through (0, 0) and (2, 4). Another passes through (1, 1) and (3, 5).

How are the lines related?

A. Perpendicular

B. The same line

C. Parallel

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

654

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q3/5

True or false?

Lines with slopes $\frac{2}{3}$ and $-\frac{3}{2}$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

655

QUESTION ID

S21-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q4/5

True or false?

The lines $y = x$ and $y = -x$ are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

656

QUESTION ID

S21-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 21 • FORWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c • EXPRESSIONS & EQUATIONS

Write and solve systems of linear equations for real-world problems

MAIN	Grade 8 main question	**8.EE.C.8.c**
BACKWARD 1	Solve word problems with one unknown	**7.EE.B.4.a**
BACKWARD 2	Use variables to represent quantities in a situation	**6.EE.B.6**
FORWARD 1	Create equations in two variables to model and solve problems	**HSA-CED.A.2**
FORWARD 2	Represent constraints with a system of inequalities	**HSA-CED.A.3**

SET 22 • 8.EE.C.8.c

MAIN

Grade 8 main question • Grade 8 • Expressions & Equations • 8.EE.C.8.c

A school sells 200 tickets. Adult tickets cost $6 and student tickets cost $4. The sales total $940.

Write a system and solve it to find how many of each ticket were sold.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

659

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

The system a + s = 200 and 6a + 4s = 940, and the answer 70 adult and 130 student tickets

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plan A costs $15 plus $0.05 per minute. Plan B costs $0.10 per minute.

Write a system and solve it to find when the plans cost the same.

System and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

660

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

The system c = 15 + 0.05m and c = 0.10m, and the answer 300 minutes (both cost $30)

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q2/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all.

Which system models this?

A. $d - q = 25$ and $0.10d + 0.25q = 4$

B. $d + q = 25$ and $0.10d + 0.25q = 4$

C. $d + q = 4$ and $0.10d + 0.25q = 25$

D. $d + q = 25$ and $0.25d + 0.10q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

661

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $d + q = 25$ and $0.10d + 0.25q = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q3/5

Jo has 25 coins, all dimes and quarters, worth $4.00 in all: d + q = 25 and 0.10d + 0.25q = 4.

How many of each coin does Jo have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

662

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

15 dimes and 10 quarters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-M-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q4/5

True or false?

Two numbers have a sum of 40 and a difference of 12. The larger number is 26.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

663

QUESTION ID

S22-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 22 • MAIN • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 1

Solve word problems with one unknown • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A taxi charges $3 plus $2 per mile. A ride costs $19.

How many miles long was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

665

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q1/5

Concert tickets cost $9 each, plus a $5 fee for the whole order. An order costs $50.

How many tickets were bought?

Tickets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

666

QUESTION ID

S22-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q2/5

The sum of a number and 3 times the number is 48.

What is the number?

A. 16

B. 45

C. 12

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

667

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q3/5

True or false?

If 2x + 7 = 25, then x = 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

668

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q4/5

True or false?

Items cost $4 each, and shipping is $6. A $30 order has 7 items.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

669

QUESTION ID

S22-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4n + 6 = 30 gives n = 6 items.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

BACKWARD 2

Use variables to represent quantities in a situation • Grade 6 • Expressions & Equations • 6.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Write an expression for the value, in dollars, of d dimes.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

671

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.10d

GRADING NOTE 0.1d is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Adult tickets cost $6.

Write an expression for the cost of a adult tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

672

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

There are 25 coins: d dimes and the rest quarters.

Which expression gives the number of quarters?

A. d − 25

B. 25d

C. 25 − d

D. d + 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

673

QUESTION ID

S22-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25 − d

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

True or false?

q quarters are worth 0.25q dollars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

674

QUESTION ID

S22-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

There are s students and 3 more teachers than students. There are 3s teachers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

675

QUESTION ID

S22-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are s + 3 teachers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 1

Create equations in two variables to model and solve problems • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

A company's cost is C = 2,000 + 15x and its revenue is R = 40x for x items.

Write and solve an equation to find the break-even number of items.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

677

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 2,000 + 15x = 40x and the answer x = 80 items

GRADING NOTE Both parts are required: a valid equation (or system) and the correct solution. Any valid model is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q1/5

A boat travels 24 miles downstream in 2 hours and 24 miles back upstream in 3 hours. Let b be the boat's speed in still water and c the speed of the current.

Write a system of two equations.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

678

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2(b + c) = 24 and
3(b − c) = 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q2/5

A boat's trips give the system $2(b + c) = 24$ and $3(b - c) = 24$.

What is the boat's speed b in still water?

A. 10 miles per hour

B. 2 miles per hour

C. 12 miles per hour

D. 8 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

679

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q3/5

True or false?

For the system 2(b + c) = 24 and 3(b − c) = 24, the current's speed c is 2 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

680

QUESTION ID

S22-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q4/5

True or false?

The break-even point is where revenue is twice the cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

681

QUESTION ID
S22-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE At break-even, revenue equals cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F1-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 22 • FORWARD 1 • Q5/5

SET 22 • 8.EE.C.8.c

FORWARD 2

Represent constraints with a system of inequalities • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

A bakery has 30 pounds of flour. Each batch of bread uses 2 pounds and each batch of muffins uses 0.5 pound.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

683

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$2b + 0.5m \leq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q1/5

A bakery has at most 20 hours of oven time. Each batch of bread takes 1 hour and each batch of muffins takes 2 hours.

Write an inequality for b batches of bread and m batches of muffins.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

684

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

b + 2m ≤ 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q2/5

A bakery's limits are $2b + 0.5m \leq 30$ (flour) and $b + 2m \leq 20$ (oven hours).

Can it make 10 batches of bread and 5 batches of muffins?

A. No; both limits are broken.

B. No; the flour runs out.

C. Yes; both limits are met.

D. No; the oven time runs out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

685

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; both limits are met.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q3/5

True or false?

In a model of batches of bread b and muffins m, the constraints $b \geq 0$ and $m \geq 0$ make sense because batches cannot be negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

686

QUESTION ID

S22-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q4/5

True or false?

The point (12, 6) meets both limits $2b + 0.5m \leq 30$ and $b + 2m \leq 20$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

687

QUESTION ID
S22-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $12 + 2(6) = 24$, which is more than 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S22-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 22 • FORWARD 2 • Q5/5

SET 23 • 8.F.A.1 • FUNCTIONS

Decide whether a relation is a function

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Find the output of a rule	**6.EE.C.9**
BACKWARD 2	Read points in the first quadrant	**5.G.A.2**
FORWARD 1	Use function notation and find the domain and range	**HSF-IF.A.1**
FORWARD 2	Find the inverse of a linear function	**HSF-BF.B.4.a**

SET 23 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

True or false?

The table shows y as a function of x.

x	1	2	3	4
y	5	5	7	9

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

690

QUESTION ID

S23-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q1/5

Is the relation in the table a function? Explain.

x	2	3	3	5
y	4	6	8	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

691

QUESTION ID

S23-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 3 has two different outputs (6 and 8).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q2/5

Which set of ordered pairs is a function?

A. (5, 1), (6, 2), (5, 3)

B. (0, 0), (0, 1), (1, 1)

C. (1, 2), (1, 3), (2, 4)

D. (1, 4), (2, 4), (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

692

QUESTION ID

S23-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4), (2, 4), (3, 4)

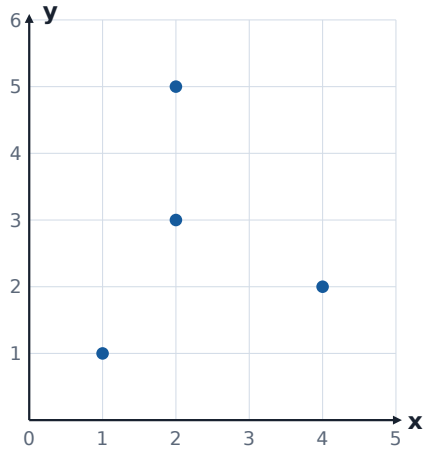
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q3/5

Do the points on the graph show y as a function of x? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

693

QUESTION ID

S23-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The input x = 2 has two outputs (3 and 5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q4/5

True or false?

A rule assigns each month of the year to the students in a class born in that month. This rule is a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

694

QUESTION ID

S23-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One month can be paired with several students, so an input has more than one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 23 • MAIN • Q5/5

SET 23 • 8.F.A.1

BACKWARD 1

Find the output of a rule • Grade 6 • Expressions & Equations • 6.EE.C.9

The rule is y = 2x + 1.

What is y when x = 4?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

696

QUESTION ID

S23-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q1/5

The table follows a pattern.

What is y when x = 4?

x	1	2	3
y	4	7	10

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

697

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q2/5

The rule is: output = input × 3.

What is the output for an input of 7?

A. 4

B. 73

C. 21

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

698

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q3/5

True or false?

For $y = x - 5$, an input of $x = 9$ gives $y = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

699

QUESTION ID

S23-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q4/5

True or false?

For y = 4x, an input of x = 3 gives y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

700

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

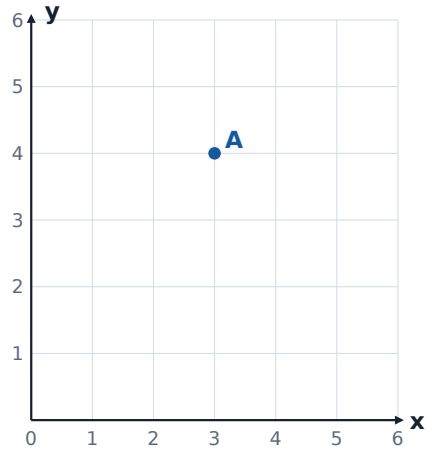
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 8.F.A.1

BACKWARD 2

Read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

QUESTION ID

S23-B2-Q1

TYPE

Short answer

CORRECT ANSWER

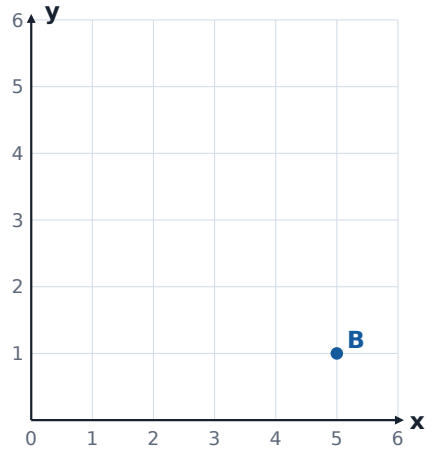
(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

703

QUESTION ID

S23-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q2/5

Which point is 2 units to the right of the origin and 5 units up?

A. (5, 5)

B. (5, 2)

C. (2, 2)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

704

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q3/5

True or false?

The point (0, 3) is on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

705

QUESTION ID

S23-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q4/5

True or false?

The point (4, 0) is 4 units up from the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

706

QUESTION ID

S23-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 8.F.A.1

FORWARD 1

Use function notation and find the domain and range • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

708

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q1/5

A function f is made of the pairs (1, 3), (2, 5), and (4, 9).

What is its range?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

709

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3, 5, and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q2/5

The output of f is 7 when the input is 2.

Which statement says this in function notation?

A. $f = 2 \times 7$

B. $f(7) = 2$

C. $2f = 7$

D. $f(2) = 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

710

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. f(2) = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q3/5

True or false?

If $f(x) = x^2$, then $f(-3) = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

711

QUESTION ID

S23-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

A function can assign two different outputs to the same input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

712

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 8.F.A.1

FORWARD 2

Find the inverse of a linear function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 2x + 3$

Write the inverse function.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

714

QUESTION ID

S23-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE $\frac{x}{2} - \frac{3}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q1**

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q1/5

$f(x) = 2x + 3$

Solve $f(x) = 11$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

715

QUESTION ID

S23-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q2/5

$g(x) = \frac{x}{5}$

What is the inverse function?

A. $g^{-1}(x) = x + 5$

B. $g^{-1}(x) = x - 5$

C. $g^{-1}(x) = 5x$

D. $g^{-1}(x) = \frac{5}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

716

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $g^{-1}(x) = 5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q3/5

True or false?

If $f(2) = 7$ and f has an inverse, then $f^{-1}(7) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

717

QUESTION ID

S23-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q4/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

718

QUESTION ID

S23-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(x) = x^2$ gives the same output for 2 and −2, so it has no inverse on all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S23-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.4.a • Set 23 • FORWARD 2 • Q5/5

SET 24 • 8.F.A.1 • FUNCTIONS

Graph a function as the set of ordered pairs of inputs and outputs

MAIN	Grade 8 main question	**8.F.A.1**
BACKWARD 1	Make ordered pairs from an equation	**6.EE.C.9**
BACKWARD 2	Graph points from a pattern	**5.G.A.2**
FORWARD 1	Choose a domain that fits a function's situation	**HSF-IF.B.5**
FORWARD 2	Graph quadratic functions and show their key features	**HSF-IF.C.7.a**

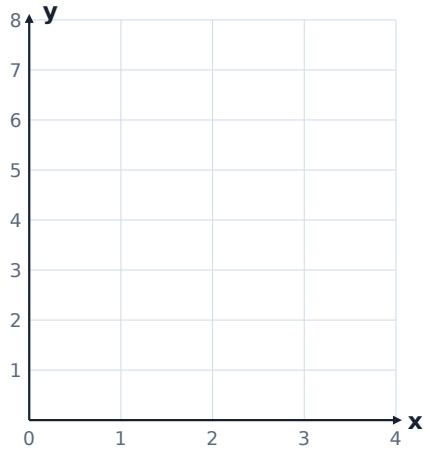
SET 24 • 8.F.A.1

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.1

The function y = 2x + 1 has the inputs 0, 1, 2, and 3.

Plot the ordered pairs (input, output) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

721

QUESTION ID

S24-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 1), (1, 3), (2, 5), and (3, 7)

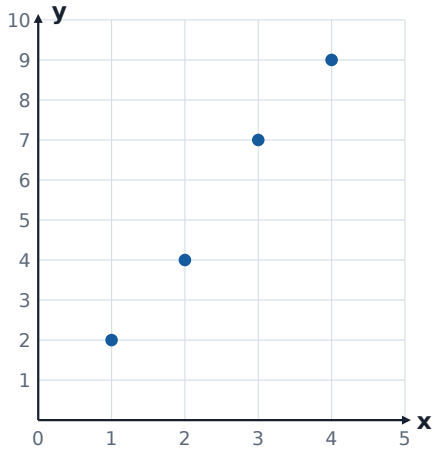
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q1/5

The points show a function.

What is the output when the input is 3?



Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

722

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q2/5

The graph of a function contains the point (4, 10).

Which statement is true?

A. When the input is 4, the output is 10.

B. When the input is 10, the output is 4.

C. The graph has only one point.

D. The function adds 4 to every input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

723

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. When the input is 4, the output is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q3/5

The rule is "output = input squared."

Write the ordered pairs (input, output) for the inputs −2, −1, 0, 1, and 2.

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

724

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 4), (−1, 1), (0, 0), (1, 1), (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q4/5

True or false?

The graph of a function can contain both (2, 5) and (2, 8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

725

QUESTION ID

S24-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 cannot have two outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-M-Q5

Grade 8 • Functions • 8.F.A.1 • Set 24 • MAIN • Q5/5

SET 24 • 8.F.A.1

BACKWARD 1

Make ordered pairs from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For y = 3x, what are the values of y when x = 0, 1, and 2?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

727

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, and 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For y = x + 4, what is y when x = 5?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

728

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q2/5

Which pair (x, y) fits y = 5x?

A. (2, 7)

B. (10, 2)

C. (2, 10)

D. (5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

729

QUESTION ID

S24-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q3/5

True or false?

The pair (3, 9) fits y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

730

QUESTION ID

S24-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

The pair (4, 6) fits y = x + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

731

QUESTION ID

S24-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

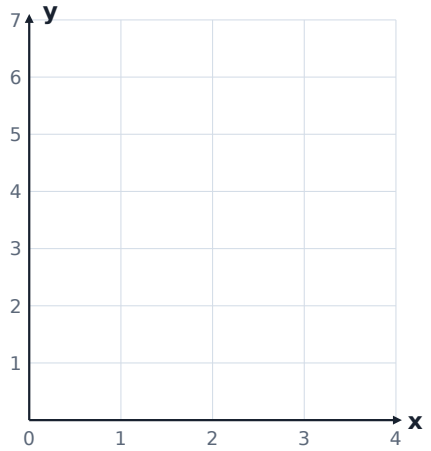
Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 8.F.A.1

BACKWARD 2

Graph points from a pattern • Grade 5 • Geometry • 5.G.A.2

Plot the points (1, 2), (2, 4), and (3, 6) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

733

QUESTION ID

S24-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), and (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q1/5

The y-values start at 1 and increase by 2: (1, 1), (2, 3), (3, 5), ...

What is the point for x = 4?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

734

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q2/5

Which point is 3 units to the right of the origin and 1 unit up?

A. (3, 1)

B. (3, 3)

C. (1, 3)

D. (1, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

735

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

The point (2, 6) is 2 units to the right of the origin and 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

736

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q4/5

True or false?

To plot (5, 2), you move 5 units up and 2 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

737

QUESTION ID

S24-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 8.F.A.1

FORWARD 1

Choose a domain that fits a function's situation • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5

A function gives the cost of n concert tickets at $12 each. A buyer may buy at most 8 tickets.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

739

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The whole numbers 0 through 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q1/5

h(t) gives the height of a ball t seconds after it is thrown, until it lands at t = 3.

What is a sensible domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

740

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \leq t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q2/5

p is the number of people in a car that seats 5.

Which domain makes sense for p?

A. $-5 \leq p \leq 5$

B. 0, 1, 2, 3, 4, 5

C. All real numbers

D. $p \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

741

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0, 1, 2, 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q3/5

True or false?

For the area of a square as a function of its side length, the domain contains only positive numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

742

QUESTION ID

S24-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q4/5

True or false?

A function that gives the number of buses needed for n students should have all real numbers as its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

743

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F1-Q5

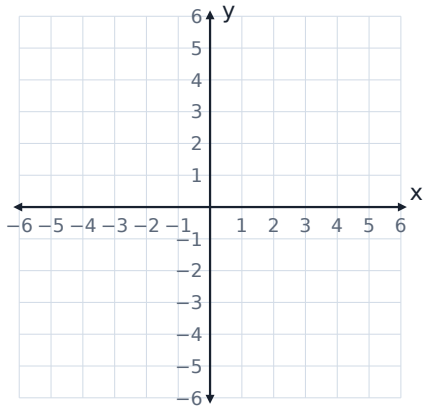
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.5 • Set 24 • FORWARD 1 • Q5/5

SET 24 • 8.F.A.1

FORWARD 2

Graph quadratic functions and show their key features • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$ on the coordinate plane. Mark the vertex and the x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

745

QUESTION ID

S24-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A U-shaped parabola with its vertex at (0, −4), crossing the x-axis at (−2, 0) and (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q1/5

What is the vertex of the graph of $y = (x - 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

746

QUESTION ID

S24-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q2/5

Where does the graph of $y = x^2 - 9$ cross the x-axis?

A. At (3, 0) and (−3, 0)

B. At (9, 0) and (−9, 0)

C. At (0, −9)

D. At (3, 0) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

747

QUESTION ID

S24-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. At (3, 0) and (−3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -x^2$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

748

QUESTION ID

S24-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 2x + 3$ has a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

749

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S24-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 24 • FORWARD 2 • Q5/5

SET 25 • 8.F.A.2 • FUNCTIONS

Compare two functions represented in different ways

MAIN	Grade 8 main question	**8.F.A.2**
BACKWARD 1	Find a unit rate from a table, graph, or equation	**7.RP.A.2.b**
BACKWARD 2	Find the starting value of a relationship	**6.EE.C.9**
FORWARD 1	Compare average rates of change of linear and nonlinear function…	**HSF-IF.B.6**
FORWARD 2	Describe the effect of transformations on the graph of a function	**HSF-BF.B.3**

SET 25 • 8.F.A.2

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.2

f(x) = 3x + 2. The table shows g.

Which function has the greater rate of change?

x	0	2	4
g(x)	1	9	17

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

752

QUESTION ID

S25-M-Q1

TYPE

Short answer

CORRECT ANSWER

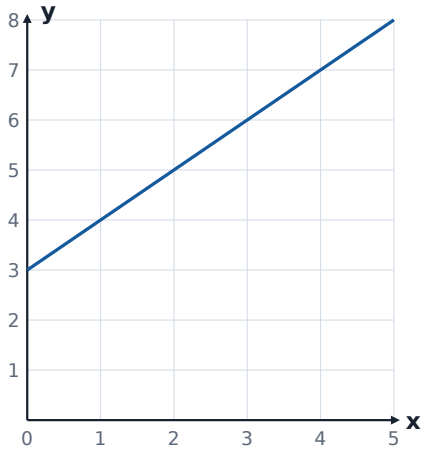
g (its rate is 4; f's is 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q1

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q1/5

The graph shows f.
$g(x) = -2x + 5$.

Which function has the greater initial value?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

753

QUESTION ID

S25-M-Q2

TYPE

Short answer

CORRECT ANSWER

g (5 is greater than 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q2/5

Function A is y = 0.5x + 10. Function B is a line through (0, 4) and (2, 8).

Which statement is true?

A. B has the greater rate of change; A has the greater initial value.

B. B has the greater rate of change and the greater initial value.

C. A has the greater rate of change and the greater initial value.

D. They have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

754

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. B has the greater rate of change; A has the greater initial value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q3

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q3/5

Car A travels d = 45t miles in t hours. The table shows Car B.

Which car is faster, and by how much?

Hours	1	2	3
Miles	50	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

755

QUESTION ID

S25-M-Q4

TYPE

Short answer

CORRECT ANSWER

Car B, by 5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-M-Q4

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q4/5

True or false?

A function with x-values 0, 1, 2 and y-values 6, 4, 2 decreases faster than $y = -x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

QUESTION ID

S25-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 8 • Functions • 8.F.A.2 • Set 25 • MAIN • Q5/5

SET 25 • 8.F.A.2

BACKWARD 1

Find a unit rate from a table, graph, or equation • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the unit rate?

x	2	4	6
y	5	10	15

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

758

QUESTION ID

S25-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q1/5

What is the unit rate in y = 7x?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

759

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q2/5

A proportional graph passes through (0, 0) and (3, 12).

What is the unit rate?

A. 36

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

760

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (1, 9) and (2, 18) has a unit rate of 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

761

QUESTION ID

S25-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q4/5

True or false?

The unit rate in y = 0.5x is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

762

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 8.F.A.2

BACKWARD 2

Find the starting value of a relationship • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

For y = 4x + 7, what is y when x = 0?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

764

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q1/5

For the table, what is y when x = 0?

x	0	1	2
y	9	11	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

765

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q2/5

Which equation gives y = 5 when x = 0?

A. y = 5x + 2

B. y = 2x + 5

C. y = 5x

D. y = x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

766

QUESTION ID

S25-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q3/5

True or false?

For y = 3x + 1, y = 1 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

767

QUESTION ID

S25-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

For y = 6x, y = 6 when x = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

768

QUESTION ID

S25-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 8.F.A.2

FORWARD 1

Compare average rates of change of linear and nonlinear functions over intervals • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$. The table shows g.

Find the average rate of change of each function from $x = 0$ to $x = 3$.

x	0	1	2	3
g(x)	0	3	6	9

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

770

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Both are 3: f goes from 0 to 9 and g goes from 0 to 9 over 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q1/5

f(x) = 2^x and g(x) = 3x + 5.

Find the average rate of change of each function from x = 0 to x = 4. Which is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

771

QUESTION ID

S25-F1-Q2

TYPE

Short answer

CORRECT ANSWER

f:
(16 − 1) ÷ 4 = 3.75;
g: 3. f has the greater average rate of change.

GRADING NOTE Both rates and the comparison are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q2/5

$f(x) = x^2$ and $g(x) = 4x$.

Which statement about the average rates of change from $x = 1$ to $x = 3$ is true?

A. f's rate is greater.

B. f's rate is 9 and g's rate is 12.

C. g's rate is greater.

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

772

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal (both are 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q3/5

True or false?

The average rate of change of $f(x) = -x^2 + 4$ from $x = 0$ to $x = 2$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

773

QUESTION ID

S25-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q4/5

True or false?

The average rate of change of a linear function depends on which interval you use.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

774

QUESTION ID

S25-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A linear function has the same rate over every interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 25 • FORWARD 1 • Q5/5

SET 25 • 8.F.A.2

FORWARD 2

Describe the effect of transformations on the graph of a function • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3

How does the graph of g(x) = f(x) + 3 compare with the graph of f?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

776

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It is the graph of f shifted up 3 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q1/5

How does the graph of $g(x) = (x - 2)^2$ compare with the graph of $y = x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

777

QUESTION ID

S25-F2-Q2

TYPE

Short answer

CORRECT ANSWER

It is shifted 2 units to the right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q2/5

Which function shifts the graph of y = |x| down 4 units?

A. y = |x| − 4

B. y = |x + 4|

C. y = 4|x|

D. y = |x − 4|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

778

QUESTION ID

S25-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = |x| − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q3/5

True or false?

The graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

779

QUESTION ID

S25-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q4/5

True or false?

The graph of y = f(x + 5) is the graph of y = f(x) shifted 5 units to the right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

780

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 5 units to the left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S25-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.B.3 • Set 25 • FORWARD 2 • Q5/5

SET 26 • 8.F.A.3 • FUNCTIONS

Decide whether a function is linear or nonlinear

MAIN	Grade 8 main question	**8.F.A.3**
BACKWARD 1	Decide whether a table shows a proportional relationship	**7.RP.A.2.a**
BACKWARD 2	Find how much y changes for each change in x	**6.EE.C.9**
FORWARD 1	Show that linear functions grow by equal differences and expone..	**HSF-LE.A.1.a**
FORWARD 2	Use factoring to find the zeros and maximum of a quadratic functi..	**HSF-IF.C.8.a**

SET 26 • 8.F.A.3

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.A.3

Which equation is a linear function?

A. $y = 4x - 7$

B. $y = x^2 + 1$

C. $y = 2^x$

D. $y = \frac{3}{x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

783

QUESTION ID

S26-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4x − 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q1/5

Is the function in the table linear? Explain.

x	0	1	2	3
y	1	2	5	10

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

784

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. y increases by 1, then 3, then 5, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q2

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q2/5

$A = s^2$ gives the area of a square with side length s.

Is A a linear function of s? Use three points to support your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

785

QUESTION ID

S26-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. (1, 1), (2, 4), and (3, 9) are not on one line: the area increases by 3, then by 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-M-Q3

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q3/5

Is the function in the table linear? Explain.

x	0	2	4	6
y	7	4	1	−2

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

786

QUESTION ID

S26-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. y decreases by 3 each time x increases by 2, a constant rate of change of −1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q4/5

True or false?

y = 3(x + 2) is a linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

787

QUESTION ID

S26-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 8 • Functions • 8.F.A.3 • Set 26 • MAIN • Q5/5

SET 26 • 8.F.A.3

BACKWARD 1

Decide whether a table shows a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	1	2	3
y	4	8	12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

789

QUESTION ID

S26-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	3
y	3	5	7

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

790

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The ratios 3, 2.5, and about 2.33 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q2/5

Which table shows a proportional relationship?

A. x: 0, 1, 2 and y: 1, 2, 3

B. x: 1, 2, 3 and y: 1, 4, 9

C. x: 1, 2, 3 and y: 2, 3, 4

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

791

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 2, 4, 6 and y: 3, 6, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q3/5

The pairs (5, 2) and (10, 4) are from a proportional relationship.

What is the constant of proportionality (y per x)?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

792

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q4/5

True or false?

y = x + 1 is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

793

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 8.F.A.3

BACKWARD 2

Find how much y changes for each change in x • Grade 6 • Expressions & Equations • 6.EE.C.9

How much does y increase each time x increases by 1?

x	0	1	2	3
y	5	8	11	14

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

795

QUESTION ID

S26-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q1/5

How much does y increase each time x increases by 2?

x	0	2	4
y	1	5	9

Increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

796

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q2/5

In y = 6x, how much does y increase when x increases by 1?

A. 1

B. 7

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

797

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q3/5

True or false?

In the table with x-values 1, 2, 3 and y-values 10, 20, 30, y increases by 10 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

798

QUESTION ID

S26-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q4/5

True or false?

In y = x + 9, y increases by 9 each time x increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

799

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 8.F.A.3

FORWARD 1

Show that linear functions grow by equal differences and exponential functions by equal factors • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

Show that f(x) = 4x + 1 grows by the same amount over every interval of length 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

801

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(x + 2) − f(x) = 4(x + 2) + 1 − (4x + 1) = 8, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

Show that g(x) = 3^x grows by the same factor over every interval of length 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

802

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

g(x + 1) ÷ g(x)
= 3^(x + 1) ÷ 3^x
= 3, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 2x^2$

B. $y = x + 5$

C. $y = 5x + 2$

D. $y = 5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

803

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5(2)^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

True or false?

The linear function $f(x) = mx + b$ changes by $m(x_2 - x_1)$ from x_1 to x_2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

804

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

True or false?

$y = x^2$ grows by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

805

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

SET 26 • 8.F.A.3

FORWARD 2

Use factoring to find the zeros and maximum of a quadratic function • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$ feet after t seconds.

Use the factored form to find when the ball is at height 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

807

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 and t = 2 seconds

GRADING NOTE Both times are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

A ball's height is $h(t) = -16t(t - 2)$ feet after t seconds. Its greatest height is halfway between the zeros.

When does it reach its greatest height, and what is that height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

808

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At t = 1 second; 16 feet

GRADING NOTE Both the time and the height are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

$f(x) = x^2 - 6x + 8 = (x - 2)(x - 4)$

What are the zeros of f?

A. 2 and 4

B. −2 and −4

C. 8 only

D. 6 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

809

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

True or false?

The graph of a quadratic function is symmetric about a vertical line through its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

True or false?

The zeros of f(x) = (x − 3)(x + 5) are 3 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

811

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The zeros are 3 and −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 8.F.B.4 • FUNCTIONS

Find the rate of change and initial value from two points, a table, or a graph

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Find a constant rate from a table	**7.RP.A.2.b**
BACKWARD 2	Subtract integers	**7.NS.A.1.c**
FORWARD 1	Construct linear and exponential functions from two points	**HSF-LE.A.2**
FORWARD 2	Write arithmetic and geometric sequences recursively and explicit..	**HSF-BF.A.2**

SET 27 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A linear function passes through (2, 7) and (5, 16).

What are its rate of change and its initial value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

814

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 1

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

The table shows a linear function.

What are its rate of change and its initial value?

x	1	3	5
y	10	16	22

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

815

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 7

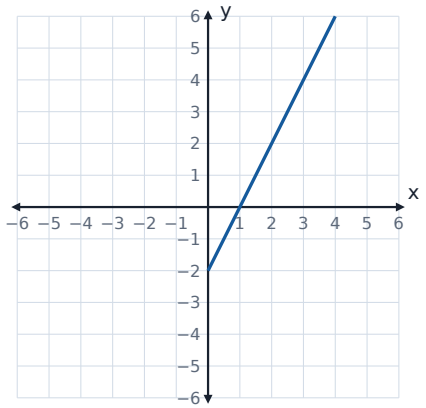
GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

The graph shows a linear function.

What are its rate of change and its initial value?



A. Rate 4, initial value −2

B. Rate −2, initial value 2

C. Rate 2, initial value −2

D. Rate 2, initial value 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

816

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Rate 2, initial value −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

A line passes through (−1, 8) and (3, 0).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

817

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −2x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

True or false?

A linear function with f(0) = 5 and f(2) = 1 has a rate of change of −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

818

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

SET 27 • 8.F.B.4

BACKWARD 1

Find a constant rate from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant rate (y per x)?

x	3	6	9
y	12	24	36

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

820

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

What is the constant rate (y per x)?

Hours	2	5	8
Miles	7	17.5	28

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

821

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

A proportional table contains the pair (4, 10).

What is the constant rate (y per x)?

A. 0.4

B. 40

C. 2.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

822

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (2, 1) and (6, 3) has a constant rate of 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

823

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

True or false?

A proportional table with the pairs (5, 15) and (8, 24) has a constant rate of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

824

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 8.F.B.4

BACKWARD 2

Subtract integers • Grade 7 • The Number System • 7.NS.A.1.c

Subtract.

$6 - (-2)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

826

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

Subtract.

$-3 - 5$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

827

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

Subtract.

$-4 - (-9)$

A. 5

B. 13

C. -13

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

828

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

True or false?

$2 - 7 = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

829

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

True or false?

$-6 - (-6) = -12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 8.F.B.4

FORWARD 1

Construct linear and exponential functions from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (1, 4) and (3, 10).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

832

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

Write the exponential function f(x) = a · b^x whose graph passes through (0, 5) and (1, 15).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

833

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

Which exponential function passes through (0, 2) and (2, 18)?

A. f(x) = 8x + 2

B. f(x) = 2 · 3^x

C. f(x) = 18 · 2^x

D. f(x) = 2 · 9^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

834

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. f(x) = 2 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

True or false?

The linear function through (0, −1) and (4, 7) is f(x) = 2x − 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

835

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

True or false?

The exponential function through (0, 4) and (1, 2) is f(x) = 4 · 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

836

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The function is f(x) = 4 · (0.5)^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

SET 27 • 8.F.B.4

FORWARD 2

Write arithmetic and geometric sequences recursively and explicitly • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula for the arithmetic sequence 7, 11, 15, ..., starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

838

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4n + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

Write a recursive formula for the sequence 7, 11, 15, ...

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

839

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a(1) = 7;

a(n) = a(n − 1) + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

Which explicit formula gives 3, 6, 12, 24, ..., starting with n = 1?

A. a(n) = 2 · 3^(n − 1)

B. a(n) = 3 · 2^(n − 1)

C. a(n) = 3 + 2n

D. a(n) = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

840

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a(n) = 3 · 2^(n − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function whose domain is the counting numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

841

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

The sequence 2, 5, 8, 11, ... has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

842

QUESTION ID
S27-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It has a common difference of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 8.F.B.4 • FUNCTIONS

Construct a linear function to model a situation

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Write an equation for a situation	**6.EE.C.9**
BACKWARD 2	Solve a two-step equation from a model	**7.EE.B.4.a**
FORWARD 1	Write a function for a situation and use it to answer a question	**HSF-BF.A.1.a**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 28 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A plumber charges $65 for a visit plus $40 per hour.

Write a function for the cost C after h hours.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

845

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

40h + 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

A candle is 30 cm tall and burns down 2.5 cm each hour.

Write a function for its height h after t hours.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

846

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

30 − 2.5t

GRADING NOTE −2.5t + 30 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

A phone plan costs $20 per month after a $50 activation fee.

Which function gives the total cost C after m months?

A. C = 20m − 50

B. C = 20m + 50

C. C = 50m + 20

D. C = 70m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

847

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. C = 20m + 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

The cost of a taxi ride is a linear function of the miles m. A 3-mile ride costs $11 and a 7-mile ride costs $21.

Write the function.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

848

QUESTION ID

S28-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5m + 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

True or false?

A tank holds 500 liters and drains 20 liters each minute. The function V = 500 − 20t gives the liters V after t minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

849

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

SET 28 • 8.F.B.4

BACKWARD 1

Write an equation for a situation • Grade 6 • Expressions & Equations • 6.EE.C.9

Tickets cost $9 each.

Write an equation for the cost c of t tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

851

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 9t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

Mia has $15 and saves $5 each week.

Write an equation for her savings s after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

852

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

s = 5w + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

A pool holds 100 gallons and fills at 10 gallons per minute.

Which equation gives the gallons g after m minutes?

A. g = 110m

B. g = 100m + 10

C. g = 10m + 100

D. g = 10m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

853

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g = 10m + 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

True or false?

Apples cost $3 per pound. The equation c = 3p gives the cost c of p pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

854

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A game costs a $10 fee plus $2 per game. The equation c = 10g + 2 gives the cost c of g games.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

855

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is c = 2g + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 8.F.B.4

BACKWARD 2

Solve a two-step equation from a model • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for h.

40h + 65 = 185

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

857

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

Solve for t.

$30 - 2.5t = 10$

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

858

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

Solve for m.

20m + 50 = 170

A. m = 3

B. m = 11

C. m = 8.5

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

859

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

True or false?

t = 4 is the solution of 500 − 20t = 420.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

860

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

True or false?

h = 5 is the solution of 12h + 8 = 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

861

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is h = 3.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 8.F.B.4

FORWARD 1

Write a function for a situation and use it to answer a question • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a

A gym charges $25 per month, but the first month is free.

Write a function for the total cost C after m months, where m ≥ 1.

C(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

863

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

25(m − 1)

GRADING NOTE 25m − 25 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

A savings account starts with $300, and $40 is added each week.

Write a function S(w), then find when the balance reaches $700.

Function and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

S(w) = 40w + 300; 10 weeks

GRADING NOTE Grade both: the function and 10 weeks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

A pool is 4 feet deep and drains 0.25 foot each hour.

Which function gives the depth after t hours, and when is the pool empty?

A. d(t) = 4 − 0.25t; after 16 hours

B. d(t) = 0.25t + 4; after 16 hours

C. d(t) = 4t − 0.25; after 1 hour

D. d(t) = 4 − 0.25t; after 4 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

865

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. d(t) = 4 − 0.25t; after 16 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

True or false?

A rental costs $30 plus $0.20 per mile. The function C(m) = 0.2m + 30 gives the cost for m miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

866

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

True or false?

A balance of $200 decreases by $15 each month. The function B(m) = 15m + 200 gives the balance after m months.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

867

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The function is B(m) = 200 − 15m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

SET 28 • 8.F.B.4

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve C = 40h + 65 for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

869

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(C − 65)/40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

Solve $F = \frac{9}{5}C + 32$ for C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

870

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(5/9)(F − 32)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

Solve $y = mx + b$ for b.

A. $b = \frac{y}{mx}$

B. $b = mx - y$

C. $b = y + mx$

D. $b = y - mx$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

871

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b = y − mx

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

True or false?

For $t \neq 0$, solving $d = rt$ for r gives $r = \frac{d}{t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

872

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

True or false?

For $l \neq 0$, solving $A = lw$ for w gives $w = A - l$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

873

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $w = A/l$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 8.F.B.4 • FUNCTIONS

Interpret the rate of change and initial value of a linear function in context

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Identify the independent and dependent variables	**6.EE.C.9**
FORWARD 1	Interpret the slope and intercept of a linear model of data	**HSS-ID.C.7**
FORWARD 2	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**

SET 29 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

Renting a bike costs C = 4h + 10 dollars for h hours.

What do the 4 and the 10 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

876

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

4: the cost goes up $4 for each hour. 10: a $10 fee (the cost at 0 hours).

GRADING NOTE Grade both numbers.

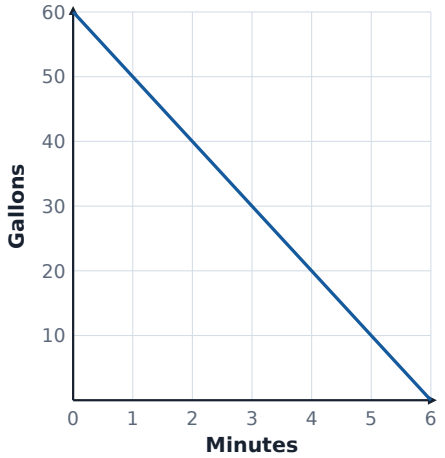
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

The graph shows the water in a tank.

What do the rate of change and the initial value mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

877

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

The tank starts with 60 gallons (initial value) and drains 10 gallons per minute (rate of change −10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

A plant's height is h = 0.5w + 12 centimeters after w weeks.

What does 0.5 mean?

A. The plant grows for 0.5 week.

B. The plant grows 0.5 cm each week.

C. The plant was 0.5 cm tall at the start.

D. The plant grows 12 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

878

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The plant grows 0.5 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

The table shows the money in a savings account.

What does the rate of change mean?

Weeks	0	4	8
Dollars	150	210	270

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

879

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

$15 is added each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

True or false?

For C = 4h + 10, the initial value means the bike costs $4 before any hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

880

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

SET 29 • 8.F.B.4

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of gallons (x) and miles (y) passes through (1, 28).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

882

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The car travels 28 miles on 1 gallon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

A proportional graph shows hours worked (x) and pay (y).

What does the point (0, 0) mean?

A. 0 hours of work earn $0.

B. The worker starts with $0 per hour.

C. The pay is $0 per hour.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

883

QUESTION ID

S29-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 0 hours of work earn $0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and laps (y) passes through (6, 3).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

884

QUESTION ID

S29-B1-Q3

TYPE

Short answer

CORRECT ANSWER

3 laps are swum in 6 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

885

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

True or false?

On a graph of tickets (x) and cost (y), the point (2, 18) means each ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

886

QUESTION ID
S29-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It means 2 tickets cost $18 ($9 each).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 8.F.B.4

BACKWARD 2

Identify the independent and dependent variables • Grade 6 • Expressions & Equations • 6.EE.C.9

In C = 4h + 10, which variable is the independent variable?

Variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

888

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

"Pay depends on the hours worked." Which quantity is the dependent variable?

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

889

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Pay

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

In d = 60t, what is d?

A. The rate

B. The independent variable

C. A constant

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

890

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

True or false?

In a table of hours worked and pay, the hours are usually the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

891

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

True or false?

In y = 3x + 2, x depends on y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

892

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 8.F.B.4

FORWARD 1

Interpret the slope and intercept of a linear model of data • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A line of fit predicts a test score y = 4.5x + 60 for x hours of study.

What does 4.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

894

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Each extra hour of study goes with about 4.5 more points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

A line of fit predicts a test score y = 4.5x + 60 for x hours of study.

What does 60 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

895

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The predicted score for 0 hours of study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

A model gives a used car's value as $y = 18{,}000 - 1{,}500x$ dollars after x years.

What does $-1{,}500$ mean?

A. The car is 1,500 years old.

B. The value drops about $1,500 each year.

C. The car costs $1,500.

D. The value rises $1,500 each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

896

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The value drops about $1,500 each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

True or false?

A model gives a child's height as y = 2.1x + 30 cm at age x months. The intercept 30 is the predicted height at 0 months.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

True or false?

A slope of −3 in a linear model means y increases by 3 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

898

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

SET 29 • 8.F.B.4

FORWARD 2

Interpret the parameters of linear and exponential functions • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

P(t) = 2,000(1.05)^t gives an account balance after t years.

What do 2,000 and 1.05 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

900

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 is the starting amount; 1.05 means the balance grows 5% each year.

GRADING NOTE Grade both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

V(t) = 25,000(0.85)^t gives a car's value after t years.

What is the yearly percent change?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

901

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 15% decrease each year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

A phone plan costs C(m) = 0.12m + 9.99 dollars for m minutes.

What does 9.99 represent?

A. A fixed monthly charge

B. The number of minutes

C. The total cost

D. The cost per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

902

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A fixed monthly charge

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

True or false?

In f(t) = 100(2)^t, the quantity doubles each time t increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

903

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

True or false?

In f(t) = 50(0.9)^t, the quantity grows by 90% each period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

904

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 10% each period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 8.F.B.5 • FUNCTIONS

Describe the relationship shown by a graph

MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Interpret points on a graph of a situation	**5.G.A.2**
BACKWARD 2	Compare speeds as unit rates	**6.RP.A.3.b**
FORWARD 1	Interpret the key features of a graph in context	**HSF-IF.B.4**
FORWARD 2	Graph and evaluate piecewise-defined functions	**HSF-IF.C.7.b**

SET 30 • 8.F.B.5

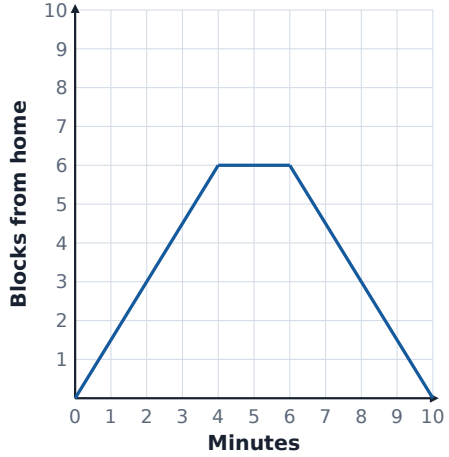
MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

The graph shows Ana's walk.

Describe what happens from 4 to 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

907

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

She stays 6 blocks from home; she stops.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

The graph shows Ana's walk.

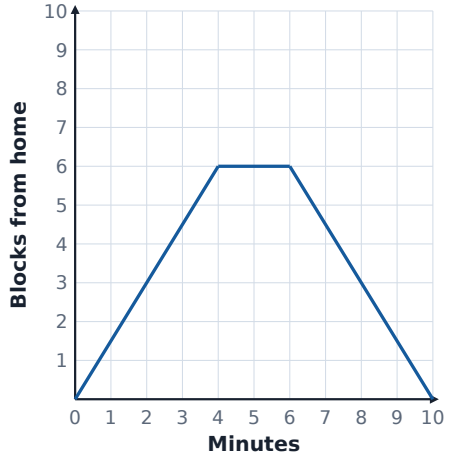
During which interval is Ana moving toward home?

A. 0 to 4 minutes

B. 6 to 10 minutes

C. 4 to 6 minutes

D. 0 to 6 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

908

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 6 to 10 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

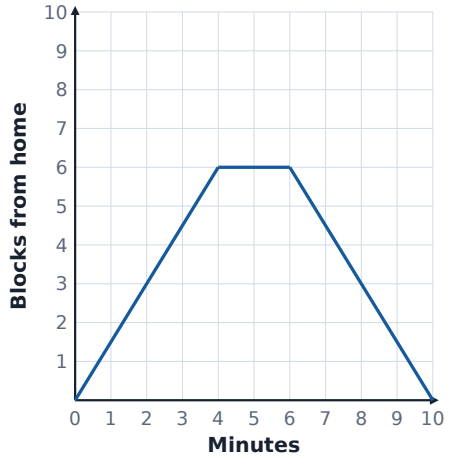
ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

The graph shows Ana's walk.

Is the relationship linear or nonlinear over the whole walk? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

909

QUESTION ID

S30-M-Q3

TYPE

Short answer

CORRECT ANSWER

Nonlinear. The graph is not one straight line; the rate of change is different on different parts.

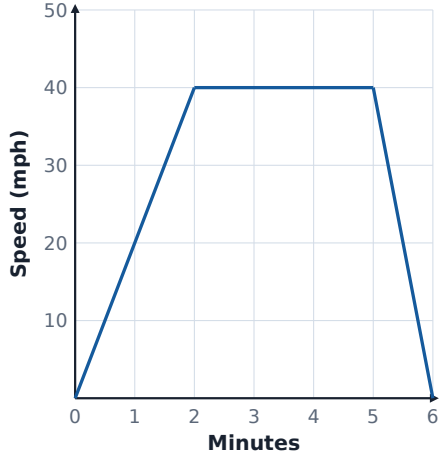
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

The graph shows a car's speed.

Describe how the speed changes over the 6 minutes.



Description: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

910

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

The car speeds up for 2 minutes, travels at a steady 40 mph until minute 5, then slows to a stop.

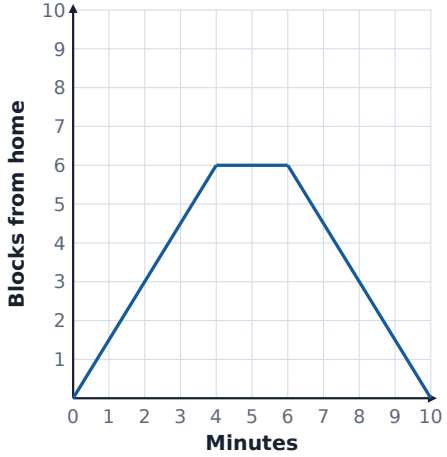
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

True or false?

In the graph of Ana's walk, she walks faster going home than going away from home.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

911

QUESTION ID

S30-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both trips cover 6 blocks in 4 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

SET 30 • 8.F.B.5

BACKWARD 1

Interpret points on a graph of a situation • Grade 5 • Geometry • 5.G.A.2

A graph shows minutes (x) and blocks from home (y). What does the point (3, 5) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

913

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

After 3 minutes, the walker is 5 blocks from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

A graph shows days (x) and plant height in centimeters (y). What does the point (10, 4) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

914

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

After 10 days, the plant is 4 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

A graph shows hours (x) and miles driven (y).

Which point shows 120 miles after 2 hours?

A. (2, 120)

B. (60, 2)

C. (2, 60)

D. (120, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

915

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 120)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

True or false?

On a graph of time (x) and distance (y), the point (0, 0) can mean the start of a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

916

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (4, 1) on a graph of hours (x) and miles (y) means 4 miles in 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

917

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 8.F.B.5

BACKWARD 2

Compare speeds as unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana walks 6 blocks in 4 minutes at a steady pace.

What is her speed in blocks per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

919

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5 blocks per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

Which is faster: 12 miles in 3 hours, or 10 miles in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

920

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10 miles in 2 hours (5 mph, compared with 4 mph)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

Which rate is the greatest?

A. 60 miles in 1 hour

B. 100 miles in 2 hours

C. 20 miles in 0.5 hour

D. 150 miles in 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

921

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 miles in 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

True or false?

Walking 3 miles in 1 hour is faster than walking 5 miles in 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

922

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

True or false?

Running 8 km in 40 minutes is the same rate as running 2 km in 20 minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

923

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 8.F.B.5

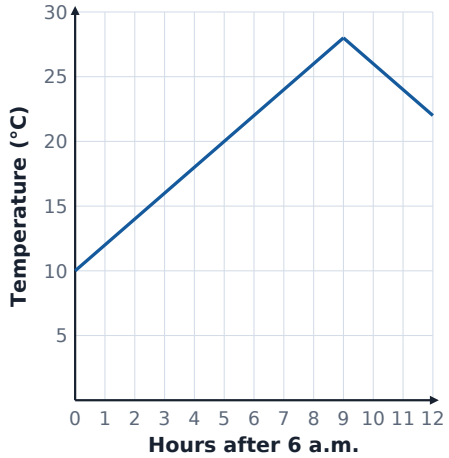
FORWARD 1

Interpret the key features of a graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the temperature during a day.

When is the temperature highest, and what is it?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

925

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9 hours after 6 a.m. (3 p.m.); 28°C

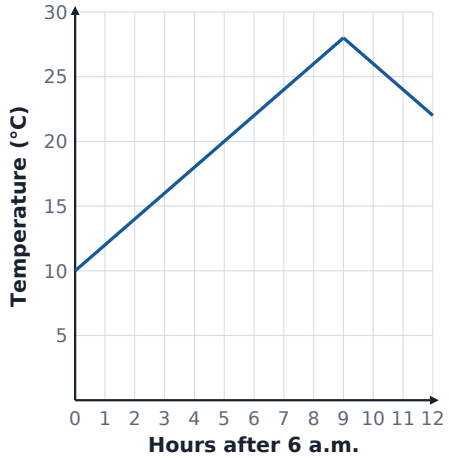
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

The graph shows the temperature during a day.

On what interval is the temperature increasing?

Interval: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

926

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

From 0 to 9 hours after 6 a.m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

The graph shows the temperature during a day.

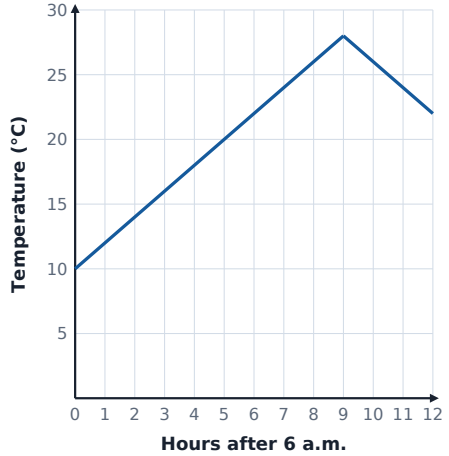
What does the y-intercept mean?

A. It was 28°C at 3 p.m.

B. It was 10°C at 6 a.m.

C. It was 22°C at 6 p.m.

D. The temperature rose for 9 hours.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

927

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. It was 10°C at 6 a.m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

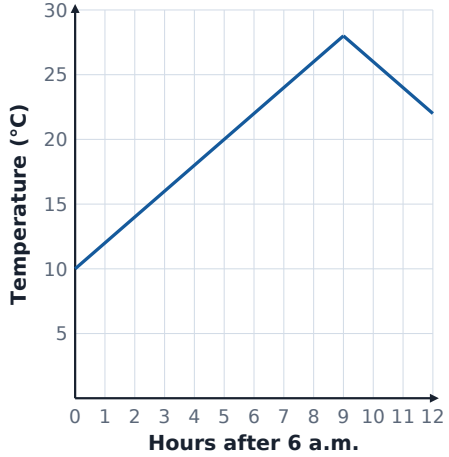
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

True or false?

In the graph, the temperature is decreasing from 9 to 12 hours after 6 a.m.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

928

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

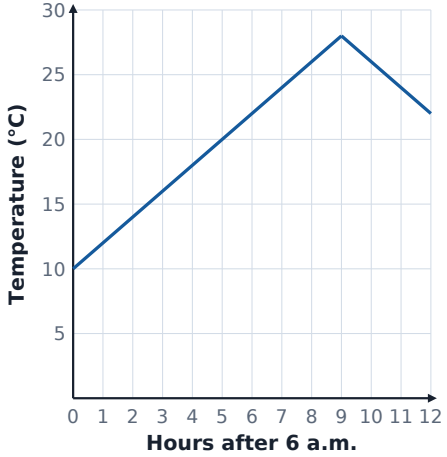
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph shows that the temperature never dropped below 22°C.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

929

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It started at 10°C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 8.F.B.5

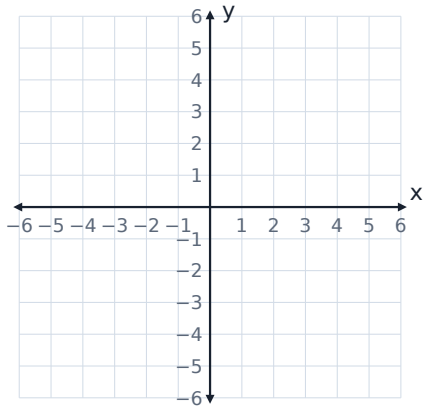
FORWARD 2

Graph and evaluate piecewise-defined functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the coordinate plane.

$f(x) = x + 2$ for $x < 0$

$f(x) = 2$ for $x \geq 0$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

931

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The line $y = x + 2$ for $x < 0$ (an open end at (0, 2) is acceptable) and the horizontal line $y = 2$ for $x \geq 0$, with a closed point at (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

$f(x) = 3$ for $x \leq 2$, and $f(x) = x + 1$ for $x > 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

932

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after that.

Which function gives the cost for m miles?

A. C(m) = 2m + 3 for all m

B. C(m) = 3m for all m

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

D. C(m) = 3 + 2m for m ≤ 1 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

933

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

True or false?

For $f(x) = |x|$, $f(-4) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

934

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

True or false?

A piecewise-defined function can never have an unbroken graph.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

935

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example, f(x) = |x| is piecewise-defined and unbroken.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

SET 31 • 8.F.B.5 • FUNCTIONS

Sketch a graph from a description of a relationship

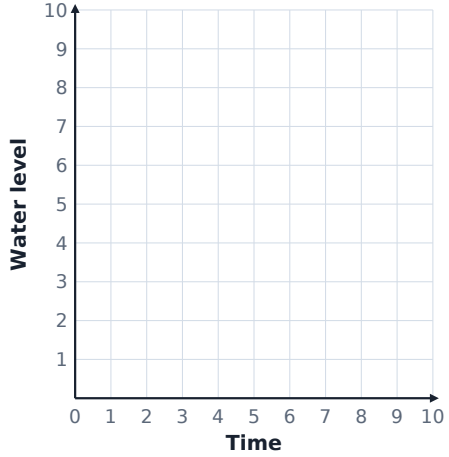
MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Graph a relationship from a table	**6.EE.C.9**
BACKWARD 2	Plot points that represent a situation	**5.G.A.2**
FORWARD 1	Sketch graphs that show given key features	**HSF-IF.B.4**
FORWARD 2	Graph exponential functions and describe their end behavior	**HSF-IF.C.7.e**

SET 31 • 8.F.B.5

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

Sketch a graph of the water level in a bathtub over time: it fills at a steady rate, stays the same while someone bathes, and then drains quickly.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

938

QUESTION ID

S31-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

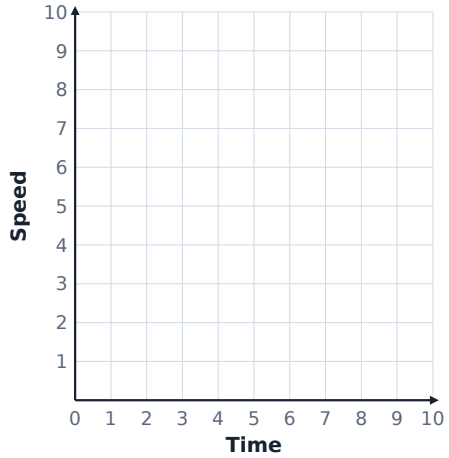
A rising straight segment from 0, then a horizontal segment, then a steeper falling segment back to 0. Any lengths are accepted if the order and steepness are right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

Sketch a graph of a car's speed over time: it speeds up, travels at a steady speed, and then slows down to a stop.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

939

QUESTION ID

S31-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

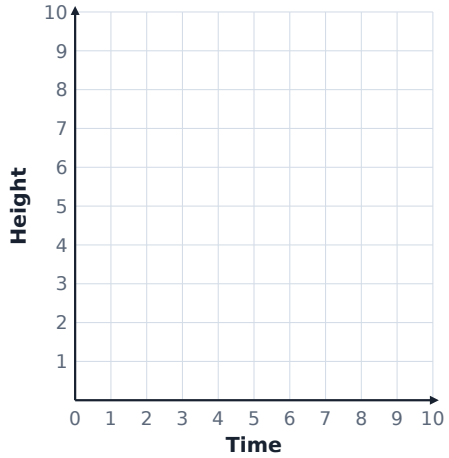
A rising segment from 0, a horizontal segment, then a falling segment back to 0. Any lengths are accepted if the order is right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

Sketch the height of a ball thrown straight up over time: it rises, slows down, reaches a highest point, and falls back to the ground.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

940

QUESTION ID

S31-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

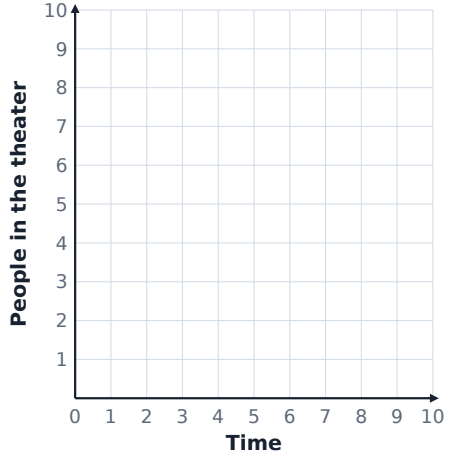
A curve that rises, levels off at one highest point, and falls back to height 0, like an upside-down U. Starting at ground level or at the throwing height is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

Sketch the number of people in a theater from 6:00 to 10:00. It fills quickly before a 7:00 show, stays full, and empties after the show ends at 9:30.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

941

QUESTION ID

S31-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A steep rise before 7:00, a horizontal segment from about 7:00 to 9:30, then a steep fall after 9:30 back to about 0. Exact times are not required if the order is right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

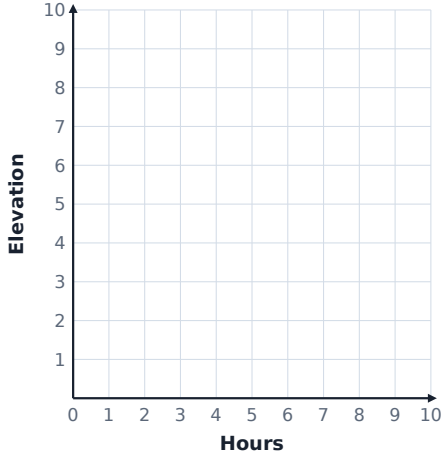
ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

Sketch a graph of a hike: the hiker climbs a steep trail for 2 hours, then a gentle slope for 1 hour, then rests for 1 hour.

Is elevation a linear function of time for the whole hike? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

942

QUESTION ID

S31-M-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A steeper rising segment from 0 to 2 hours, a less steep rising segment from 2 to 3 hours, and a horizontal segment from 3 to 4 hours

Written: No. The rate of change is different on each part (steep, gentle, then zero), so the whole graph is not one line.

GRADING NOTE Grade both: the three-part graph in the right order and steepness, and "no" with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

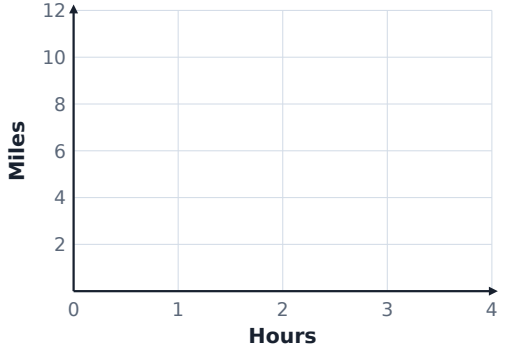
SET 31 • 8.F.B.5

BACKWARD 1

Graph a relationship from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Graph the pairs in the table.

Hours	0	1	2	3
Miles	0	4	8	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

944

QUESTION ID

S31-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 0), (1, 4), (2, 8), and (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

A hiker walks 4 miles each hour, so miles = 4 × hours.

How many miles does she walk in 2.5 hours?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

945

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

Which table shows y increasing by the same amount each time?

A. x: 0, 1, 2 and y: 0, 3, 4

B. x: 0, 1, 2 and y: 5, 5, 6

C. x: 0, 1, 2 and y: 6, 3, 9

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

946

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A table where y is 5 for every x shows y not changing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

947

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

True or false?

A table with y-values 2, 4, 3 shows y always increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

948

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

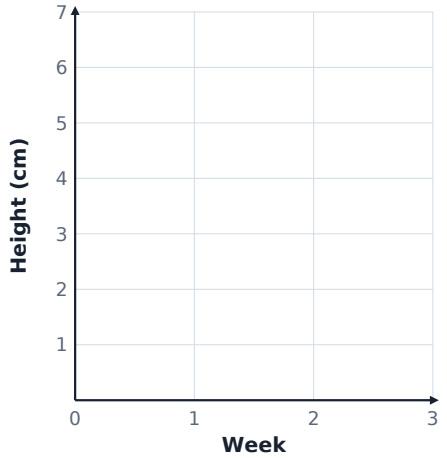
SET 31 • 8.F.B.5

BACKWARD 2

Plot points that represent a situation • Grade 5 • Geometry • 5.G.A.2

A plant is 2 cm tall at week 0, 4 cm at week 1, and 6 cm at week 2.

Plot the points (week, height).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

950

QUESTION ID

S31-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 2), (1, 4), and (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

A plant is 2 cm tall at week 0 and grows 2 cm each week.

Which point shows its height at week 3?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

951

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

On a graph of weeks (x) and plant height in cm (y), what does the point (2, 6) mean?

A. The plant grows 2 cm.

B. At week 6 the plant is 2 cm tall.

C. The plant grows 6 cm each week.

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

952

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

On a graph of weeks (x) and height (y), the point (0, 2) shows the height at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

953

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

True or false?

On a graph of weeks (x) and height in cm (y), the point (1, 4) means the plant is 1 cm tall at week 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

954

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

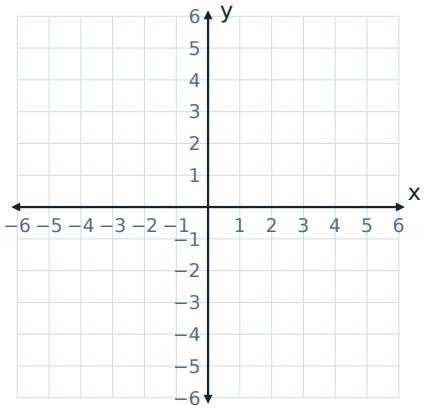
Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 8.F.B.5

FORWARD 1

Sketch graphs that show given key features • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

Sketch a function that increases for 0 < x < 3, has a maximum value of 5 at x = 3, and decreases for x > 3.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

956

QUESTION ID

S31-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

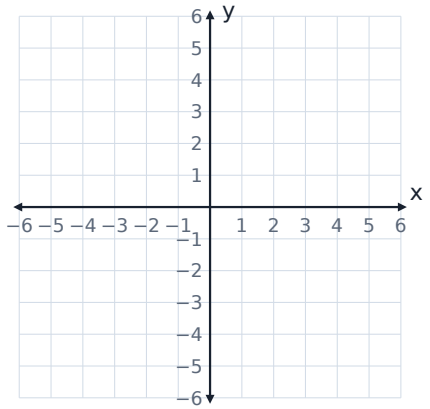
Any graph that rises up to the point (3, 5) and falls after it, with 5 as its highest value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

Sketch a function whose graph crosses the y-axis at 2 and that increases for all x shown.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

957

QUESTION ID

S31-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Any always-increasing graph through (0, 2); a line or a curve is accepted

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

Which description matches $y = -(x - 2)^2 + 4$?

A. A maximum of 4 at $x = 2$; increasing for $x < 2$

B. A minimum of 4 at $x = 2$

C. A maximum of 2 at $x = 4$

D. Increasing for every x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

958

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 4 at x = 2; increasing for x < 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

A graph with a maximum at x = 3 decreases for some x-values greater than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

959

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

A function that is always decreasing can have its maximum in the middle of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

960

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

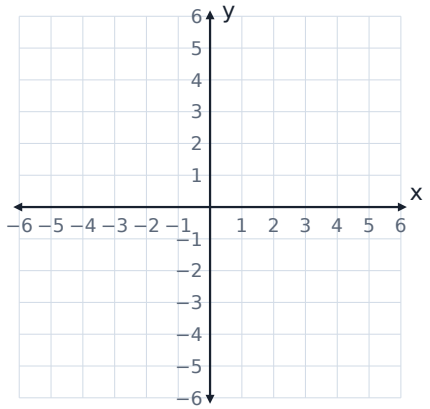
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • 8.F.B.5

FORWARD 2

Graph exponential functions and describe their end behavior • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph y = 2^x for x from −2 to 2 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

962

QUESTION ID

S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The points (−2, 1/4), (−1, 1/2), (0, 1), (1, 2), and (2, 4) joined by a smooth increasing curve that gets close to the x-axis on the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

What is the y-intercept of the graph of y = 5(0.5)^x?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

963

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

For y = 3^x, what happens to y as x decreases without bound?

A. y gets closer to 3.

B. y decreases without bound.

C. y increases.

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

964

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph of y = 2^x never touches the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

965

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

True or false?

The function y = (0.5)^x is increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

966

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

SET 32 • 8.G.A.1.a • GEOMETRY

Verify that rigid motions take lines to lines and segments to segments of the same length

MAIN	Grade 8 main question	**8.G.A.1.a**
BACKWARD 1	Reflect points across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
FORWARD 1	Define rotations, reflections, and translations precisely	**HSG-CO.A.4**
FORWARD 2	Use rigid motions to decide whether figures are congruent	**HSG-CO.B.6**

SET 32 • 8.G.A.1.a

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.a

Segment AB has endpoints A(1, 2) and B(5, 2). It is translated 3 units down.

What are the endpoints of the image, and how long is the image?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

969

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

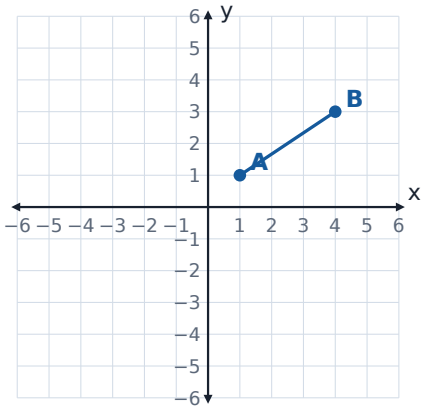
A'(1, −1) and B'(5, −1); it is 4 units long, the same as AB.

GRADING NOTE Grade both the endpoints and the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

Reflect segment AB across the y-axis. Draw the image A'B'.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

970

QUESTION ID

S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The segment from A'(−1, 1) to B'(−4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

A segment 7 cm long is rotated 90° about a point.

How long is its image?

A. 3.5 cm

B. 7 cm

C. 14 cm

D. It depends on the center of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

971

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

True or false?

The image of a straight line under a translation is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

True or false?

A reflection can map a segment 5 units long to a segment 10 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

973

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Reflections keep lengths the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, −4) is reflected across the x-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

975

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

The point (−2, 5) is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

976

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

Which point is the reflection of (6, 1) across the y-axis?

A. (1, 6)

B. (6, −1)

C. (−6, −1)

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

977

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The reflection of (−3, −3) across the x-axis is (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

978

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

True or false?

The reflection of (4, 2) across the x-axis is (−4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

979

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 2

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

981

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Find the distance between (1, −1) and (5, −1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

982

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which two points are 6 units apart?

A. (−2, 3) and (2, 3)

B. (0, 1) and (0, 5)

C. (3, −2) and (3, 2)

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

983

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

(0, −3) and (0, 4) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

984

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

(−5, 1) and (2, 1) are 3 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

985

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 7 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 1

Define rotations, reflections, and translations precisely • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A reflection across line m maps point P to point P'.

How is line m related to segment PP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

987

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Line m is the perpendicular bisector of PP'.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

A rotation of 90° about point O maps P to P'.

What is true about OP, OP', and angle POP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

988

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

OP = OP', and angle POP' measures 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

In a translation, the segments that join each point to its image are...

A. perpendicular to each other

B. all through one point

C. different lengths

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

989

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

True or false?

A rotation keeps every point the same distance from the center of rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

990

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

True or false?

In a reflection, a point (not on the line) and its image are on the same side of the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

991

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 2

Use rigid motions to decide whether figures are congruent • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is mapped onto triangle DEF by a rotation followed by a translation.

What can you conclude about the triangles?

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

993

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

They are congruent; their corresponding sides and angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

Explain why two figures are congruent when a sequence of rigid motions maps one onto the other.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

994

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rigid motions keep all lengths and angle measures, so every part of one figure matches the corresponding part of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

Which transformation does NOT always map a figure onto a congruent figure?

A. A translation

B. A rotation

C. A dilation with a scale factor of 2

D. A reflection

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

995

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A dilation with a scale factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

True or false?

If a translation maps segment AB onto segment CD, then AB = CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

996

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

True or false?

Two triangles with the same area must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

997

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

SET 33 • 8.G.A.1.b • GEOMETRY

Verify that rigid motions take angles to angles of the same measure

MAIN	Grade 8 main question	**8.G.A.1.b**
BACKWARD 1	Find unknown angles with supplementary, complementary, and v…	**7.G.B.5**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Specify a sequence of transformations that maps a figure onto a…	**HSG-CO.A.5**
FORWARD 2	Prove theorems about parallelograms	**HSG-CO.C.11**

SET 33 • 8.G.A.1.b

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.b

Angle ABC measures 65°. It is reflected across a line.

What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

1000

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

Triangle PQR has angles of 40°, 60°, and 80°. It is rotated 180° about the origin.

What are the angle measures of the image?

Measures: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

1001

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

40°, 60°, and 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

A right angle is translated 5 units up.

What is its image?

A. A right angle

B. A straight angle

C. An obtuse angle

D. An acute angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

1002

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

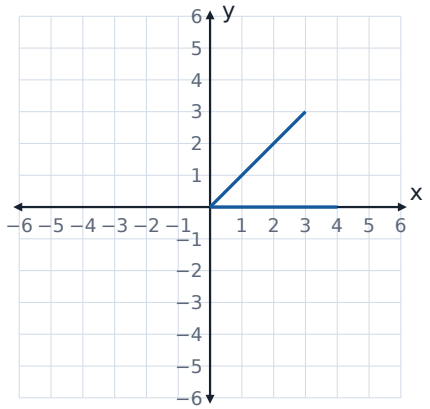
A. A right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

The angle has its vertex at the origin. Rotate it 90° counterclockwise about the origin and draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

1003

QUESTION ID

S33-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

An angle with its vertex at the origin and rays through (0, 4) and (−3, 3); it still measures 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

True or false?

A rotation can map a 30° angle onto a 60° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

1004

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rotations keep angle measures the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 1

Find unknown angles with supplementary, complementary, and vertical angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 65°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

115°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines intersect. One angle measures 40°.

What does the vertical angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

What is the complement of a 25° angle?

A. 65°

B. 75°

C. 155°

D. 25°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Supplementary angles add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 40°, 60°, and 80°.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

A triangle has a 90° angle.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

Which angle measures make an obtuse triangle?

A. 60°, 60°, 60°

B. 30°, 40°, 110°

C. 45°, 45°, 90°

D. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30°, 40°, 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

True or false?

An acute triangle has three angles that each measure less than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 1

Specify a sequence of transformations that maps a figure onto another • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle A(1, 1), B(3, 1), C(1, 4) is mapped onto A'(−1, −1), B'(−3, −1), C'(−1, −4).

Describe a transformation or sequence of transformations that does this.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

1018

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin.

GRADING NOTE A reflection across the x-axis followed by a reflection across the y-axis (or the reverse) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

Which single transformation maps (2, 3) to (2, −3) and (5, 1) to (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

1019

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which rotation about the origin maps (4, 0) to (0, 4)?

A. 270° counterclockwise

B. 180°

C. 90° clockwise

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

1020

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across a line and then reflecting the image across the same line gives back the original figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

1021

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A translation 3 units right followed by a translation 3 units left moves the figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

1022

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The figure returns to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 2

Prove theorems about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram. Prove that angle A and angle B are supplementary. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5

1024

QUESTION ID

S33-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: AD ∥ BC (definition of a parallelogram), and AB is a transversal; angles A and B are same-side interior angles, so they are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, angle A = 70°.

Find the other three angles and give a reason for each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5

1025

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Angle B = 110° and angle D = 110° (consecutive angles are supplementary); angle C = 70° (opposite angles are congruent).

GRADING NOTE Grade the angles and the reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5

Which statement about every parallelogram can be proved?

A. All of its angles are right angles.

B. Its diagonals are congruent.

C. Its diagonals bisect each other.

D. Its diagonals are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5

1026

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Its diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5

True or false?

Opposite sides of a parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5

1027

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5

True or false?

A quadrilateral with one pair of parallel sides must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5

1028

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 8.G.A.1.c • GEOMETRY

Verify that rigid motions take parallel lines to parallel lines

MAIN	Grade 8 main question	**8.G.A.1.c**
BACKWARD 1	Identify parallel lines and segments	**4.G.A.1**
BACKWARD 2	Draw polygons in the coordinate plane and find side lengths	**6.G.A.3**
FORWARD 1	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**
FORWARD 2	Use coordinates to prove simple geometric theorems	**HSG-GPE.B.4**

SET 34 • 8.G.A.1.c

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.c

Lines p and q are parallel. Both are rotated 90° about the same point.

Are the images parallel? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

1031

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

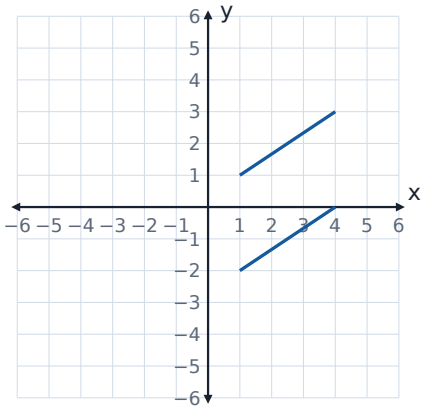
Yes. Rigid motions take parallel lines to parallel lines; both lines turn by the same angle, so the images never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

The two segments are parallel. Translate both segments 4 units to the left and draw the images.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

1032

QUESTION ID

S34-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Segments from (−3, 1) to (0, 3) and from (−3, −2) to (0, 0); the images are still parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

A parallelogram is reflected across a line.

What is its image?

A. A rectangle, in every case

B. A figure with no parallel sides

C. A parallelogram congruent to the original

D. A trapezoid with only one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

1033

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A parallelogram congruent to the original

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

The line y = 2x + 1 is translated 3 units up.

What is the equation of the image? Is the image parallel to the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

1034

QUESTION ID

S34-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x + 4; yes, the slope is still 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

True or false?

A rigid motion can map two parallel lines onto two lines that intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

1035

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 1

Identify parallel lines and segments • Grade 4 • Geometry • 4.G.A.1

Name a quadrilateral that has two pairs of parallel sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

1037

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

GRADING NOTE Any parallelogram (square, rhombus, rectangle) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

How many pairs of parallel sides does a trapezoid that is not a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

Which capital letter contains a pair of parallel segments?

A. V

B. H

C. T

D. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. H

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

True or false?

Parallel lines never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

1040

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

True or false?

Perpendicular lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

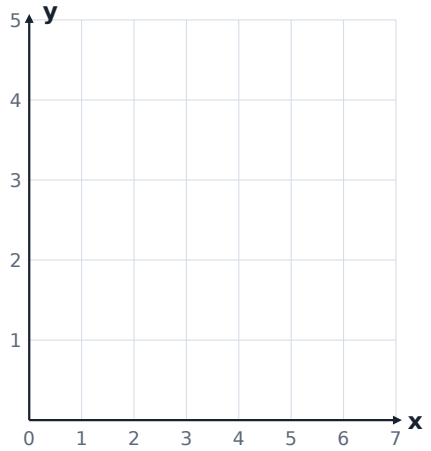
Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 2

Draw polygons in the coordinate plane and find side lengths • Grade 6 • Geometry • 6.G.A.3

Draw the quadrilateral with vertices (1, 1), (5, 1), (6, 4), and (2, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S34-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The quadrilateral with those four vertices (a parallelogram)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

Which side is parallel to the side from (1, 1) to (5, 1)?

A. From (1, 1) to (6, 4)

B. From (2, 4) to (6, 4)

C. From (1, 1) to (2, 4)

D. From (5, 1) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. From (2, 4) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

True or false?

The side from (2, 4) to (6, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

True or false?

The segment from (1, 1) to (1, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

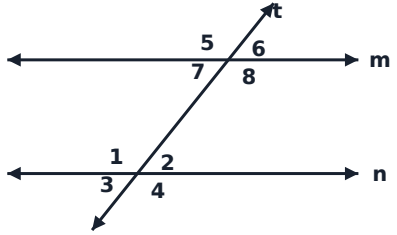
SET 34 • 8.G.A.1.c

FORWARD 1

Prove theorems about angles formed by parallel lines • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

1049

QUESTION ID

S34-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 2 = angle 6 (corresponding angles, m ∥ n); angle 6 = angle 7 (vertical angles); so angle 2 = angle 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. One angle measures 72°.

What does the same-side interior angle measure? Give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

1050

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

108°; same-side interior angles are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

Which statement can be proved from "corresponding angles formed by parallel lines are congruent" and "vertical angles are congruent"?

A. Alternate interior angles are congruent.

B. The transversal is parallel to the lines.

C. The lines are perpendicular.

D. All the angles are right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

1051

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

True or false?

If alternate interior angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

1052

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

True or false?

Same-side interior angles formed by parallel lines are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

1053

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

FORWARD 2

Use coordinates to prove simple geometric theorems • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Show that the quadrilateral with vertices (0, 0), (4, 0), (5, 3), and (1, 3) is a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

1055

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The sides from (0, 0) to (4, 0) and from (1, 3) to (5, 3) both have slope 0; the sides from (0, 0) to (1, 3) and from (4, 0) to (5, 3) both have slope 3. Both pairs of opposite sides are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

Is the triangle with vertices (0, 0), (4, 0), and (0, 3) a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

1056

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. One side is horizontal and one is vertical, so they are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

In order, (0, 0), (2, 0), (3, 2), and a fourth point form a parallelogram.

Which is the fourth point?

A. (0, 3)

B. (3, 0)

C. (1, 2)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

1057

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

True or false?

A quadrilateral whose opposite sides have equal slopes is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

1058

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

True or false?

The points (0, 0), (1, 1), and (2, 3) lie on one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

1059

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 8.G.A.2 • GEOMETRY

Describe a sequence of rigid motions that shows two figures are congruent

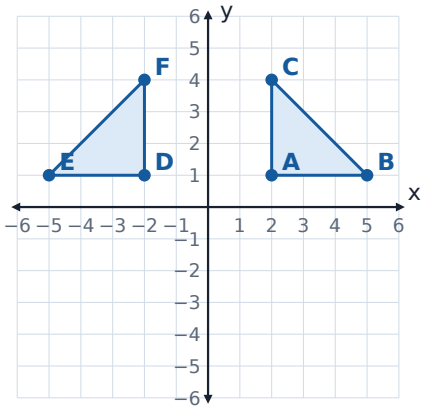
MAIN	Grade 8 main question	**8.G.A.2**
BACKWARD 1	Draw a polygon from its vertices	**6.G.A.3**
BACKWARD 2	Use signs to locate points across the axes	**6.NS.C.6.b**
FORWARD 1	Use corresponding parts to explain triangle congruence	**HSG-CO.B.7**
FORWARD 2	Use triangle congruence criteria	**HSG-CO.B.8**

SET 35 • 8.G.A.2

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.2

Describe a rigid motion that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

1062

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

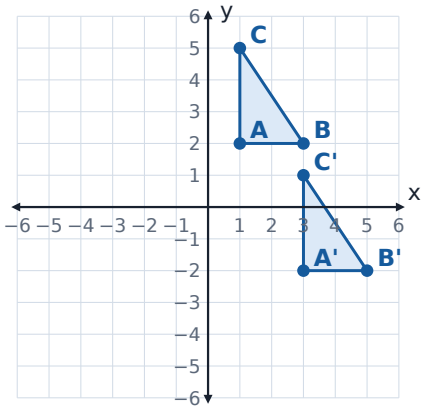
A reflection across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

Describe a rigid motion that maps triangle ABC onto triangle A'B'C'.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

1063

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

A translation 2 units right and 4 units down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

Figure P is mapped onto figure Q by a 90° rotation followed by a translation.

Which statement is true?

A. P and Q are congruent.

B. Q is larger than P.

C. P and Q are not congruent because they face different directions.

D. P and Q have different angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

1064

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. P and Q are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

Explain why a triangle and its image after a dilation with a scale factor of 2 are not congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

1065

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

The dilation doubles every side length, but rigid motions (which show congruence) keep lengths the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

1065

True or false?

Two figures are congruent if a sequence of rotations, reflections, and translations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

1066

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

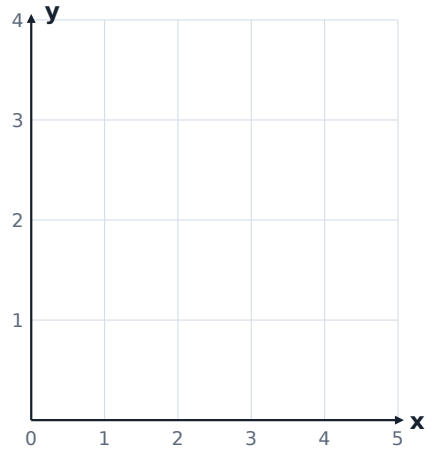
Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

SET 35 • 8.G.A.2

BACKWARD 1

Draw a polygon from its vertices • Grade 6 • Geometry • 6.G.A.3

Draw the triangle with vertices (1, 1), (4, 1), and (1, 3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with those three vertices

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

A triangle has vertices (1, 1), (4, 1), and (1, 3).

How long is the side from (1, 1) to (4, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

1069

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

A rectangle has vertices (0, 0), (5, 0), and (5, 2).

What is the fourth vertex?

A. (0, 2)

B. (2, 0)

C. (5, 5)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

True or false?

The side from (1, 1) to (1, 3) is 2 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The points (2, 1), (2, 5), and (6, 1) form a triangle with a side 3 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its sides along the grid lines are 4 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 8.G.A.2

BACKWARD 2

Use signs to locate points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (5, 2) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

The point $(-1, -4)$ is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(-1, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

Points (3, 2) and (3, −2) differ only in the sign of the y-coordinate.

How are they related?

A. They are reflections across the x-axis.

B. They are reflections across the y-axis.

C. They are the same point.

D. They are 3 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They are reflections across the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

True or false?

(−4, 1) and (4, 1) are reflections of each other across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

1077

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

True or false?

(2, −6) and (−2, 6) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 8.G.A.2

FORWARD 1

Use corresponding parts to explain triangle congruence • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7

Triangle ABC is congruent to triangle DEF, with AB = 5 and angle B = 40°.

Which side and which angle of DEF must equal these?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

1080

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

DE = 5 and angle E = 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

Two triangles have all corresponding sides and all corresponding angles congruent.

What can you conclude? Explain with rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

1081

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The triangles are congruent; a sequence of rigid motions maps one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

Triangle PQR is congruent to triangle XYZ.

Which statement must be true?

A. PQ = YZ

B. QR = YZ

C. Angle P = angle Z

D. PR = XY

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

1082

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. QR = YZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

True or false?

Corresponding parts of congruent triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

1083

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

True or false?

If triangle ABC is congruent to triangle DEF, then AC = EF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

1084

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE AC = DF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

SET 35 • 8.G.A.2

FORWARD 2

Use triangle congruence criteria • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8

Two triangles have AB = DE, angle A = angle D, and AC = DF.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

1086

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

Two triangles have angle A = angle D, AB = DE, and angle B = angle E.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

1087

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

Which is NOT a valid congruence criterion for triangles?

A. AAA

B. ASA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

1088

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. AAA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

True or false?

SSS means that three pairs of equal sides show two triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

1089

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

True or false?

Two sides and an angle that is not between them (SSA) always prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

1090

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 8.G.A.3 • GEOMETRY

Translate figures on the coordinate plane

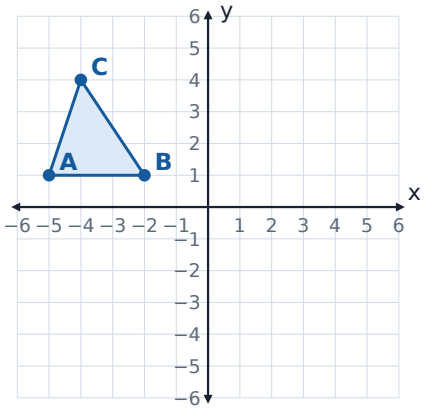
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Find and plot points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Add integers	**7.NS.A.1.b**
FORWARD 1	Describe transformations as functions and decide which preserve…	**HSG-CO.A.2**
FORWARD 2	Find the point that divides a segment in a given ratio	**HSG-GPE.B.6**

SET 36 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Translate triangle ABC 4 units right and 2 units down. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

1093

QUESTION ID

S36-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, −1), B'(2, −1), and C'(0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

The point P(3, −2) is translated 5 units left and 4 units up.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

1094

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

Which rule describes a translation 6 units down?

A. $(x, y) \to (x, y - 6)$

B. $(x, y) \to (x, y + 6)$

C. $(x, y) \to (x + 6, y)$

D. $(x, y) \to (x - 6, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

1095

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x, y) \to (x, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

A translation maps A(2, 5) to A'(−1, 7).

Where does it map B(4, 0)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

1096

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

True or false?

The rule (x, y) → (x + 3, y − 1) moves every point 3 units right and 1 unit down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

1097

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

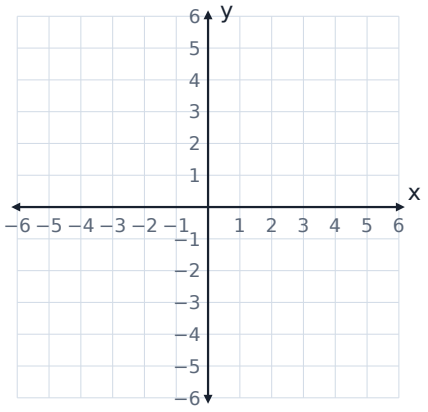
Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

SET 36 • 8.G.A.3

BACKWARD 1

Find and plot points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot the points (−3, 2) and (4, −5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S36-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

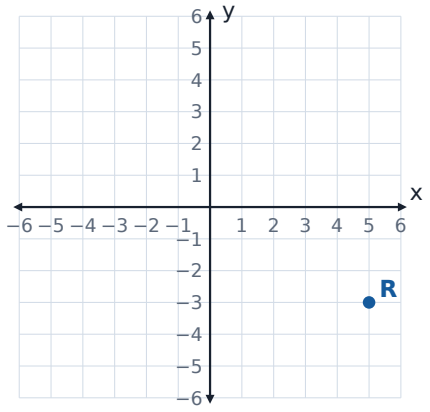
Points at (−3, 2) and (4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

What are the coordinates of point R?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

Which point is 2 units to the left of (1, 4)?

A. (1, 6)

B. (−1, 4)

C. (1, 2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

Moving 3 units up from (2, −1) gives (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

1102

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

Moving 4 units left from (0, 0) gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 8.G.A.3

BACKWARD 2

Add integers • Grade 7 • The Number System • 7.NS.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$-4 + 4$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

Add.

$3 + (-5)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

Add.

$-1 + 6$

A. 5

B. -5

C. 7

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

True or false?

$-2 + (-3) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

1108

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

True or false?

$5 + (-8) = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 8.G.A.3

FORWARD 1

Describe transformations as functions and decide which preserve distance • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2

Write the rule for a translation 2 units left and 7 units up.

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

1111

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x, y) \rightarrow (x - 2, y + 7)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

Does the rule (x, y) → (2x, y) keep distances the same? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

1112

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. It doubles horizontal distances; the segment from (0, 0) to (1, 0) becomes 2 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

Which transformation keeps all distances and angles the same?

A. $(x, y) \to (x, 2y)$

B. $(x, y) \to (x^2, y)$

C. $(x, y) \to (3x, 3y)$

D. $(x, y) \to (x + 4, y - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

1113

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x + 4, y − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

A translation is a function whose inputs and outputs are points of the plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

1114

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

True or false?

The rule (x, y) → (x + 1, y + 1) changes the length of every segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

1115

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 8.G.A.3

FORWARD 2

Find the point that divides a segment in a given ratio • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the midpoint of the segment from (2, 4) to (8, 10).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5

1117

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5

Find the point that is $\frac{1}{3}$ of the way from A(0, 0) to B(9, 6).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5

1118

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5

Point P is on segment AB from A(1, 1) to B(7, 4), with AP : PB = 2 : 1.

What are the coordinates of P?

A. (3, 2)

B. (4, 2.5)

C. (6, 3)

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5

1119

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5

True or false?

The midpoint of the segment from (−4, 2) to (6, −2) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5

1120

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5

True or false?

The point halfway from (0, 0) to (10, 4) is (10, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5

1121

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The midpoint is (5, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 8.G.A.3 • GEOMETRY

Reflect figures across the x-axis and the y-axis

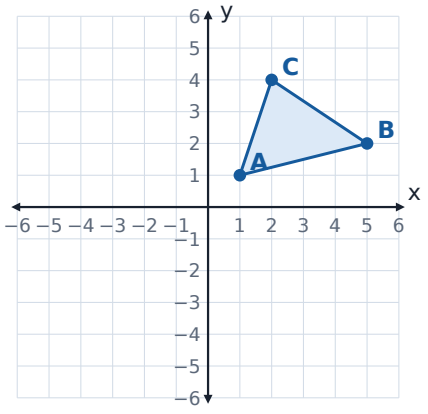
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Locate points that are reflections across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the opposite of a number	**6.NS.C.6.a**
FORWARD 1	Describe the rotations and reflections that carry a polygon onto it…	**HSG-CO.A.3**
FORWARD 2	Draw the image of a figure after a sequence of transformations	**HSG-CO.B.6**

SET 37 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Reflect triangle ABC across the x-axis. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

1124

QUESTION ID

S37-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(1, −1), B'(5, −2), and C'(2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

The point $(-3, 5)$ is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

1125

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

Which rule describes a reflection across the x-axis?

A. (x, y) → (−x, y)

B. (x, y) → (−x, −y)

C. (x, y) → (y, x)

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

1126

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

A segment has endpoints (2, 3) and (5, 1). It is reflected across the y-axis.

What are the endpoints of the image?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

1127

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 3) and (−5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

True or false?

Reflecting (4, −6) across the x-axis gives (−4, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

1128

QUESTION ID
S37-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The image is (4, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

SET 37 • 8.G.A.3

BACKWARD 1

Locate points that are reflections across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (7, −2) is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

The point (−6, −3) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

Reflecting a point across the y-axis changes...

A. the sign of the y-coordinate

B. neither sign

C. the sign of the x-coordinate

D. both signs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the sign of the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

True or false?

(0, 5) is its own reflection across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

True or false?

The reflection of (−1, 8) across the y-axis is (−1, −8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 8.G.A.3

BACKWARD 2

Find the opposite of a number • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of −7?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

Which pair are opposites?

A. −4 and 4

B. 4 and $\frac{1}{4}$

C. 0 and 1

D. −4 and −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

The opposite of 0 is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 8.G.A.3

FORWARD 1

Describe the rotations and reflections that carry a polygon onto itself • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3

How many lines of symmetry does a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

1142

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

Which rotations (between 0° and 360°) about its center carry a regular hexagon onto itself?

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

1143

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°, 120°, 180°, 240°, and 300°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

Which figure is carried onto itself by a 180° rotation but has no line of symmetry?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. An equilateral triangle

D. An isosceles trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

1144

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A parallelogram that is not a rectangle or a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A rectangle that is not a square has exactly 2 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

1145

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

An isosceles trapezoid is carried onto itself by a 90° rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

1146

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

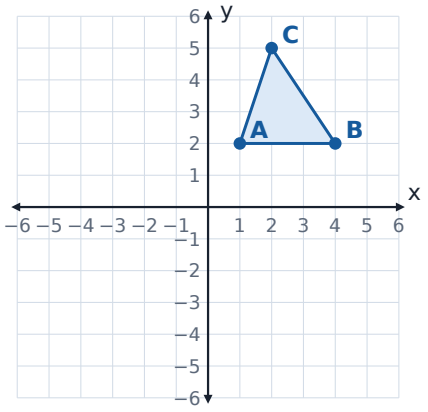
High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 8.G.A.3

FORWARD 2

Draw the image of a figure after a sequence of transformations • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Reflect triangle ABC across the x-axis, then reflect that image across the y-axis. Draw the final image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

1148

QUESTION ID

S37-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A''(−1, −2), B''(−4, −2), and C''(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

What single transformation has the same effect as a reflection across the x-axis followed by a reflection across the y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

1149

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

The point (3, 1) is reflected across the line y = x.

What is its image?

A. (3, −1)

B. (−1, −3)

C. (−3, 1)

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

1150

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

True or false?

Reflecting a figure across one line and then across a parallel line has the same effect as a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

1151

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

True or false?

Doing a translation and a reflection in the other order always gives the same image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

1152

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 8.G.A.3 • GEOMETRY

Rotate figures 90° and 180° about the origin

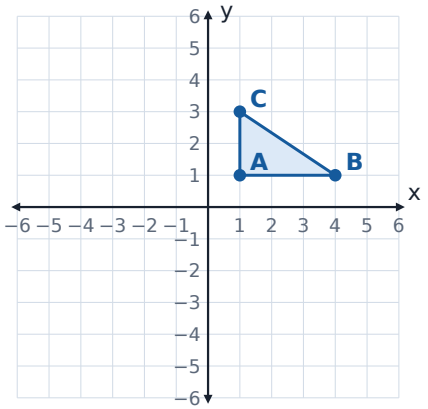
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Relate turns to angle measures	**4.MD.C.5.a**
BACKWARD 2	Locate a point reflected across both axes	**6.NS.C.6.b**
FORWARD 1	Describe a rotation in terms of distance and angle	**HSG-CO.A.4**
FORWARD 2	Use relationships among central angles, inscribed angles, and rad…	**HSG-C.A.2**

SET 38 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Rotate triangle ABC 90° counterclockwise about the origin. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

1155

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, 1), B'(−1, 4), and C'(−3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

The point (5, −2) is rotated 180° about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

1156

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

Which rule describes a 90° clockwise rotation about the origin?

A. (x, y) → (y, x)

B. (x, y) → (y, −x)

C. (x, y) → (−y, x)

D. (x, y) → (−x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

1157

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

The point (2, 3) is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

1158

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

True or false?

Rotating $(-4, 1)$ by $180°$ about the origin gives $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

1159

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

SET 38 • 8.G.A.3

BACKWARD 1

Relate turns to angle measures • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in a quarter turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

How many degrees are in a half turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

How many degrees are in a three-quarter turn?

A. 360°

B. 180°

C. 270°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

True or false?

A full turn is 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

True or false?

A half turn is 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 8.G.A.3

BACKWARD 2

Locate a point reflected across both axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, 4) is reflected across the x-axis and then across the y-axis.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

The point (−2, 5) is reflected across both axes.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

Reflecting a point across both axes changes...

A. the signs of both coordinates

B. only the y-coordinate

C. neither coordinate

D. only the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the signs of both coordinates

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

Reflecting (1, −6) across both axes gives (−1, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

1170

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

Reflecting (4, 4) across both axes gives (4, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 8.G.A.3

FORWARD 1

Describe a rotation in terms of distance and angle • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A rotation of 120° about point O maps A to A'.

Which lengths and which angle are determined by the rotation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

1173

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

OA = OA', and angle AOA' = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

P(6, 0) is rotated 60° counterclockwise about the origin.

How far is the image from the origin?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

1174

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

A rotation about point O maps every point P to a point P' so that...

A. P' is on the line of reflection

B. P' is twice as far from O as P

C. OP = OP' and angle POP' is the angle of rotation

D. PP' is parallel to one fixed line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

1175

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. OP = OP' and angle POP' is the angle of rotation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

True or false?

In a rotation by a nonzero angle less than 360°, the center is the only point that does not move.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

1176

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

A 90° rotation maps every horizontal segment to another horizontal segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

1177

QUESTION ID
S38-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It maps horizontal segments to vertical segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 8.G.A.3

FORWARD 2

Use relationships among central angles, inscribed angles, and radii • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2

A central angle of a circle measures 80°.

What does an inscribed angle that intercepts the same arc measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5

1179

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5

An inscribed angle intercepts a semicircle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5

1180

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5

A radius is drawn to the point where a tangent line touches a circle.

What angle do the radius and the tangent line form?

A. 90°

B. 45°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5

1181

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5

True or false?

Rotating a circle about its center by any angle carries the circle onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5

1182

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5

True or false?

An inscribed angle measures twice its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5

1183

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures half of its intercepted arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 8.G.A.3 • GEOMETRY

Dilate figures on the coordinate plane with the origin as the center

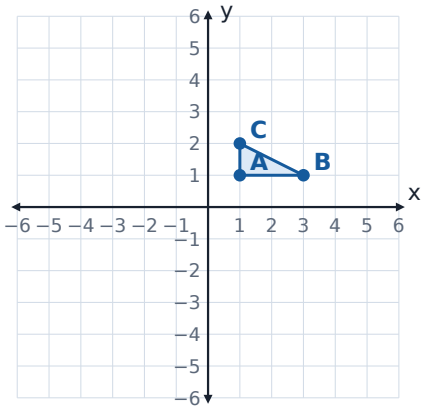
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Use a scale factor to find lengths in a scale drawing	**7.G.A.1**
BACKWARD 2	Multiply rational numbers	**7.NS.A.2.a**
FORWARD 1	Verify the properties of dilations	**HSG-SRT.A.1**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 39 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Dilate triangle ABC by a scale factor of 2 with the origin as the center. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

1186

QUESTION ID

S39-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(2, 2), B'(6, 2), and C'(2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

The point (6, −4) is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

1187

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

Which rule describes a dilation with a scale factor of 3 and the origin as the center?

A. (x, y) → (3x, y)

B. (x, y) → (x + 3, y + 3)

C. (x, y) → $(\frac{x}{3}, \frac{y}{3})$

D. (x, y) → (3x, 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

1188

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (3x, 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

A dilation with the origin as the center maps (4, 2) to (10, 5).

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

1189

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE 5/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

True or false?

A dilation with a scale factor of $\frac{1}{3}$ and the origin as the center maps (9, 6) to (3, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

1190

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

SET 39 • 8.G.A.3

BACKWARD 1

Use a scale factor to find lengths in a scale drawing • Grade 7 • Geometry • 7.G.A.1

A 4 cm segment is enlarged by a scale factor of 3.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

A 10 cm segment is reduced by a scale factor of $\frac{1}{2}$.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

A rectangle 2 cm by 5 cm is enlarged by a scale factor of 4.

What are the new dimensions?

A. 8 cm by 20 cm

B. 8 cm by 5 cm

C. 6 cm by 9 cm

D. 2 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A scale factor greater than 1 makes a figure larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A scale factor of $\frac{1}{4}$ makes each length 4 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 8.G.A.3

BACKWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$2 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

Multiply.

$\frac{1}{2} \times (-6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

Multiply.

3.5×4

A. 12

B. 0.875

C. 14

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

True or false?

$3 \times (-2) = -6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 9 = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 8.G.A.3

FORWARD 1

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with the origin as the center and a scale factor of 3 is applied to the line y = x + 1.

What is the equation of the image?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

1204

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 3 (a line parallel to the original)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

A segment 4 units long is dilated by a scale factor of 2.5.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

1205

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

A dilation is applied to a line that passes through the center of dilation.

What is the image?

A. A parallel line

B. A single point

C. A perpendicular line

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

1206

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

1207

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

A dilation with a scale factor of 2 doubles every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

1208

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Dilations keep angle measures the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 8.G.A.3

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-foot person casts a 4-foot shadow. At the same time, a tree casts a 22-foot shadow.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

1210

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

Triangle ABC is similar to triangle DEF, with AB = 8, DE = 12, and BC = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

1211

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

A segment parallel to one side of a triangle splits a second side into parts of 3 and 6. It splits the third side into a first part of 4 and a second part.

How long is the second part?

A. 2

B. 12

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

1212

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

True or false?

In similar triangles, the ratio of the perimeters equals the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

1213

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

True or false?

Triangles with sides 3, 4, 5 and 6, 8, 12 are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

1214

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 8.G.A.4 • GEOMETRY

Describe a sequence of transformations that shows two figures are similar

MAIN	Grade 8 main question	**8.G.A.4**
BACKWARD 1	Find a scale factor between scale copies	**7.G.A.1**
BACKWARD 2	Decide whether two ratios are equivalent	**7.RP.A.2.a**
FORWARD 1	Use similarity transformations to decide whether triangles are sim…	**HSG-SRT.A.2**
FORWARD 2	Use a translation and a dilation to show that any two circles are s…	**HSG-C.A.1**

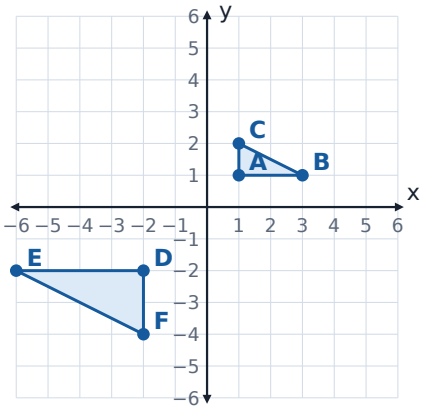
SET 40 • 8.G.A.4

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Describe a sequence of transformations that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

1217

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 and the origin as the center, followed by a 180° rotation about the origin.

GRADING NOTE Any correct sequence is accepted, for example the rotation first, then the dilation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

Explain why a 2-by-3 rectangle and a 4-by-6 rectangle are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

1218

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 (followed by rigid motions to move it into place) maps the first rectangle onto the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

Which sequence of transformations maps a figure onto a figure that is similar but not congruent?

A. A rotation, then a translation

B. A translation only

C. Two reflections

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

1219

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

Are a 3-4-5 triangle and a 5-12-13 triangle similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

1220

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The ratios of corresponding sides (3/5, 4/12, 5/13) are not equal, so no dilation and rigid motions can map one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

True or false?

Two figures are similar if a sequence of rotations, reflections, translations, and dilations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

1221

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

SET 40 • 8.G.A.4

BACKWARD 1

Find a scale factor between scale copies • Grade 7 • Geometry • 7.G.A.1

A 3 cm side becomes 12 cm in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

A 15 m side becomes 5 m in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

A triangle's sides are 2, 3, and 4. Which side lengths are a scale copy?

A. 6, 9, 12

B. 4, 5, 6

C. 2, 6, 8

D. 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6, 9, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A scale copy with a scale factor of 1 is the same size as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A rectangle 2 by 3 and a rectangle 3 by 4 are scale copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 8.G.A.4

BACKWARD 2

Decide whether two ratios are equivalent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Are 4 : 6 and 10 : 15 equivalent ratios?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; both equal 2 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

Are 3 : 5 and 9 : 12 equivalent ratios?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No; 3/5 = 0.6 and 9/12 = 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 5 : 8?

A. 25 : 48

B. 15 : 24

C. 8 : 5

D. 10 : 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15 : 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

True or false?

4 : 5 and 5 : 6 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 8.G.A.4

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 5, 7, and 9. Triangle DEF has sides 10, 14, and 18.

Are they similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

1235

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; the scale factor is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

Two triangles are similar. Explain what is true about their corresponding angles and sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

1236

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Corresponding angles are equal, and corresponding sides are proportional.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

Triangle PQR is similar to triangle STU with a scale factor of 3. PQ = 4.

What is ST?

A. 12

B. 4

C. $\frac{4}{3}$

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

1237

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

True or false?

If a dilation followed by a rotation maps one triangle onto another, the triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

1238

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

True or false?

Any two rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

1239

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 8.G.A.4

FORWARD 2

Use a translation and a dilation to show that any two circles are similar • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1

Circle A has center (0, 0) and radius 2. Circle B has center (5, 1) and radius 6.

Describe a translation followed by a dilation that maps circle A onto circle B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

1241

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Translate 5 units right and 1 unit up, then dilate about (5, 1) with a scale factor of 3.

GRADING NOTE Any correct sequence is accepted, for example a dilation about (0, 0) with scale factor 3 followed by a translation 5 units right and 1 unit up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

What scale factor of a dilation about the center maps a circle with a radius of 4 onto a circle with the same center and a radius of 10?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

1242

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE $\frac{5}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

Which statement explains why any two circles are similar?

A. A rotation alone maps any circle onto any other circle.

B. All circles have the same radius.

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

D. All circles have the same area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

1243

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

True or false?

Any two circles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

1244

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

A circle with a radius of 3 can be mapped onto a circle with a radius of 6 by a translation alone.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

1245

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A translation keeps the radius 3; a dilation is also needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 8.G.A.5 • GEOMETRY

Use the angle sum and exterior angles of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for supplementary and complementary…	**7.G.B.5**
BACKWARD 2	Find an unknown angle in a sum of angles	**4.MD.C.7**
FORWARD 1	Prove theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Define trigonometric ratios from similar right triangles	**HSG-SRT.C.6**

SET 41 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle measure 52° and 71°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

1248

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

The angles of a triangle measure x°, (x + 10)°, and (2x − 30)°.

Write and solve an equation to find x.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

1249

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation x + (x + 10) + (2x − 30) = 180 and the answer x = 50

GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

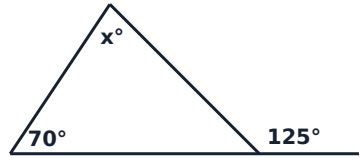
In the figure, what is the value of x?

A. 55

B. 70

C. 125

D. 35



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

1250

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

1251

QUESTION ID

S41-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The exterior angle = the sum of the two remote interior angles

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: the exterior angle and the adjacent interior angle form a linear pair, so they add to 180°; the three interior angles also add to 180°; so the exterior angle equals the sum of the other two interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

1252

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two obtuse angles alone add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

SET 41 • 8.G.A.5

BACKWARD 1

Write and solve equations for supplementary and complementary angles • Grade 7 • Geometry • 7.G.B.5

Two supplementary angles measure x° and 130°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

Two complementary angles measure 2x° and 40°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

Angles of x° and (x + 30)° are supplementary.

What is x?

A. 75

B. 150

C. 105

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Angles of 47° and 43° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

True or false?

Angles of 100° and 90° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 8.G.A.5

BACKWARD 2

Find an unknown angle in a sum of angles • Grade 4 • Measurement & Data • 4.MD.C.7

A 90° angle is split into two angles. One is 35°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

A straight angle (180°) is split into two angles. One is 110°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

An angle of 120° is split into angles of 75° and x°.

What is x?

A. 195

B. 55

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

Angles of 30° and 60° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

Angles of 80° and 80° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 8.G.A.5

FORWARD 1

Prove theorems about triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the angles of triangle ABC add to 180°. Use a line through C that is parallel to AB. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

1266

QUESTION ID

S41-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B + angle C = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: draw line l through C parallel to AB. The angles at C along l add to 180° (straight angle). The two outer ones equal angles A and B (alternate interior angles, l ∥ AB). So angle A + angle B + angle C = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

The angles of a triangle are 2x°, 3x°, and 4x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

1267

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

An isosceles triangle has base angles of 40°.

What does its third angle measure?

A. 100°

B. 40°

C. 80°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

1268

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

True or false?

The base angles of an isosceles triangle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

1269

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

1270

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 8.G.A.5

FORWARD 2

Define trigonometric ratios from similar right triangles • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is sin A?

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5

1272

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is tan A?

tan A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5

1273

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is cos A?

A. $\frac{3}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5

1274

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5

True or false?

In every right triangle with a 30° angle, the side opposite the 30° angle is half the hypotenuse, so $\sin 30° = \frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5

1275

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5

True or false?

sin A changes when a right triangle is enlarged but its angles stay the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5

1276

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Similar right triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 8.G.A.5 • GEOMETRY

Write and solve equations for angles formed by parallel lines cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for vertical and adjacent angles	**7.G.B.5**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Prove that alternate interior angles are congruent	**HSG-CO.C.9**
FORWARD 2	Use slopes to show that lines are parallel or perpendicular	**HSG-GPE.B.5**

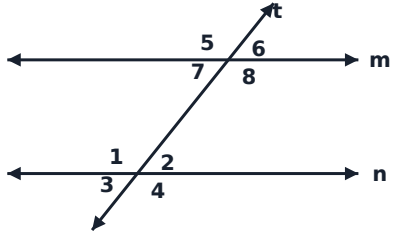
SET 42 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel and angle 2 = 52°.

What does angle 6 measure? Give the reason.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

1279

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

52°; corresponding angles are congruent.

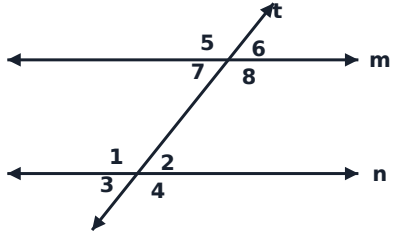
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Angle 1 = (3x + 10)° and angle 8 = 115°.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

1280

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 3x + 10 = 115 (alternate interior angles are congruent) and the answer x = 35

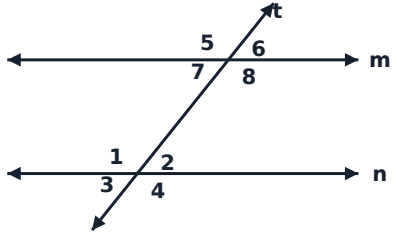
GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

In the figure, lines m and n are parallel. Angle 2 = (2x + 4)° and angle 8 = (4x − 10)°.

What is x?



A. 93 **B.** 31

C. 43 **D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

1281

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

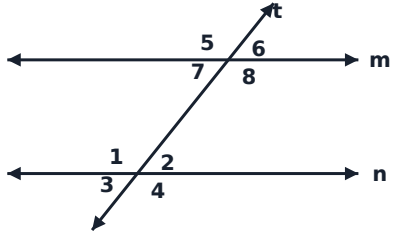
B. 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

In the figure, lines m and n are parallel and angle 4 = 128°.

What does angle 5 measure? Give the reasons.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

1282

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

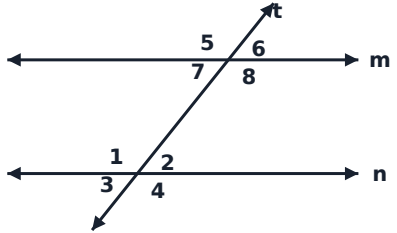
128°: angle 1 = angle 4 (vertical angles), and angle 5 = angle 1 (corresponding angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

True or false?

In the figure, angles 3 and 6 are alternate exterior angles, so they are congruent when m and n are parallel.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

1283

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

SET 42 • 8.G.A.5

BACKWARD 1

Write and solve equations for vertical and adjacent angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. Vertical angles measure (2x + 20)° and 80°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

1285

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

Two lines intersect. Adjacent angles measure x° and 3x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

Two lines intersect. Adjacent angles measure (x + 40)° and 2x°.

What is x?

A. $\frac{140}{3}$

B. 50

C. 40

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 140/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

True or false?

Vertical angles of 5x° and 75° give x = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

True or false?

Adjacent angles that form a straight line measure (x + 10)° and x°. Then x = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

1289

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 85.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 8.G.A.5

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 10 = 115

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

Solve for x.

$2x - 15 = 75$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$6x - 6 = 180$

A. x = 174

B. x = 29

C. x = 31

D. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

True or false?

x = 20 is the solution of 4x + 20 = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 25$ is the solution of $5x - 5 = 115$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

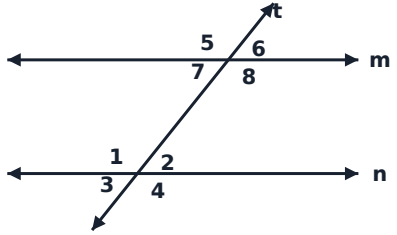
SET 42 • 8.G.A.5

FORWARD 1

Prove that alternate interior angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 1 = angle 8. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

1297

QUESTION ID

S42-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 1 = angle 8

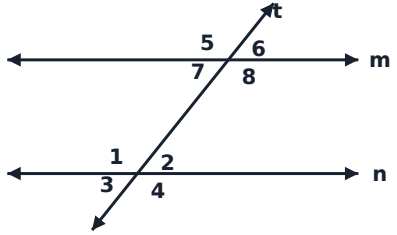
GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 1 = angle 5 (corresponding angles, m ∥ n); angle 5 = angle 8 (vertical angles); so angle 1 = angle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

1298

QUESTION ID

S42-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 + angle 8 = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 6 + angle 8 = 180° (linear pair); angle 2 = angle 6 (corresponding angles); so angle 2 + angle 8 = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

A proof says: angle 1 = angle 5 (corresponding angles). Angle 5 = angle 8 (?). So angle 1 = angle 8.

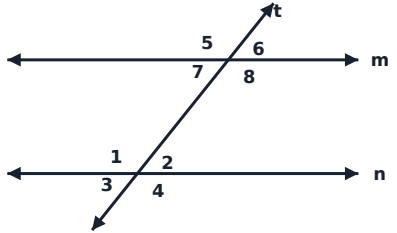
What is the missing reason?

A. Linear pairs are supplementary.

B. Vertical angles are congruent.

C. Alternate interior angles are congruent.

D. The triangle angle sum is 180°.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

1299

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

If corresponding angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

1300

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

1300

True or false?

"The angles look equal in the figure" is a valid reason in a proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

1301

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 8.G.A.5

FORWARD 2

Use slopes to show that lines are parallel or perpendicular • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A transversal crosses the lines $y = 2x + 1$ and $y = 2x - 3$.

Are the two lines parallel? Explain with slopes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

1303

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; both lines have slope 2 and different y-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

Write an equation of the line through (1, 4) that is perpendicular to $y = -\frac{1}{2}x + 3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

1304

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

Which pair of lines is perpendicular?

A. $y = x$ and $y = x + 2$

B. $y = 3x$ and $y = -3x$

C. $y = 3x + 1$ and $y = 3x - 1$

D. $y = 3x + 1$ and $y = -\frac{1}{3}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

1305

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 3x + 1$ and $y = -(1/3)x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

Two non-vertical lines are parallel exactly when their slopes are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

1306

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

The lines $y = \frac{2}{5}x$ and $y = \frac{5}{2}x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

1307

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 8.G.A.5 • GEOMETRY

Use the angle-angle criterion to show that triangles are similar

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Find a missing length in a scale copy	**7.G.A.1**
BACKWARD 2	Use the triangle angle sum to find a missing angle	**7.G.B.5**
FORWARD 1	Establish the angle-angle criterion with similarity transformations	**HSG-SRT.A.3**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 43 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 35° and 85°. Triangle DEF has angles of 35° and 60°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

1310

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The third angle of DEF is 180° − 35° − 60° = 85°, so two pairs of angles are equal (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

A 5-foot post casts a 3-foot shadow. At the same time, a tree casts a 12-foot shadow.

Explain why the two triangles are similar, and find the height of the tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

1311

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

Both triangles have a right angle and the same angle of the sun's rays, so they are similar by AA. The tree is 20 feet tall.

GRADING NOTE Grade both: the AA reason and 20 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

Which information shows that two triangles are similar by the angle-angle criterion?

A. One pair of sides is proportional.

B. All the sides are equal.

C. Two pairs of corresponding angles are equal.

D. One pair of corresponding angles is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

1312

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Two pairs of corresponding angles are equal.

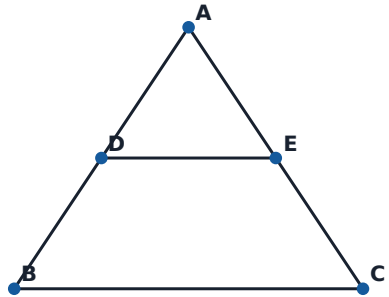
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

In the figure, segment DE is parallel to BC.

Explain why triangle ADE is similar to triangle ABC.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

1313

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

The triangles share angle A, and angle ADE = angle ABC (corresponding angles, because DE ∥ BC). So they are similar by AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

True or false?

Two triangles that each have a 90° angle and a 40° angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

1314

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

SET 43 • 8.G.A.5

BACKWARD 1

Find a missing length in a scale copy • Grade 7 • Geometry • 7.G.A.1

A triangle has sides 4, 6, and 8. In a scale copy, the side of 4 becomes 10.

What does the side of 6 become?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A drawing with a scale factor of 3 shows a 2 cm side.

How long is that side in the scale copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A 5-by-8 rectangle is scaled so the 5 becomes 15.

What does the 8 become?

A. 40

B. 24

C. 13

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

In a scale copy, every length is multiplied by the same scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 2, every angle is doubled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 8.G.A.5

BACKWARD 2

Use the triangle angle sum to find a missing angle • Grade 7 • Geometry • 7.G.B.5

A triangle has angles of 35° and 60°.

What is the third angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q1/5

A right triangle has a 40° angle.

What is the third angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q2/5

A triangle has angles of 90° and 25°.

What is the third angle?

A. 75°

B. 65°

C. 115°

D. 55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q3/5

True or false?

A triangle with angles of 50° and 60° has a third angle of 70°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q4/5

True or false?

A triangle with angles of 45° and 45° has a third angle of 45°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 8.G.A.5

FORWARD 1

Establish the angle-angle criterion with similarity transformations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Which transformations show that the triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

1328

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation that makes AB the same length as DE, followed by rigid motions that move the image onto DEF; the equal angles make the third vertices match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

Why is it enough to know two pairs of equal angles to show triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

1329

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The third angles must also be equal, because each triangle's angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

Triangles with angles 30°, 70°, 80° and 70°, 30°, 80° are...

A. congruent in every case

B. similar

C. not similar

D. impossible to compare

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

1330

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

True or false?

Two equilateral triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

1331

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

True or false?

Two isosceles triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

1332

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 8.G.A.5

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 1.5 m stick casts a 2 m shadow. A building casts a 36 m shadow at the same time.

How tall is the building?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1334

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

27 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

Triangle PQR is similar to triangle XYZ with PQ = 6, XY = 9, and QR = 8.

Find YZ.

YZ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1335

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

Two similar triangles have a scale factor of 3. The smaller one has a perimeter of 14.

What is the larger one's perimeter?

A. 17

B. 14

C. 42

D. 126

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1336

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

True or false?

A mirror can be used with similar triangles to find the height of a tall object.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1337

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

True or false?

If two triangles are similar with a scale factor of 2, their areas are also in the ratio 2 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1338

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Their areas are in the ratio 4 : 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 8.G.B.6 • GEOMETRY

Explain a proof of the Pythagorean theorem and its converse

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the area of a right triangle	**6.G.A.1**
BACKWARD 2	Find the area of a square	**3.MD.C.7.b**
FORWARD 1	Prove the Pythagorean theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Derive the equation of a circle from the Pythagorean theorem	**HSG-GPE.A.1**

SET 44 • 8.G.B.6

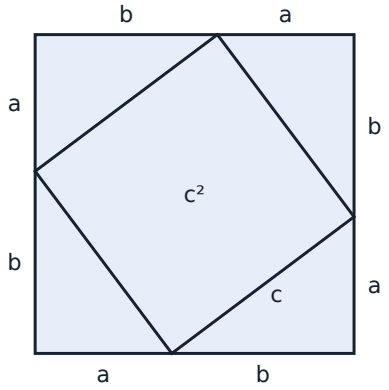
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

The figure shows a square with sides a + b, made of four right triangles and a tilted square with side c.

Explain how the figure proves that $a^2 + b^2 = c^2$.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

1341

QUESTION ID

S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

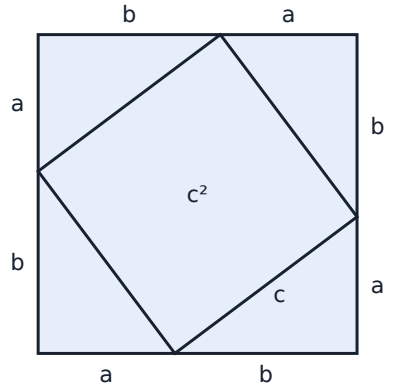
$a^2 + b^2 = c^2$

GRADING NOTE REQUIRED WORK: A complete argument: the large square's area is $(a + b)^2 = a^2 + 2ab + b^2$; it is also 4 triangles ($4 \cdot \frac{1}{2}ab = 2ab$) plus the inner square c^2; so $a^2 + 2ab + b^2 = 2ab + c^2$, and subtracting 2ab gives $a^2 + b^2 = c^2$. Any equivalent wording is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

In the figure, write the area of the large square as the sum of the areas of the four triangles and the inner square.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

1342

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 ·

$$\frac{1}{2}ab + c^2 = 2ab + c^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

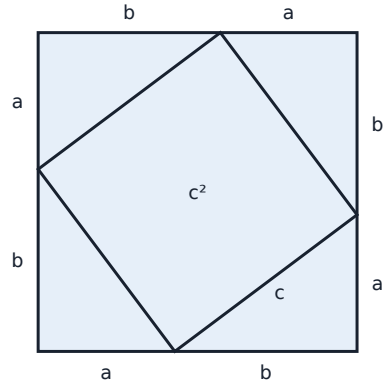
In the proof with the figure, which step comes after writing $(a + b)^2 = 2ab + c^2$?

A. Divide both sides by c

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

C. Add 2ab to both sides

D. Take the square root of each term



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

1343

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

A triangle has sides a, b, and c with $a^2 + b^2 = c^2$. Explain why it must be a right triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

1344

QUESTION ID

S44-M-Q4

TYPE

Short answer

CORRECT ANSWER

A right triangle with legs a and b has a hypotenuse of $\sqrt{(a^2 + b^2)} = c$. So the two triangles have the same three sides and are congruent (SSS), and the given triangle has a right angle too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

1344

True or false?

The Pythagorean theorem $a^2 + b^2 = c^2$ is true for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

1345

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true only for right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

SET 44 • 8.G.B.6

BACKWARD 1

Find the area of a right triangle • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs of 3 and 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

A right triangle has legs of 5 and 12.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

30 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

What is the total area of four right triangles that each have legs of 2 and 6?

A. 24

B. 12

C. 48

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

A right triangle with legs of 8 and 3 has an area of 12 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

A right triangle with legs of 6 and 6 has an area of 36 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 8.G.B.6

BACKWARD 2

Find the area of a square • Grade 3 • Measurement & Data • 3.MD.C.7.b

A square has sides of 7 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

49 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

A square has sides of 9 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

81 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

A square has sides of 5 units.

What is its area?

A. 15 square units

B. 25 square units

C. 10 square units

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

True or false?

A square with 4-unit sides has an area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

A square with 6-unit sides has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 8.G.B.6

FORWARD 1

Prove the Pythagorean theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In right triangle ABC with the right angle at C, altitude CD is drawn to hypotenuse AB.

Why is $AC^2 = AD \cdot AB$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

1359

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Triangle ACD is similar to triangle ABC (AA), so $AC/AB = AD/AC$, which gives $AC^2 = AD \cdot AB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

In the same triangle, $BC^2 = BD \cdot AB$. Add this to $AC^2 = AD \cdot AB$. What do you get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

1360

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$AC^2 + BC^2 = AD \cdot AB + BD \cdot AB = AB(AD + BD) = AB \cdot AB = AB^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

Which criterion shows that triangle ACD is similar to triangle ABC?

A. ASA

B. AA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

1361

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

The altitude to the hypotenuse divides a right triangle into two triangles similar to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

1362

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

The altitude to the hypotenuse always cuts the hypotenuse into two equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

1363

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 8.G.B.6

FORWARD 2

Derive the equation of a circle from the Pythagorean theorem • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

1365

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

Write the equation of the circle with center (2, −1) and radius 3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

1366

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 + (y + 1)^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

Which point is on the circle $x^2 + y^2 = 100$?

A. (5, 5)

B. (10, 10)

C. (6, 8)

D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

1367

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

True or false?

The equation $(x - h)^2 + (y - k)^2 = r^2$ comes from the Pythagorean theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

1368

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

True or false?

The circle $x^2 + y^2 = 16$ has a radius of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

1369

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its radius is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 8.G.B.6 • GEOMETRY

Use the converse of the Pythagorean theorem to decide whether a triangle is a right triangle

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the squares of whole numbers	**6.EE.A.1**
BACKWARD 2	Evaluate an expression with exponents	**6.EE.A.2.c**
FORWARD 1	Classify a triangle as acute, right, or obtuse from its side lengths	**HSG-CO.C.10**
FORWARD 2	Use coordinates to show that a triangle is a right triangle	**HSG-GPE.B.4**

SET 45 • 8.G.B.6

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides 9, 12, and 15 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

1372

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

Is a triangle with sides 7, 8, and 11 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

1373

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. $7^2 + 8^2 = 49 + 64 = 113$, but $11^2 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

Which side lengths form a right triangle?

A. 6, 8, 12

B. 20, 21, 29

C. 5, 7, 9

D. 10, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

1374

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20, 21, 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

A carpenter checks a corner. Points 3 feet and 4 feet from the corner along the two edges are 5 feet apart.

Is the corner a right angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

1375

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. $3^2 + 4^2 = 5^2$, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

True or false?

A triangle with sides 2, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

1376

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 + 9 = 13$, but $4^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

SET 45 • 8.G.B.6

BACKWARD 1

Find the squares of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

Find the value.

11^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

13^2

A. 26

B. 169

C. 139

D. 196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

$20^2 = 400$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

$9^2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

1382

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 8.G.B.6

BACKWARD 2

Evaluate an expression with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $a^2 + b^2$ when a = 6 and b = 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

Evaluate $a^2 + b^2$ when a = 5 and b = 12.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

Evaluate $c^2 - a^2$ when $c = 10$ and $a = 6$.

A. 16

B. 136

C. 64

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

True or false?

When a = 3 and b = 4, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

True or false?

When a = 2 and b = 3, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 8.G.B.6

FORWARD 1

Classify a triangle as acute, right, or obtuse from its side lengths • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

Is a triangle with sides 5, 6, and 8 acute, right, or obtuse? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

1390

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Obtuse; $5^2 + 6^2 = 61$, which is less than $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

Is a triangle with sides 6, 7, and 8 acute, right, or obtuse? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

1391

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Acute; $6^2 + 7^2 = 85$, which is greater than $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

A triangle has sides 8, 15, and 17.

What kind of triangle is it?

A. Obtuse

B. Acute

C. Right

D. Not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

1392

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

True or false?

If $a^2 + b^2 < c^2$ for the longest side c, the triangle is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

1393

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

True or false?

A triangle with sides 4, 4, and 7 is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

1394

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^2 + 4^2 = 32 < 49$, so it is obtuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 8.G.B.6

FORWARD 2

Use coordinates to show that a triangle is a right triangle • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Is the triangle with vertices (1, 1), (4, 1), and (4, 5) a right triangle? Use side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

1396

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The sides are 3, 4, and 5, and $3^2 + 4^2 = 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

Use slopes to decide whether the triangle with vertices (0, 0), (2, 4), and (6, 2) has a right angle at (2, 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

1397

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The slope from (2, 4) to (0, 0) is 2 and the slope from (2, 4) to (6, 2) is −1/2; their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

Which triangle is a right triangle?

A. (0, 0), (2, 2), (4, 1)

B. (0, 0), (3, 1), (1, 3)

C. (1, 1), (2, 3), (3, 1)

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

1398

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

True or false?

The triangle with vertices (0, 0), (5, 0), and (0, 5) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

1399

QUESTION ID

S45-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

True or false?

The triangle with vertices (0, 0), (4, 0), and (2, 1) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in two dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Find the squares of decimals and whole numbers	**6.EE.A.1**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve right triangles with the Pythagorean theorem and trigonom...	**HSG-SRT.C.8**
FORWARD 2	Apply geometric methods to solve design problems	**HSG-MG.A.3**

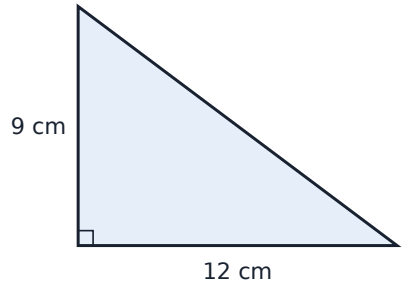
SET 46 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

Find the length of the hypotenuse.

Hypotenuse: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

1403

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

A 13-foot ladder leans against a wall. Its base is 5 feet from the wall.

How high up the wall does it reach?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

1404

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

A rectangular screen is 48 inches wide and 36 inches tall.

How long is its diagonal?

A. 60 inches

B. 84 inches

C. 72 inches

D. 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

1405

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

A right triangle has a hypotenuse of 17 and a leg of 8.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

1406

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

True or false?

A square with 10 cm sides has a diagonal of about 14.1 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

1407

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

SET 46 • 8.G.B.7

BACKWARD 1

Find the squares of decimals and whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

14^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

Find the value.

2.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

17^2

A. 189

B. 279

C. 289

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 289

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1.5^2 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 8.G.B.7

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 64 = 289

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

Solve for x.

25 + x = 169

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

Solve for x.

81 + x = 100

A. x = 19

B. x = 181

C. x = 9

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

True or false?

x = 15 is the solution of x + 49 = 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

x = 46 is the solution of 64 + x = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is x = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 8.G.B.7

FORWARD 1

Solve right triangles with the Pythagorean theorem and trigonometric ratios • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has a hypotenuse of 10 and a 30° angle. sin 30° = 0.5.

How long is the side opposite the 30° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

1421

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

A ramp rises 3 feet over a horizontal distance of 4 feet.

How long is the ramp?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

1422

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

A kite string is 50 m long and makes a right triangle with the ground. The kite is 30 m above the ground.

How far is the kite horizontally from the person?

A. 58.3 m

B. 40 m

C. 20 m

D. 80 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

1423

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

True or false?

A right triangle with two 45° angles and legs of 7 has a hypotenuse of $7\sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

1424

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

True or false?

tan 45° = 2

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

1425

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE tan 45° = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 8.G.B.7

FORWARD 2

Apply geometric methods to solve design problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular field is 120 m by 50 m.

How much shorter is walking along the diagonal than walking along two sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

1427

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40 m (170 m compared with 130 m)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

A TV is sold as a 50-inch TV (its diagonal). It is 40 inches wide.

How tall is it?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

1428

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

30 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

A cable runs from the top of a 12 m pole to a stake 5 m from the base of the pole.

How long is the cable?

A. 13 m

B. 17 m

C. √119 m

D. 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

1429

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

True or false?

A 10-foot ladder whose base is 6 feet from a wall reaches 8 feet up the wall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

1430

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

True or false?

The diagonal of a 9-by-12 rectangle is 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

1431

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diagonal is 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in three dimensions

MAIN	Grade 8 main question	8.G.B.7
BACKWARD 1	Identify the faces, edges, and vertices of a prism	6.G.A.4
BACKWARD 2	Describe cross sections of right prisms	7.G.A.3
FORWARD 1	Use the Pythagorean theorem to solve problems in three dimensio	HSG-SRT.C.8
FORWARD 2	Identify solids made by rotating two-dimensional figures	HSG-GMD.B.4

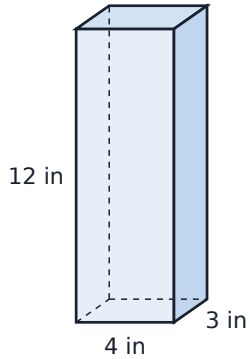
SET 47 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

A box is 4 inches long, 3 inches wide, and 12 inches tall.

How long is the diagonal from a bottom corner to the opposite top corner?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

1434

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

13 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

A cube has 6 cm edges.

Find the diagonal of one face, and then the diagonal from one corner of the cube to the opposite corner. Give exact values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

1435

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

Face diagonal 6√2 cm; cube diagonal 6√3 cm

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

Will a 26-inch stick fit inside a box that is 24 by 6 by 8 inches, from one corner to the opposite corner?

A. No; the box is too short.

B. Yes; the face diagonal is 30 inches.

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

D. No; the diagonal is only 24 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

1436

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

A right square pyramid has its top directly above the center of its base. Its base edges are 10 m and its height is 12 m.

What is the slant height (from the top to the midpoint of a base edge)?

Slant height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

1437

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

True or false?

The corner-to-corner diagonal of a 2 by 3 by 6 box is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

1438

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

SET 47 • 8.G.B.7

BACKWARD 1

Identify the faces, edges, and vertices of a prism • Grade 6 • Geometry • 6.G.A.4

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

How many edges does a rectangular prism have?

Edges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

How many vertices does a cube have?

A. 6

B. 4

C. 12

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

Each face of a rectangular prism is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A triangular prism has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 5 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 8.G.B.7

BACKWARD 2

Describe cross sections of right prisms • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane through two opposite vertical edges.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

A cube is sliced by a plane parallel to one face, through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

A right rectangular prism is sliced parallel to its base, through its interior.

What is the cross section?

A. A circle

B. A triangle

C. A pentagon

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

True or false?

A diagonal slice through a box can make a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A slice of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 8.G.B.7

FORWARD 1

Use the Pythagorean theorem to solve problems in three dimensions • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A room is 12 ft long, 9 ft wide, and 8 ft high.

How long is the diagonal of the floor?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

1452

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

A room is 12 ft long, 9 ft wide, and 8 ft high.

How long is the diagonal from a floor corner to the opposite ceiling corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

1453

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

17 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

A box is 1 by 2 by 2.

How long is its corner-to-corner diagonal?

A. 3

B. 9

C. √5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

1454

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

True or false?

The corner-to-corner diagonal of a cube with edge s is s√3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

1455

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

True or false?

A box 3 by 4 by 5 has a corner-to-corner diagonal of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

1456

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $\sqrt{50} = 5\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 8.G.B.7

FORWARD 2

Identify solids made by rotating two-dimensional figures • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A right triangle with legs 3 and 4 is rotated around the leg of length 4.

What solid does it make, and what is its slant height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5

1458

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A cone with radius 3 and height 4; its slant height is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5

A 2-by-5 rectangle is rotated around one of its 5-unit sides.

What solid does it make?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5

1459

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A cylinder with radius 2 and height 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5

A sphere is sliced by a plane through its center.

What is the cross section?

A. A circle with the same radius as the sphere

B. An oval that is not a circle

C. A square

D. A smaller circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5

1460

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A circle with the same radius as the sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5

True or false?

Rotating a semicircle about its diameter makes a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5

1461

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5

True or false?

A cube can never have a triangular cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5

1462

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A slice through three corners next to one vertex is a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 8.G.B.8 • GEOMETRY

Find the distance between two points on the coordinate plane

MAIN	Grade 8 main question	**8.G.B.8**
BACKWARD 1	Find distances along grid lines	**6.NS.C.8**
BACKWARD 2	Find a distance as the absolute value of a difference	**7.NS.A.1.c**
FORWARD 1	Use the Pythagorean theorem with coordinates to solve problems	**HSG-SRT.C.8**
FORWARD 2	Use coordinates to find perimeters and areas of polygons	**HSG-GPE.B.7**

SET 48 • 8.G.B.8

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (7, 10).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

1465

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

Find the distance between (−3, 4) and (2, −8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

1466

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

What is the distance between (0, 0) and (4, 4), to the nearest tenth?

A. 16.0 units

B. 4.0 units

C. 5.7 units

D. 8.0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

1467

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5.7 units

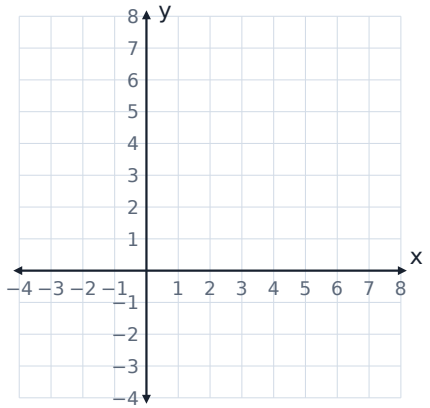
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Plot A(−2, −1) and B(4, 7). Draw a right triangle with AB as its hypotenuse, and find AB.

AB = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

1468

QUESTION ID

S48-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points A and B, joined by the hypotenuse, with a horizontal leg of 6 units and a vertical leg of 8 units

Written: 10 units

GRADING NOTE Grade both: the right triangle with legs 6 and 8, and AB = 10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

True or false?

The distance between (−1, −1) and (2, 3) is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

1469

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

SET 48 • 8.G.B.8

BACKWARD 1

Find distances along grid lines • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (−4, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

Find the distance between (3, −6) and (3, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

Which two points are 8 units apart?

A. (1, 1) and (1, 8)

B. (−3, 0) and (3, 0)

C. (−3, 0) and (5, 0)

D. (0, −4) and (0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, 0) and (5, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

True or false?

(2, −5) and (2, 5) are 10 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

True or false?

(−6, 3) and (−1, 3) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 8.G.B.8

BACKWARD 2

Find a distance as the absolute value of a difference • Grade 7 • The Number System • 7.NS.A.1.c

Find $|7 - (-2)|$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

How far apart are −4 and 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

Which expression gives the distance between −8 and −3?

A. |−8 − (−3)|

B. −8 − 3

C. |−8 + 3| − 1

D. −8 + (−3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. |−8 − (−3)|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

True or false?

The distance between −5 and 1 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

True or false?

The distance between −7 and −2 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 8.G.B.8

FORWARD 1

Use the Pythagorean theorem with coordinates to solve problems • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

On a map grid, a park is at (2, 3) and a school is at (10, 9). Each unit is 1 km.

How far apart are they in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

1483

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

A drone flies from (0, 0) to (12, 5) on a grid in meters.

How far does it fly in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

1484

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

Which point is exactly 5 units from (1, 1)?

A. (6, 1)

B. (5, 5)

C. (1, 7)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

1485

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

True or false?

Every point that is 5 units from the origin lies on a circle with radius 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

1486

QUESTION ID

S48-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

True or false?

The points (0, 0) and (3, 3) are 6 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

1487

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 8.G.B.8

FORWARD 2

Use coordinates to find perimeters and areas of polygons • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

1489

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

Find the area of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

1490

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

A rhombus has vertices (0, 0), (3, 4), (8, 4), and (5, 0).

What is its perimeter?

A. 20 units

B. 18 units

C. 16 units

D. 25 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

1491

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

True or false?

A square with vertices (0, 0), (3, 1), (2, 4), and (−1, 3) has an area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

1492

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

True or false?

The rectangle with vertices (0, 0), (4, 0), (4, 3), and (0, 3) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

1493

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its perimeter is 14 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 8.G.C.9 • GEOMETRY

Find the volume of a cylinder

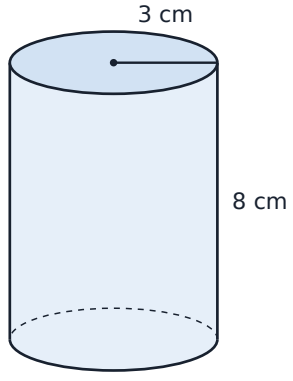
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle	**7.G.B.4**
BACKWARD 2	Find the volume of a right prism as base area times height	**7.G.B.6**
FORWARD 1	Give an informal argument for volume formulas	**HSG-GMD.A.1**
FORWARD 2	Use density with volume in a model	**HSG-MG.A.2**

SET 49 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cylinder in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

1496

QUESTION ID

S49-M-Q1

TYPE

Short answer

CORRECT ANSWER

72π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

A cylinder has a diameter of 10 inches and a height of 6 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

1497

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

471 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

A can holds 128π cubic centimeters and has a radius of 4 cm.

What is its height?

A. 8 cm

B. 2 cm

C. 16 cm

D. 32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

1498

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

A cylinder-shaped tank has a radius of 2 m and a height of 5 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

1499

QUESTION ID

S49-M-Q4

TYPE

Short answer

CORRECT ANSWER

62,800 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

1500

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the volume by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

SET 49 • 8.G.C.9

BACKWARD 1

Find the area of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 3.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9π square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

A circle has a diameter of 10 cm.

What is its area? Use 3.14 for π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

78.5 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

A circle has a radius of 4 m.

What is its area in terms of π?

A. 64π square meters

B. 4π square meters

C. 16π square meters

D. 8π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 1 has an area of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

True or false?

A circle with a diameter of 6 has an area of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The radius is 3, so the area is 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism as base area times height • Grade 7 • Geometry • 7.G.B.6

A right prism has a base area of 20 square cm and a height of 7 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

1508

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

140 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

A right triangular prism has a triangle base with area 6 square inches and a length of 10 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

1509

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

A right prism has a base area of 15 square m and a height of 4 m.

What is its volume?

A. 19 cubic meters

B. 30 cubic meters

C. 225 cubic meters

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

1510

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

True or false?

The volume of any right prism is the area of its base times its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

1511

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

A prism with a base area of 9 and a height of 2 has a volume of 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

1512

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 8.G.C.9

FORWARD 1

Give an informal argument for volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of a cylinder is the area of its base times its height.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

1514

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The cylinder is a stack of equal circles (cross sections) with area πr^2; stacking them to height h gives $\pi r^2 \cdot h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

A straight stack of coins and a slanted stack have the same coins and the same height.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

1515

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. At every height the cross sections have the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

An oblique cylinder and a right cylinder have the same radius and height.

How do their volumes compare?

A. The oblique one is larger.

B. It cannot be known.

C. The right one is larger.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

1516

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

True or false?

Cavalieri's principle compares the areas of cross sections at every height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

1517

QUESTION ID

S49-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

True or false?

Leaning a stack of paper to one side increases its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

1518

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 8.G.C.9

FORWARD 2

Use density with volume in a model • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum cylinder has a radius of 2 cm and a height of 10 cm. Aluminum has a density of 2.7 grams per cubic centimeter.

What is its mass? Use 3.14 for π.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

1520

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

339.12 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

A town has 12,000 people and covers 30 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

1521

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

400 people per square mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

A 500 cubic-centimeter block has a mass of 4,000 grams.

What is its density?

A. 8 grams per cubic centimeter

B. 2 grams per cubic centimeter

C. 0.125 gram per cubic centimeter

D. 4,500 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

1522

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

True or false?

If two objects have the same volume, the one with the greater density has the greater mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

1523

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

True or false?

Density is volume divided by mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

1524

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Density is mass divided by volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 8.G.C.9 • GEOMETRY

Find the volume of a cone

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle from its radius or diameter	**7.G.B.4**
BACKWARD 2	Multiply a fraction by a whole number	**5.NF.B.4.a**
FORWARD 1	Explain why a cone's volume is one-third of a cylinder's	**HSG-GMD.A.1**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

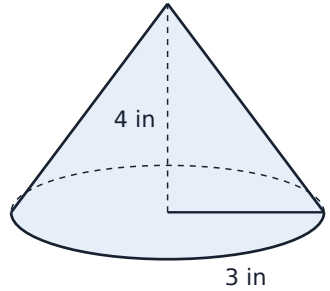
SET 50 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cone in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

1527

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

12π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

A cone has a diameter of 12 feet and a height of 10 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

1528

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

376.8 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

A cone and a cylinder have the same base and the same height. The cylinder holds 90 cubic cm.

How much does the cone hold?

A. 30 cubic cm

B. 90 cubic cm

C. 270 cubic cm

D. 45 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

1529

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

A cone has a volume of 50π cubic units and a radius of 5 units.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

1530

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

True or false?

A cone with a radius of 6 and a height of 2 has a volume of 24π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

1531

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

SET 50 • 8.G.C.9

BACKWARD 1

Find the area of a circle from its radius or diameter • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 6.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

A circle has a diameter of 4 m.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

A circle has a radius of 10 cm.

What is its area? Use 3.14 for π.

A. 314 square cm

B. 31.4 square cm

C. 62.8 square cm

D. 628 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 314 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

1536

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

True or false?

A circle with a radius of 5 has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 8.G.C.9

BACKWARD 2

Multiply a fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 36$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 90$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 48$

A. 144

B. 12

C. 16

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{3} \times 27 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 15 = 45$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 8.G.C.9

FORWARD 1

Explain why a cone's volume is one-third of a cylinder's • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cone and a cylinder have the same base and height. Water from 3 full cones exactly fills the cylinder.

What formula does this suggest for the cone's volume?

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

1545

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$$V = \frac{1}{3}\pi r^2 h$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

Explain why an oblique cone and a right cone with the same base and height have the same volume.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

1546

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

At every height their cross sections are circles with the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

A pyramid and a prism have the same base and height.

How do their volumes compare?

A. The pyramid's is $\frac{1}{3}$ of the prism's.

B. The prism's is $\frac{1}{3}$ of the pyramid's.

C. The pyramid's is $\frac{1}{2}$ of the prism's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

1547

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The pyramid's is 1/3 of the prism's.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

True or false?

A cone's volume formula has the same form as a pyramid's: $\frac{1}{3}$ × base area × height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

1548

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

True or false?

A cone has half the volume of a cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

1549

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A paper cone cup has a radius of 4 cm and a height of 9 cm.

How many full cups fill a cylinder-shaped jug with a radius of 8 cm and a height of 18 cm?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

1551

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

A cone-shaped pile of sand has a radius of 3 m and holds 30π cubic meters.

How tall is the pile?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

1552

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

A cone has a radius of 6 and a height of 8.

What is its slant height?

A. 14

B. 10

C. 48

D. √28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

1553

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

True or false?

Doubling the height of a cone doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

1554

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

True or false?

Doubling the radius of a cone doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

1555

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the volume by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 8.G.C.9 • GEOMETRY

Find the volume of a sphere

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a cube	**6.G.A.2**
BACKWARD 2	Evaluate a formula with a fraction and an exponent	**6.EE.A.2.c**
FORWARD 1	Model objects with spheres, cylinders, and cones	**HSG-MG.A.1**
FORWARD 2	Use volume formulas to solve problems with spheres	**HSG-GMD.A.3**

SET 51 • 8.G.C.9

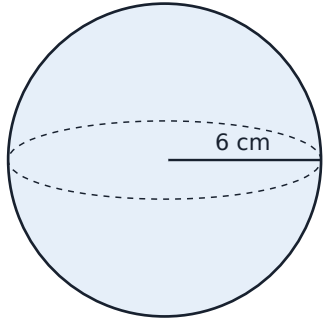
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of the sphere in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

1558

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

288π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

A ball has a diameter of 10 cm.

Find its volume to the nearest tenth. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

1559

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

523.3 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

A hemisphere (half of a sphere) has a radius of 3.

What is its volume in terms of π?

A. 36π

B. 18π

C. 9π

D. 27π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

1560

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

A sphere has a volume of 36π cubic units.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

1561

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

True or false?

A sphere with a radius of 1 has a volume of $\frac{4}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

1562

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

SET 51 • 8.G.C.9

BACKWARD 1

Find the volume of a cube • Grade 6 • Geometry • 6.G.A.2

A cube has edges of 3 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

A cube has edges of $\frac{1}{2}$ foot.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/8 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

A cube has edges of 6 cm.

What is its volume?

A. 18 cubic cm

B. 216 cubic cm

C. 72 cubic cm

D. 36 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 216 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

True or false?

A cube with 5-meter edges has a volume of 15 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 8.G.C.9

BACKWARD 2

Evaluate a formula with a fraction and an exponent • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $V = \frac{4}{3}r^3$. Find V when r = 3.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

Use $V = \frac{4}{3}r^3$. Find V when r = 6.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $V = \frac{1}{2}r^3$. Find V when r = 2.

A. 6

B. 4

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

True or false?

When r = 1, $\frac{4}{3}r^3 = \frac{4}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When r = 2, $\frac{4}{3}r^3 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 32/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 8.G.C.9

FORWARD 1

Model objects with spheres, cylinders, and cones • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

An orange is modeled as a sphere with a diameter of 8 cm.

Estimate its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

1576

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{256}{3}\pi$ cubic centimeters

GRADING NOTE About 268 cubic centimeters is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

A tennis ball can holds 3 balls in a cylinder whose radius equals the ball's radius r and whose height is 6r.

What fraction of the can is filled by the balls?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

1577

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

Which shape is the best model for a scoop of ice cream?

A. A cylinder

B. A sphere

C. A cone

D. A cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

1578

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

True or false?

Modeling Earth as a sphere is a useful approximation even though Earth is slightly flattened.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

1579

QUESTION ID

S51-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

True or false?

A soccer ball is best modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

1580

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems with spheres • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A spherical tank has a radius of 3 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

1582

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

113,040 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r.

What is the ratio of the sphere's volume to the cylinder's volume?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

1583

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

Doubling a sphere's radius multiplies its volume by...

A. 6

B. 2

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

1584

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

True or false?

A sphere with volume 288π has a radius of 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

1585

QUESTION ID

S51-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

True or false?

A hemisphere with radius 6 has a volume of 288π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

1586

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a volume of 144π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 8.G.C.9 • GEOMETRY

Solve real-world problems with the volumes of cylinders, cones, and spheres

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a solid made of two prisms	**5.MD.C.5.c**
BACKWARD 2	Find the volume of a right prism	**7.G.B.6**
FORWARD 1	Model real objects with combinations of solids	**HSG-MG.A.1**
FORWARD 2	Apply volume to design and capacity problems	**HSG-MG.A.3**

SET 52 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

A grain silo is a cylinder with a radius of 6 m and a height of 10 m, topped with half of a sphere with a radius of 6 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

1589

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

504π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

An ice cream cone (radius 3 cm, height 10 cm) is filled and topped with half of a sphere of ice cream with a radius of 3 cm.

Find the total volume of ice cream in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

1590

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

48π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

A cone and a cylinder each have a radius of 2 and a height of 6.

How many full cones of water fill the cylinder?

A. 6

B. 3

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

1591

QUESTION ID

S52-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

A cylinder with a radius of 5 cm holds water. A ball with a radius of 3 cm is dropped in and sinks completely.

How many centimeters does the water level rise?

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

1592

QUESTION ID

S52-M-Q4

TYPE

Short answer

CORRECT ANSWER

1.44 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

True or false?

A sphere with radius r fits exactly in a cylinder with radius r and height 2r. The sphere's volume is $\frac{2}{3}$ of the cylinder's.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

1593

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

SET 52 • 8.G.C.9

BACKWARD 1

Find the volume of a solid made of two prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 4 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

Two cubes with 2-unit edges are joined face to face.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

A 5 by 2 by 1 prism and a 1 by 2 by 1 prism are joined with no overlap.

What is the total volume?

A. 12 cubic units

B. 7 cubic units

C. 20 cubic units

D. 10 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

Volumes of parts that do not overlap can be added to find a total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

A 3 by 3 by 3 cube and a 1 by 1 by 1 cube joined with no overlap have a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 27 + 1 = 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism • Grade 7 • Geometry • 7.G.B.6

A right rectangular prism is 6 cm by 4 cm by 5 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

1601

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

120 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q1/5

A right prism has a base area of 12 square feet and a height of 3 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

1602

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q2/5

A right triangular prism has a base area of 10 square inches and a length of 8 inches.

What is its volume?

A. 40 cubic inches

B. 80 cubic inches

C. 18 cubic inches

D. 800 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

1603

QUESTION ID

S52-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 80 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A cube with edges of 10 m has a volume of 1,000 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

1604

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A 2 by 5 by 10 box has a volume of 17 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

1605

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 8.G.C.9

FORWARD 1

Model real objects with combinations of solids • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

A capsule-shaped vitamin is a cylinder 1 cm long with a radius of 0.3 cm, with half of a sphere of radius 0.3 cm on each end.

What solids make up the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

1607

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A cylinder and two half-spheres (together, one whole sphere)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q1/5

A pencil is modeled as a cylinder with a cone at one end.

Which measurements are needed to find its volume?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

1608

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The radius, the length of the cylinder part, and the height of the cone part.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q2/5

Which solids best model a traffic cone sitting on a square base?

A. A cone on a square prism

B. A sphere on a cylinder

C. Two cylinders

D. A cube on a pyramid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

1609

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A cone on a square prism

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q3/5

True or false?

A model can use simpler shapes even when the real object is not exactly that shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

1610

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q4/5

True or false?

A water bottle is best modeled as one sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

1611

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 52 • FORWARD 1 • Q5/5

SET 52 • 8.G.C.9

FORWARD 2

Apply volume to design and capacity problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A company wants a cylinder-shaped can that holds 500π cubic cm with a radius of 5 cm.

How tall must the can be?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

1613

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q1/5

A cone-shaped funnel has a radius of 6 cm and a height of 10 cm. Water drains at 6π cubic cm per second.

How long does a full funnel take to drain?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

1614

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

20 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q2/5

Each can is a cylinder holding the same volume.

Which design uses the most metal for the curved side?

A. Radius 4 cm, height 6.25 cm

B. Radius 2.5 cm, height 16 cm

C. Radius 5 cm, height 4 cm

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

1615

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Radius 2 cm, height 25 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q3/5

True or false?

Two cans with the same volume can have different surface areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

1616

QUESTION ID

S52-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q4/5

True or false?

A taller can always holds more than a shorter can.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

1617

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 8.SP.A.1 • STATISTICS & PROBABILITY

Construct a scatter plot for bivariate measurement data

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Graph points that represent a real-world situation	**5.G.A.2**
BACKWARD 2	Display a numerical data set in a dot plot	**6.SP.B.4**
FORWARD 1	Represent two-variable data on a scatter plot and describe how t…	**HSS-ID.B.6**
FORWARD 2	Fit an exponential or quadratic function to data and use it to make	**HSS-ID.B.6.a**

SET 53 • 8.SP.A.1

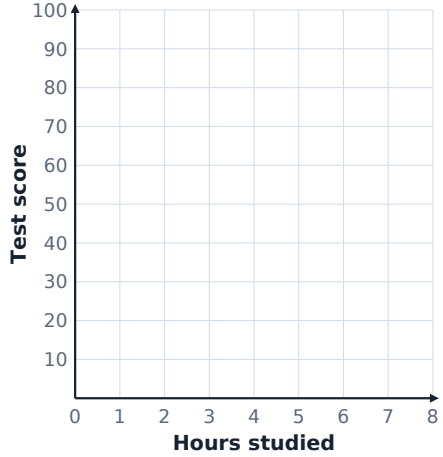
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

Seven students recorded (hours studied, test score):

(1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), (7, 90)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

1620

QUESTION ID

S53-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 50), (2, 60), (3, 60), (4, 70), (5, 80), (6, 80), and (7, 90)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

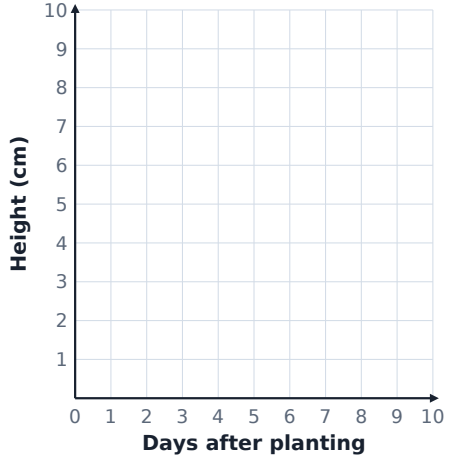
ID S53-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q1/5

A bean plant was measured on different days (days after planting, height in cm):

(2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), (10, 8)

Make a scatter plot of the data on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

1621

QUESTION ID

S53-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (2, 1), (4, 3), (5, 4), (6, 4), (8, 6), (9, 7), and (10, 8)

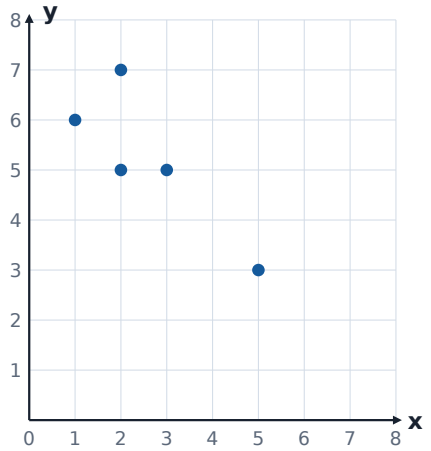
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q2/5

The scatter plot should show the data (1, 6), (2, 5), (3, 5), (5, 3), and (7, 2), but one point is plotted wrong.

Which point is wrong, and where should it be?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

1622

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

The pair (7, 2) is plotted at (2, 7). It should be 7 units right and 2 units up, at (7, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q3/5

Ten students recorded their shoe sizes and heights to make a scatter plot.

What does each point on the scatter plot show?

A. How many students have each shoe size

B. One student's shoe size and height

C. The mean shoe size of the class

D. One student's shoe size only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

1623

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. One student's shoe size and height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A scatter plot of hours slept and test scores puts two points on the grid for each student, one for each variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

1624

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each student is one point: (hours slept, test score).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • MAIN • Q5/5

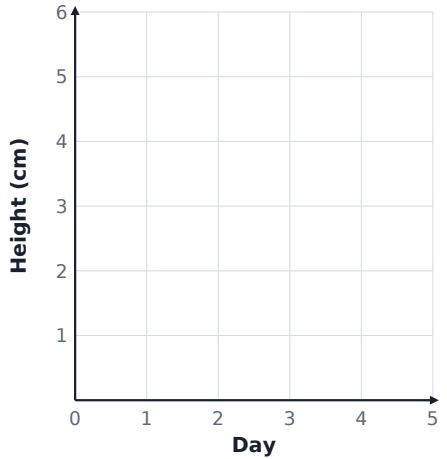
SET 53 • 8.SP.A.1

BACKWARD 1

Graph points that represent a real-world situation • Grade 5 • Geometry • 5.G.A.2

A plant was 2 cm tall on day 1, 4 cm tall on day 2, and 5 cm tall on day 3.

Plot the points (day, height) on the grid.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

1626

QUESTION ID

S53-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), and (3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q1/5

A graph shows points (day, height in cm).

What does the point (4, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

1627

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

On day 4, the height was 3 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q2/5

Points on a graph are (hours, miles).

Which point shows 5 hours and 2 miles?

A. (5, 5)

B. (2, 5)

C. (5, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

1628

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q3/5

True or false?

To plot (3, 6), start at the origin, move 3 units right, then 6 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

1629

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The point (0, 4) is on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

1630

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (0, 4) is on the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 8.SP.A.1

BACKWARD 2

Display a numerical data set in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The numbers of pets of 10 students are 0, 1, 1, 2, 2, 2, 3, 3, 4, and 1.

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

1632

QUESTION ID

S53-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Dots: 1 above 0, 3 above 1, 3 above 2, 2 above 3, 1 above 4, and none above 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many students are shown in the dot plot?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

1633

QUESTION ID

S53-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

Which display shows each value of a data set as a dot above a number line?

A. A box plot

B. A bar graph

C. A histogram

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

1634

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

In a dot plot, a value that appears 3 times has 3 dots stacked above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

1635

QUESTION ID

S53-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

True or false?

A dot plot shows two different measurements for each student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

1636

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A dot plot shows one variable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 1

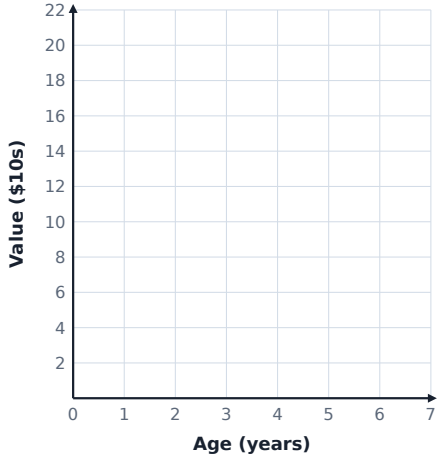
Represent two-variable data on a scatter plot and describe how the variables are related • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Six used bikes have these (age in years, value in tens of dollars):

(1, 20), (2, 17), (3, 15), (4, 12), (5, 10), (6, 8)

Make a scatter plot of the data. Then describe how age and value are related.

Relationship: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

1638

QUESTION ID

S53-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (1, 20), (2, 17), (3, 15), (4, 12), (5, 10), and (6, 8)

Written: As age increases, value decreases: a negative linear association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q1/5

A scatter plot of hours of sleep and reaction time shows points falling from left to right.

Describe how the variables are related.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

1639

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

As sleep increases, reaction time tends to decrease (a negative association).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q2/5

Which pair of variables would most likely show a positive association on a scatter plot?

A. Height and arm span of adults

B. Outside temperature and heating bill

C. Shoe size and test score

D. Age of a car and its value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

1640

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Height and arm span of adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q3/5

True or false?

A scatter plot whose points rise from left to right shows a positive association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

1641

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q4/5

True or false?

If two variables have no association, their scatter plot points lie on a horizontal line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

1642

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE With no association, the points are scattered with no trend.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 53 • FORWARD 1 • Q5/5

SET 53 • 8.SP.A.1

FORWARD 2

Fit an exponential or quadratic function to data and use it to make predictions • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A bacteria count is 100 at hour 0, 200 at hour 1, 400 at hour 2, and 800 at hour 3.

Write a function that fits the data.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

1644

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

100 · 2^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q1/5

A bacteria count fits y = 100 · 2^x after x hours.

Predict the count at hour 6.

Count: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

1645

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6,400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q2/5

Which function fits the data exactly?

x	0	1	2	3	4
y	0	1	4	9	16

A. $y = x^2$

B. $y = 4x$

C. $y = x + 4$

D. $y = 2^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

1646

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q3/5

True or false?

The data (0, 5), (1, 10), (2, 20), and (3, 40) fit $y = 5 \cdot 2^x$ exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q4/5

1647

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q4/5

True or false?

Data whose y-values go up by the same amount at each step are best fit by an exponential function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

1648

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Equal differences mean a linear function fits.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 53 • FORWARD 2 • Q5/5

SET 54 • 8.SP.A.1 • STATISTICS & PROBABILITY

Describe clustering, outliers, and positive, negative, linear, or nonlinear association in a scatter plot

MAIN	Grade 8 main question	**8.SP.A.1**
BACKWARD 1	Describe a data distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Describe the overall pattern of data and any striking deviations	**6.SP.B.5.c**
FORWARD 1	Distinguish between correlation and causation	**HSS-ID.C.9**
FORWARD 2	Interpret the correlation coefficient of a linear fit	**HSS-ID.C.8**

SET 54 • 8.SP.A.1

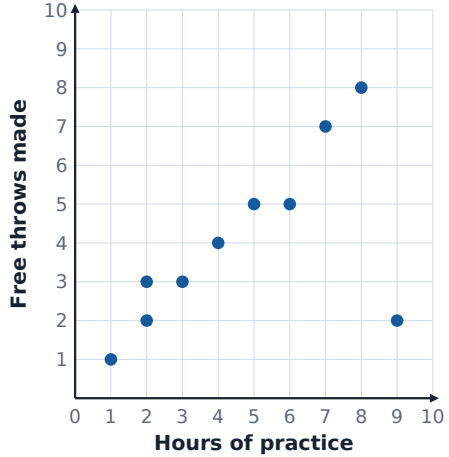
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Describe the association in the scatter plot (positive, negative, or none; linear or nonlinear), and name any outlier.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

1651

QUESTION ID
S54-M-Q1

TYPE
Short answer

CORRECT ANSWER
A positive linear association; the point (9, 2) is an outlier.

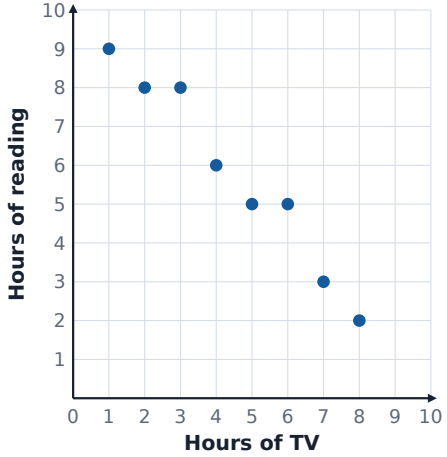
GRADING NOTE Both the association and the outlier are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q1/5

Describe the association in the scatter plot.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

1652

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

A negative linear association: as hours of TV increase, hours of reading decrease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

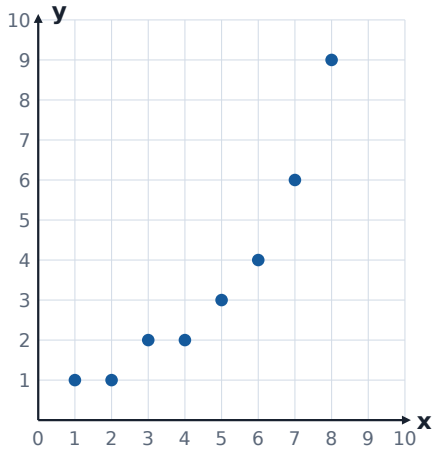
Which best describes the association in the scatter plot?

A. Positive and linear

B. No association

C. Negative and nonlinear

D. Positive and nonlinear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

1653

QUESTION ID

S54-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Positive and nonlinear

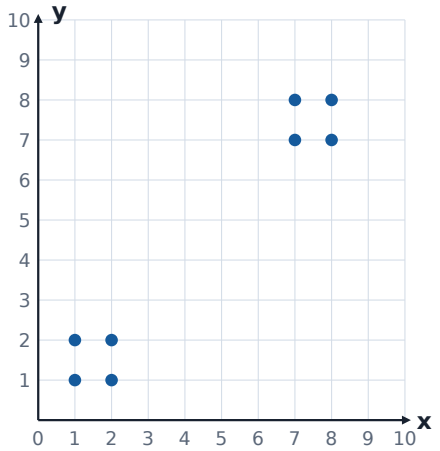
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q3/5

The points in the scatter plot form two groups.

Describe the two clusters.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

1654

QUESTION ID

S54-M-Q4

TYPE

Short answer

CORRECT ANSWER

One cluster has x and y from 1 to 2; the other has x and y from 7 to 8.

GRADING NOTE Any description that locates both clusters is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q4/5

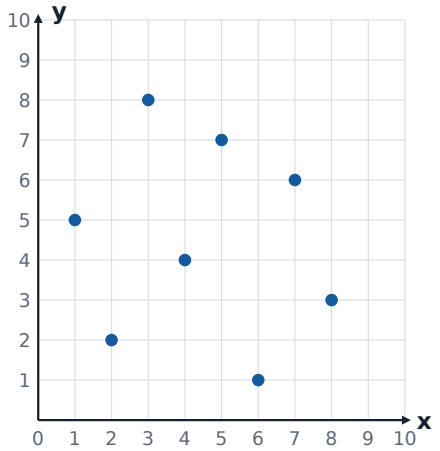
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The scatter plot shows a negative association.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

1655

QUESTION ID

S54-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The points show no association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 54 • MAIN • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 1

Describe a data distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is the data in the dot plot symmetric or skewed?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

1657

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric, centered at 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of the data in the dot plot?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

1658

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which value is far from the rest of the data?

3, 4, 4, 5, 5, 6, 18

A. 18

B. 6

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

1659

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

The data set 2, 3, 3, 4, 4, 4, 5, 5, 6 is symmetric about 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

1660

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

The range of a data set measures its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

1661

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range measures spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 8.SP.A.1

BACKWARD 2

Describe the overall pattern of data and any striking deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the median.

4, 6, 7, 7, 8, 9, 30

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

1663

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q1/5

Which value is a striking deviation from the pattern?

4, 6, 7, 7, 8, 9, 30

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

1664

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q2/5

The data are 10, 12, 12, 13, 14, and 40.

Which measure of center is changed more by the value 40?

A. The mean

B. The median

C. Both are changed equally

D. Neither is changed

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

1665

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q3/5

True or false?

Removing a value that is much larger than the others lowers the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

1666

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q4/5

True or false?

An outlier always changes the median by a large amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

1667

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The median depends on the middle value, so it changes little.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 1

Distinguish between correlation and causation • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more ice-cream sales also have more sunburns.

Does buying ice cream cause sunburns? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

1669

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No. Both go up in hot, sunny weather; a third variable explains the association.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q1/5

A study finds that students who sleep more tend to have higher grades.

Can you conclude that more sleep causes higher grades? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

1670

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The data show an association, not a cause; a randomized experiment is needed to show cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q2/5

Which kind of study can show that one variable causes a change in another?

A. An observational study

B. A randomized experiment

C. A survey of volunteers

D. A scatter plot of survey data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

1671

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q3/5

True or false?

Two variables can be strongly associated without one causing the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

1672

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q4**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q4/5

True or false?

A strong correlation shows that one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

1673

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 54 • FORWARD 1 • Q5/5

SET 54 • 8.SP.A.1

FORWARD 2

Interpret the correlation coefficient of a linear fit • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

The correlation coefficient of a data set is $r = -0.92$.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

1675

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A strong negative linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q1/5

Which shows the stronger linear association, r = 0.45 or r = −0.81? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

1676

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

r = −0.81, because its absolute value is closer to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q2/5

Points lie almost exactly on a line that rises from left to right.

Which value of r fits?

A. −0.98

B. 0.98

C. 0

D. 0.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

1677

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.98

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q3/5

True or false?

The correlation coefficient r is always between −1 and 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

1678

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q4/5

True or false?

r = −0.9 shows a weaker linear association than r = 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S54-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q5/5

1679

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE |−0.9| is greater than |0.5|, so r = −0.9 is stronger.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S54-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 8.SP.A.2 • STATISTICS & PROBABILITY

Informally fit a straight line to a scatter plot and judge how closely it fits

MAIN	Grade 8 main question	**8.SP.A.2**
BACKWARD 1	Decide from a table or graph whether a relationship is proportion…	**7.RP.A.2.a**
BACKWARD 2	Write an equation for a relationship shown in a table or graph	**6.EE.C.9**
FORWARD 1	Fit a linear function to data that suggest a linear association	**HSS-ID.B.6.c**
FORWARD 2	Judge the fit of a line by finding and plotting residuals	**HSS-ID.B.6.b**

SET 55 • 8.SP.A.2

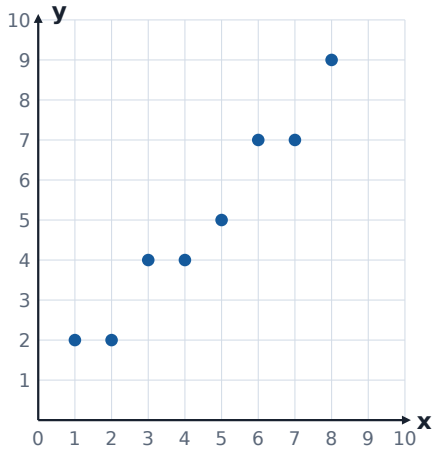
MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Draw a straight line that fits the data. Then write an equation of your line.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

1682

QUESTION ID

S55-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line close to the points, with about as many points above it as below it, for example through (0, 0.5) and (8, 8.5)

Written: The equation of the line drawn, about y = x + 0.5

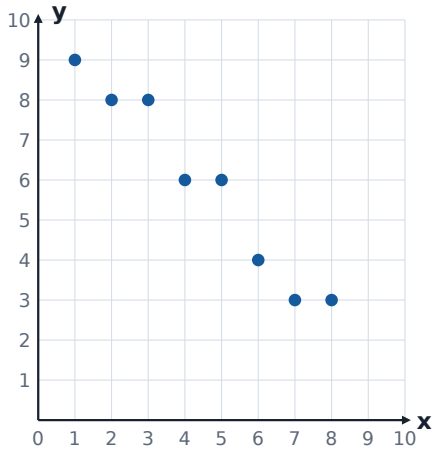
GRADING NOTE Slopes from about 0.8 to 1.2 and y-intercepts from about 0 to 1.5 are accepted when they match the line drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Draw a straight line that fits the data. Use your line to estimate y when x = 9.



Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

1683

QUESTION ID

S55-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line close to the points, with about as many points above it as below it, for example through (0, 10) and (8, 2)

Written: About 1 to 2, read from the line drawn (y = −x + 10 gives 1)

GRADING NOTE Accept an estimate that matches the student's own line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q2/5

Two students drew lines for the same scatter plot. Line A has every point above it. Line B has about half the points above it and half below, all close to it.

Which line fits better? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

1684

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

Line B, because the points are close to it and balanced on both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q3/5

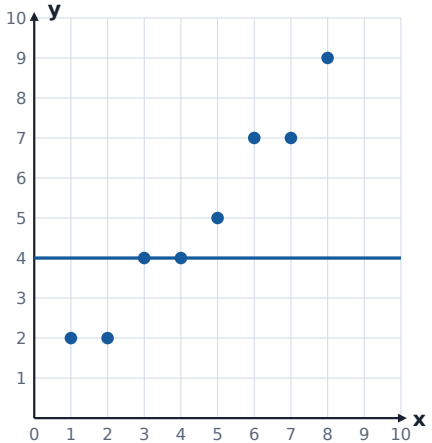
Why is the line on the scatter plot a poor fit?

A. It is not long enough.

B. It does not follow the upward trend of the points.

C. It starts at the y-axis.

D. It passes through too many points.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

1685

QUESTION ID

S55-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. It does not follow the upward trend of the points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q4/5

True or false?

A line that fits a scatter plot well must pass through every point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

1686

QUESTION ID

S55-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A good line is close to the points; it may pass through none of them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

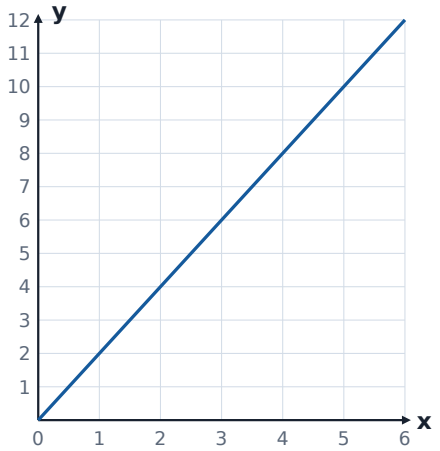
Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 55 • MAIN • Q5/5

SET 55 • 8.SP.A.2

BACKWARD 1

Decide from a table or graph whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Is the relationship shown in the graph proportional? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

1688

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The graph is a straight line through the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q1/5

Is the relationship in the table proportional? Explain.

x	1	2	4
y	3	6	13

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

1689

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

No. 3 ÷ 1 = 3 and 6 ÷ 2 = 3, but 13 ÷ 4 is not 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q2/5

Which graph shows a proportional relationship?

A. A curve through (0, 0) and (2, 4)

B. A straight line through (0, 2) and (2, 6)

C. A horizontal line through (0, 3)

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

1690

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A straight line through (0, 0) and (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q3/5

True or false?

A straight line through (0, 0) and (3, 9) shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

1691

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every straight-line graph shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

1692

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The line must also pass through the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 8.SP.A.2

BACKWARD 2

Write an equation for a relationship shown in a table or graph • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

1694

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q1/5

Write an equation for y in terms of x.

x	0	1	2	3
y	2	3	4	5

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

1695

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q2/5

A line passes through (0, 0), (1, 5), and (2, 10).

Which equation does it show?

A. y = 5x

B. y = 10x

C. x = 5y

D. y = x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

1696

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. y = 5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q3/5

True or false?

The points (1, 3), (2, 6), and (3, 9) fit y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

1697

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q4/5

True or false?

The points (0, 1), (1, 2), and (2, 3) fit y = 2x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

1698

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They fit y = x + 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 1

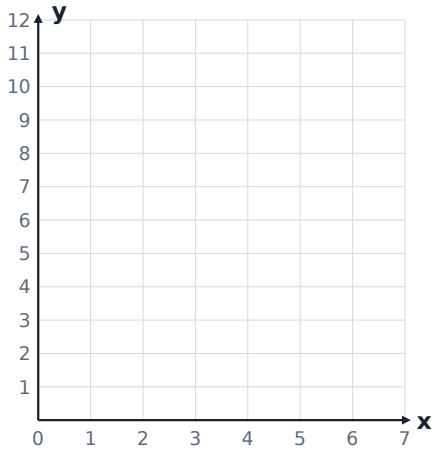
Fit a linear function to data that suggest a linear association • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

ANSWER KEY Reduced copy of the question page, with its answer

The data are (1, 3), (2, 5), (3, 6), (4, 8), (5, 10), and (6, 11).

Make a scatter plot, draw a line that fits the data, and write an equation of your line.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

1700

QUESTION ID

S55-F1-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (1, 3), (2, 5), (3, 6), (4, 8), (5, 10), and (6, 11), and a straight line close to them with points on both sides

Written: The equation of the line drawn, about y = 1.6x + 1.5

GRADING NOTE Slopes from about 1.4 to 1.8 are accepted when they match the line drawn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q1/5

A line of fit passes through (2, 5) and (6, 13).

Write its equation.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

1701

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q2**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q2/5

A line of fit for (hours, dollars) data is y = 12x + 8.

What does it predict for 5 hours?

A. $100

B. $68

C. $25

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

1702

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q3/5

True or false?

A line of fit through (0, 4) and (4, 12) has a slope of 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

1703

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q4/5

True or false?

A line of fit should always be drawn through the first and last data points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

1704

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It should follow the whole trend, not just two points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 55 • FORWARD 1 • Q5/5

SET 55 • 8.SP.A.2

FORWARD 2

Judge the fit of a line by finding and plotting residuals • High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit is y = 2x + 1. A data point is (3, 8).

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

1706

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit is y = 2x + 1. A data point is (5, 9).

Find the residual (actual y minus predicted y).

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

1707

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q2/5

A residual plot shows points scattered above and below 0 with no pattern.

What does this suggest?

A. A curve must be used.

B. Every prediction is exact.

C. The line is a poor fit.

D. The line is a good fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

1708

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The line is a good fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q3/5

True or false?

A residual of 0 means the data point lies on the line of fit.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

1709

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q4/5

True or false?

A residual plot with a clear U shape shows that a linear model fits well.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

1710

QUESTION ID

S55-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A U shape suggests a curve fits better.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 55 • FORWARD 2 • Q5/5

SET 56 • 8.SP.A.3 • STATISTICS & PROBABILITY

Use the equation of a linear model to interpret the slope and intercept and to make predictions

MAIN	Grade 8 main question	**8.SP.A.3**
BACKWARD 1	Explain what a point on the graph of a proportional relationship m…	**7.RP.A.2.d**
BACKWARD 2	Evaluate a formula for given values of its variable	**6.EE.A.2.c**
FORWARD 1	Interpret the slope and intercept of a linear model in context	**HSS-ID.C.7**
FORWARD 2	Evaluate reports and predictions based on data	**HSS-IC.B.6**

SET 56 • 8.SP.A.3

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A line of fit for the height h (inches) of a plant after w weeks is h = 1.5w + 4.

What do the slope and the intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

1713

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: the plant grows about 1.5 inches per week. Intercept: it was about 4 inches tall at week 0.

GRADING NOTE Both meanings are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q1/5

A line of fit for the gas g (gallons) left in a car's tank after m miles is $g = 14 - 0.04m$.

Predict the gas left after 200 miles.

Gas left: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

1714

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q2/5

A model for the water level w (feet) of a pond t days after the rain stops is $w = 12 - 0.3t$.

What does -0.3 mean?

A. The level rises about 0.3 foot each day.

B. The pond is empty after 0.3 day.

C. The level starts at 0.3 foot.

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

1715

QUESTION ID

S56-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The level drops about 0.3 foot each day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A model for the score s after h hours of practice is $s = 6h + 58$.

Predict the score after 5 hours of practice.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

1716

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

88

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q4/5

True or false?

In the model y = 3.2x + 15 for the cost of a taxi ride of x miles, 15 is the cost per mile.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

1717

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 15 is the starting fee; 3.2 is the cost per mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 56 • MAIN • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 1

Explain what a point on the graph of a proportional relationship means • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph of cost (dollars) against pounds is proportional and passes through (1, 3).

What does (1, 3) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

1719

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 pound costs $3; this is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q1/5

A proportional graph of miles against hours passes through (0, 0).

What does (0, 0) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

1720

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

In 0 hours, 0 miles are traveled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q2/5

A proportional graph of (gallons, cost) passes through (1, 4).

What is the unit rate?

A. $5 per gallon

B. $4 per gallon

C. $0.25 per gallon

D. $1 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

1721

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4 per gallon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q3/5

True or false?

On a proportional graph of (hours, pages), the point (1, 20) means 20 pages are read per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

1722

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q4/5

True or false?

On a proportional graph of (tickets, cost), the point (2, 18) shows the unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

1723

QUESTION ID
S56-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The unit rate is shown at x = 1: (1, 9).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 8.SP.A.3

BACKWARD 2

Evaluate a formula for given values of its variable • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1.5w + 4 when w = 6.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

1725

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 300 − 40t when t = 3.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

1726

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 6h + 58 when h = 2.

A. 62

B. 128

C. 66

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

1727

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q3/5

True or false?

When x = 4, 3x + 5 = 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

1728

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3 × 4 + 5 = 17.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When t = 2, 10 − 3t = 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q5/5

1729

GRADING NOTE 10 − 6 = 4.

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 1

Interpret the slope and intercept of a linear model in context • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for the number n of ice pops sold on a day with a high temperature of t °F is n = 4t − 200.

What does the slope mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

1731

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 4 more ice pops are sold for each 1 °F rise in temperature.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q1/5

A model for ice-pop sales is $n = 4t - 200$, where t is the temperature in °F.

Does the intercept -200 make sense in this situation? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

1732

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. It would mean −200 ice pops sold at 0 °F, but sales cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q2/5

A model for a phone's battery level b (percent) after h hours of use is b = 100 − 12.5h.

What does 100 mean?

A. The battery lasts 100 hours.

B. The battery charges 100% each hour.

C. The battery loses 100% each hour.

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

1733

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The battery starts at 100%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q3/5

True or false?

In the model y = 0.8x + 2, y increases by about 0.8 for each increase of 1 in x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

1734

QUESTION ID

S56-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q4/5

True or false?

In a linear model, the intercept always has a sensible meaning in the situation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

1735

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 56 • FORWARD 1 • Q5/5

SET 56 • 8.SP.A.3

FORWARD 2

Evaluate reports and predictions based on data • High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A model of children's heights for ages 2 to 10 is $h = 2.5a + 31$ inches at age a.

A report uses it to predict the height of a 40-year-old. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

1737

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

No. It predicts 131 inches (about 11 feet). The model fits only ages 2 to 10, so using it far outside the data is not reliable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q1/5

A line of fit was made from data collected between 60 °F and 90 °F.

A report uses it to predict sales at 70 °F. Is that reasonable? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

1738

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 70 °F is inside the range of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q2/5

A report says: "People who own more books live longer, so buying books makes you live longer."

What is the main flaw?

A. It counts too few books.

B. It reports a positive association.

C. It treats an association as a cause.

D. It uses a line of fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

1739

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. It treats an association as a cause.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q3/5

True or false?

Predictions far outside the range of the data are less reliable than predictions inside it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

1740

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q4/5

True or false?

A survey of volunteers always represents the whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

1741

QUESTION ID

S56-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 8.SP.A.4 • STATISTICS & PROBABILITY

Construct a two-way table summarizing data on two categorical variables

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Report the number of observations in a data set	**6.SP.B.5.a**
BACKWARD 2	Solve "how many more" and "how many in all" problems with a sc…	**3.MD.B.3**
FORWARD 1	Find joint, marginal, and conditional relative frequencies in a two…	**HSS-ID.B.5**
FORWARD 2	Use a two-way table to find conditional probabilities and decide…	**HSS-CP.A.4**

SET 57 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

60 students were asked whether they have a curfew and whether they have chores. 36 have a curfew. 24 of the students with a curfew have chores. 8 of the students without a curfew have chores.

Complete the two-way table.

	Chores	No chores	Total
Curfew			
No curfew			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

1744

QUESTION ID

S57-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Curfew: 24 chores, 12 no chores, 36 total. No curfew: 8 chores, 16 no chores, 24 total. Totals: 32 chores, 28 no chores, 60 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q1/5

50 students were asked whether they prefer summer or winter. 22 are in Grade 7 and 28 are in Grade 8. 15 of the Grade 7 students and 18 of the Grade 8 students prefer summer.

Complete the two-way table.

	Summer	Winter	Total
Grade 7			
Grade 8			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

1745

QUESTION ID

S57-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Grade 7: 15 summer, 7 winter, 22 total. Grade 8: 18 summer, 10 winter, 28 total. Totals: 33 summer, 17 winter, 50 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q2/5

90 people were asked whether they exercise daily and whether they eat breakfast. 40 exercise daily. 30 of the people who exercise daily eat breakfast. 50 of the 90 people eat breakfast.

Complete the two-way table.

	Breakfast	No breakfast	Total
Exercises			
Does not			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

1746

QUESTION ID

S57-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Exercises: 30 breakfast, 10 no breakfast, 40 total. Does not: 20 breakfast, 30 no breakfast, 50 total. Totals: 50 breakfast, 40 no breakfast, 90 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q3/5

Ten students answered two questions: "Do you have a pet?" and "Do you have a sibling?" Their answers (pet, sibling) are:

yes–yes, yes–no, no–no, yes–yes, no–yes, yes–yes, no–no, no–yes, yes–no, no–no

Complete the two-way table.

	Sibling	No sibling	Total
Pet			
No pet			
Total			

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

1747

QUESTION ID

S57-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Pet: 3 sibling, 2 no sibling, 5 total. No pet: 2 sibling, 3 no sibling, 5 total. Totals: 5 sibling, 5 no sibling, 10 in all.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q4/5

True or false?

A completed two-way table for 60 students can have row totals of 36 and 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

1748

QUESTION ID
S57-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The row totals must add to 60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • MAIN • Q5/5

SET 57 • 8.SP.A.4

BACKWARD 1

Report the number of observations in a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In a survey, 12 students chose soccer, 9 chose basketball, and 7 chose tennis.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

1750

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many observations are shown in the dot plot?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

1751

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q2/5

Each of 25 students named a favorite fruit. 5 different fruits were named.

How many observations are in the data set?

A. 5

B. 125

C. 25

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

1752

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

If 14 students answer yes and 11 answer no, there are 25 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

1753

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The data set 3, 3, 5, 7 has 3 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

1754

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 4 observations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 57 • BACKWARD 1 • Q5/5

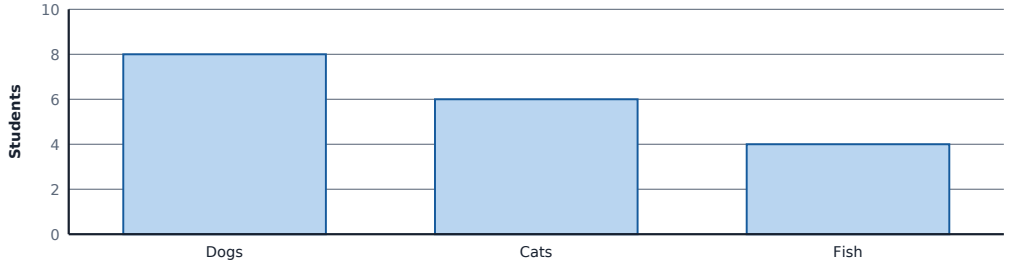
SET 57 • 8.SP.A.4

BACKWARD 2

Solve "how many more" and "how many in all" problems with a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How many more students chose dogs than fish?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

1756

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

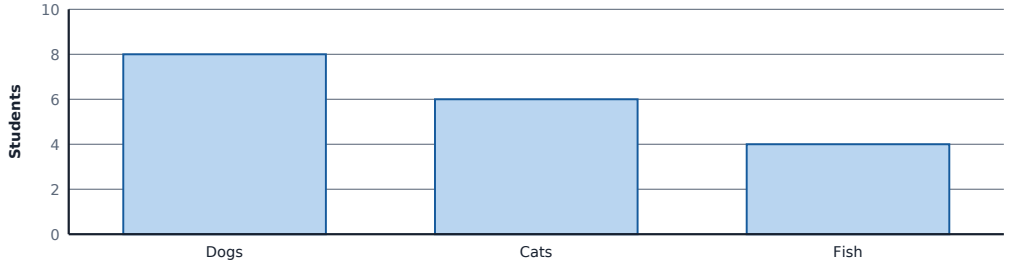
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many students chose a pet in all?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

1757

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

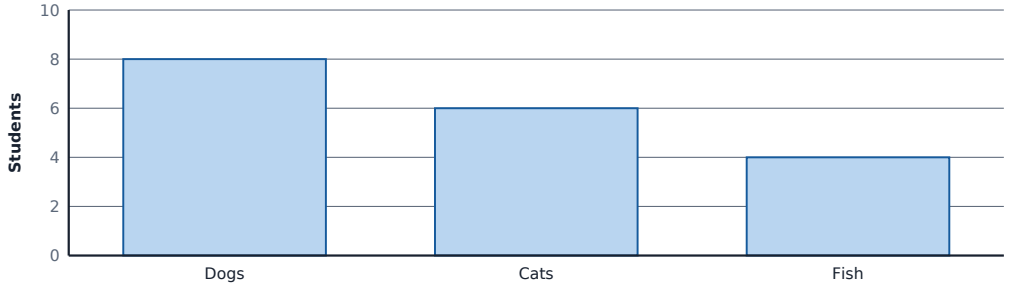
18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

How many fewer students chose fish than cats?



A. 4 **B.** 6 **C.** 10 **D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

1758

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

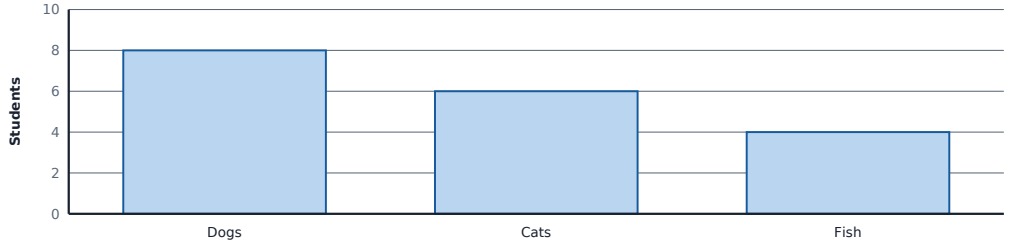
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Fewer students chose cats than fish.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

1759

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 6 chose cats and 4 chose fish.

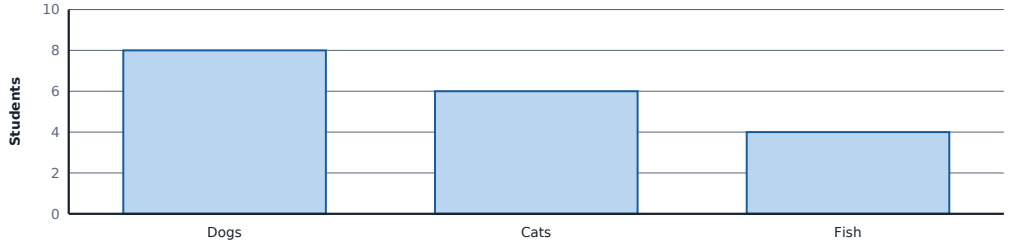
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The same number of students chose dogs and cats.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

1760

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 1

Find joint, marginal, and conditional relative frequencies in a two-way table • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

What is the joint relative frequency of students who have a curfew and have chores?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

1762

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.4

GRADING NOTE $\frac{24}{60}$ or 40% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q1**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q1/5

What is the marginal relative frequency of students who have chores? Round to the nearest hundredth.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

1763

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.53

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q2/5

What is the conditional relative frequency of having chores among students with no curfew?

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. $\frac{2}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{3}$ **D.** $\frac{2}{15}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

1764

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q3**

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The marginal relative frequency of students with a curfew is 0.36.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

1765

QUESTION ID

S57-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $\frac{36}{60} = 0.6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q4/5

True or false?

The joint relative frequencies in the four inner cells add to 2.

	Chores	No chores	Total
Curfew	24	12	36
No curfew	8	16	24
Total	32	28	60

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

1766

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 1 • Q5/5

SET 57 • 8.SP.A.4

FORWARD 2

Use a two-way table to find conditional probabilities and decide whether events are independent • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

A student from the table is chosen at random.

Find P(in club | Grade 8).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

1768

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE $\frac{18}{60}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q1**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are being in the club and being in Grade 7 independent? Explain with conditional probabilities.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

1769

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. P(club | Grade 7) = $\frac{12}{40}$ = 0.3 and P(club | Grade 8) = $\frac{18}{60}$ = 0.3, both equal to P(club) = 0.3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q2/5

A student from the table is chosen at random.

Find P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. 0.6 **B.** 0.12 **C.** 0.4 **D.** 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

1770

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q3/5

True or false?

For a student chosen at random from the table, P(not in club) = 0.7.

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

1771

QUESTION ID

S57-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For the table, P(in club | Grade 7) equals P(Grade 7 | in club).

	In club	Not in club	Total
Grade 7	12	28	40
Grade 8	18	42	60
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

1772

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 0.3 and 0.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 8.SP.A.4 • STATISTICS & PROBABILITY

Use relative frequencies in a two-way table to describe a possible association

MAIN	Grade 8 main question	**8.SP.A.4**
BACKWARD 1	Find a percent of a quantity and the percent one number is of ano…	**6.RP.A.3.c**
BACKWARD 2	Use ratio language to describe two quantities	**6.RP.A.1**
FORWARD 1	Recognize conditional probability and independence in everyday…	**HSS-CP.A.5**
FORWARD 2	Find a conditional probability as the fraction of B's outcomes that…	**HSS-CP.B.6**

SET 58 • 8.SP.A.4

MAIN

Grade 8 main question • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

For each row, what percent of the people use a phone at night?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

1775

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

Sleeps 8+ hours: 30%. Sleeps under 8 hours: 60%.

GRADING NOTE Both percents are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q1/5

Is there evidence of an association between sleep and phone use at night? Explain using the row relative frequencies.

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

1776

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. 60% of those who sleep under 8 hours use a phone at night, but only 30% of those who sleep 8+ hours do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q2/5

Which conclusion do the row relative frequencies support?

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. There is a strong association because 45 is greater than 30.

B. Grade 7 students are more likely to eat breakfast.

C. Grade 8 students are more likely to eat breakfast.

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

1777

QUESTION ID

S58-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. There is no association: 60% of each grade eat breakfast.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent of the people who use a phone at night sleep under 8 hours?

	Phone at night	No phone at night	Total
Sleeps 8+ hours	12	28	40
Sleeps under 8 hours	36	24	60
Total	48	52	100

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

1778

QUESTION ID

S58-M-Q4

TYPE

Short answer

CORRECT ANSWER

75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Grade 8 students are more likely to eat breakfast than Grade 7 students, because 45 is greater than 30.

	Breakfast	No breakfast	Total
Grade 7	30	20	50
Grade 8	45	30	75
Total	75	50	125

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

1779

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both grades have 60%; compare relative frequencies, not counts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 58 • MAIN • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 1

Find a percent of a quantity and the percent one number is of another • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

What percent of 40 is 12?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

1781

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q1/5

Find 60% of 75.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

1782

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q2/5

36 out of 48 is what percent?

A. 75%

B. 48%

C. 12%

D. 36%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

1783

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

15 out of 60 is 15%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

1784

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 15 out of 60 is 25%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

20% of 50 is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

1785

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20% of 50 is 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 8.SP.A.4

BACKWARD 2

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

In a group, 12 people use a phone at night and 28 do not.

Write the ratio of users to non-users in simplest form.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

1787

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3 : 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q1/5

A class has 18 students who walk to school and 12 who ride.

For every 3 students who walk, how many ride?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

1788

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q2/5

In a survey, 30 people eat breakfast and 20 do not.

Which statement describes the ratio?

A. For every 3 who eat breakfast, 2 do not.

B. For every 1 who eats breakfast, 30 do not.

C. For every 3 who eat breakfast, 5 do not.

D. For every 2 who eat breakfast, 3 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

1789

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. For every 3 who eat breakfast, 2 do not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q3/5

True or false?

The ratio 24 : 16 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

1790

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q4/5

True or false?

The ratio 10 : 15 is equivalent to 3 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

1791

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10 : 15 is equivalent to 2 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 1

Recognize conditional probability and independence in everyday situations • High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain the difference between P(sleeps under 8 hours | uses a phone at night) and P(uses a phone at night | sleeps under 8 hours).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

1793

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The first is the chance that a night phone user sleeps under 8 hours; the second is the chance that a person who sleeps under 8 hours uses a phone at night. The condition (the group you look in) is different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q1/5

A fair coin is flipped twice, and the flips are independent.

Is P(second flip is heads | first flip is heads) equal to P(second flip is heads)? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

1794

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Both are $\frac{1}{2}$; the first flip does not change the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q2/5

Which pair of events is most likely independent?

A. Smoking and getting lung disease

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

C. Rain today and carrying an umbrella today

D. Studying more and getting a higher grade

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

1795

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Rolling a 6 on a fair number cube and flipping heads on a fair coin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

If two events are independent, knowing that one happened does not change the probability of the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

1796

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

1797

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 8.SP.A.4

FORWARD 2

Find a conditional probability as the fraction of B's outcomes that are also in A • High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is even, what is the probability that it is greater than 3?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

1799

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins are flipped, and the flips are independent.

Given that at least one coin shows heads, what is the probability that both show heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

1800

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q2/5

Of 100 people, 48 use a phone at night, and 36 of those sleep under 8 hours.

A night phone user is chosen at random. What is P(sleeps under 8 hours | uses a phone at night)?

A. 0.75

B. 0.36

C. 0.48

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

1801

QUESTION ID

S58-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q3/5

True or false?

A card is drawn at random from cards numbered 1 to 10.

Given that the number is odd, the probability that it is 5 or less is $\frac{3}{5}$.

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

1802

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q4/5

True or false?

A fair number cube numbered 1 to 6 is rolled.

Given that the roll is less than 3, the probability that it is 1 is $\frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5

1803

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only 1 and 2 are possible, so it is $\frac{1}{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q5**

High School (Grade 10 in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 58 • FORWARD 2 • Q5/5